

Exploring Relations among Fairness Notions in Discrete Fair Division*

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Abstract

Fair allocation of indivisible items among agents is a fundamental and extensively studied problem. However, *fairness* does not have a single universally accepted definition, leading to many competing fairness notions. Some of these notions are considered stronger or more desirable, but they are also more difficult to guarantee. In this work, we examine 22 different fairness notions and organize them into a hierarchy. Formally, we say that a notion F_1 *implies* another notion F_2 if every F_1 -fair allocation is also F_2 -fair. We give a near-complete picture of implications among fairness notions: for almost every pair of notions, we either prove an implication or give a counterexample demonstrating that the implication does not hold. Although some of these results are already known, many are new. We examine multiple settings, including the allocation of goods, chores, and mixed manna, and different valuation classes like additive, submodular, and subadditive. We believe this work clarifies the relative strengths and applicability of these notions, providing a foundation for future research in fair division. Moreover, we develop an *inference engine* to automate part of our work. It is available as a user-friendly web application and may have broader applications beyond fair division.

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1 Introduction

The problem of fairly allocating items among multiple agents has garnered significant attention in multi-agent systems and game theory. It has many real-world applications like dividing inheritance, distributing natural resources among countries or states, allocating public housing, divorce settlements, and assigning research papers to reviewers.

Research in fair division began with the study of *divisible* resources, such as land, water, and cake [Ste49, Str80, Var74]. *Fairness* was formally defined primarily in two ways: *envy-freeness* (EF) and *proportionality* (PROP). EF ensures that each agent believes they received the best bundle compared to others, while PROP guarantees that each agent’s value for their own bundle is at least $1/n$ times their value for the entire set of items, where n is the number of agents. By the 1980s, EF and PROP allocations were shown to exist and be efficiently computable [Ste49, Str80, Var74].

These positive results no longer hold for indivisible items. For instance, when 5 identical goods must be divided among two agents, an EF or PROP allocation is not possible. However, one can still aim for *approximate* fairness; e.g., an allocation where one agent receives 3 goods and the other receives 2 goods is intuitively as fair as possible. However, when we move beyond such simple examples to the general setting where items are heterogeneous and agents have different preferences, formally defining (approximate) fairness becomes much more complex. As a result, many fairness notions have been proposed for the indivisible setting.

The concept of EF was relaxed to a notion called $EF1^1$ (envy-freeness up to one item), and algorithms were designed to guarantee $EF1$ allocations [Bud11, LMMS04]. A stronger relaxation of EF, called EFX (envy-free up to any item), was introduced in [CKM⁺19]. Despite significant efforts, the existence of EFX allocations remains an open problem. Consequently, relaxations of EFX [CGR⁺23, AMN20, CKMS21] as well as the existence of EFX in special cases [CGM24, PR20, ABR⁺21] have been explored.

A similar story played out for relaxations of PROP: The existence of $PROP1$ (proportional up to one item) was easy to prove [ACIW21], but stronger notions like MMS (maximin share) [KPW18] and $PROPx$ (proportional up to any item) [AMS20] were shown to be infeasible. Thus, many relaxations of these notions have been studied [KPW18, AG24, AMSS23, BGGS21b, CGR⁺23, KM25]. Additionally, enforcing polynomial-time computability, compatibility with efficiency notions (such as Pareto optimality), or other constraints [GMT14, BCE⁺17, BB18, EGK⁺24] makes these problems more challenging, often necessitating the use of weaker fairness concepts.

Hence, for the problem of fairly allocating indivisible items, a variety of fairness notions have been proposed, each offering a different level of perceived fairness. We believe that a systematic comparison of these fairness notions is essential for guiding both practical applications and future research in this area. To contribute to this objective, we present a comprehensive analysis of 22 different fairness notions, examined through the lens of *implications*.

Formally, we say that a fairness notion F_1 *implies* another notion F_2 if every F_1 -fair allocation is also F_2 -fair. Conversely, if we can identify an F_1 -fair allocation that is not F_2 -fair, we have a *counterexample*, demonstrating that F_1 does not imply F_2 . For many well-studied settings in fair allocation (e.g., additive valuations), we give a near-complete picture of the implications among fairness notions. For almost every pair of notions, we either prove that one notion implies the other, or we give a counterexample. These results establish a hierarchy of fairness notions, and the counterexamples highlight the strengths and weaknesses of each notion. See Fig. 1 for the implications in the context of additive valuations over goods and over chores, and Figs. 3–8 (pages 18–20) for other fair division settings.

¹We formally define $EF1$ and other fairness notions in Section 3.

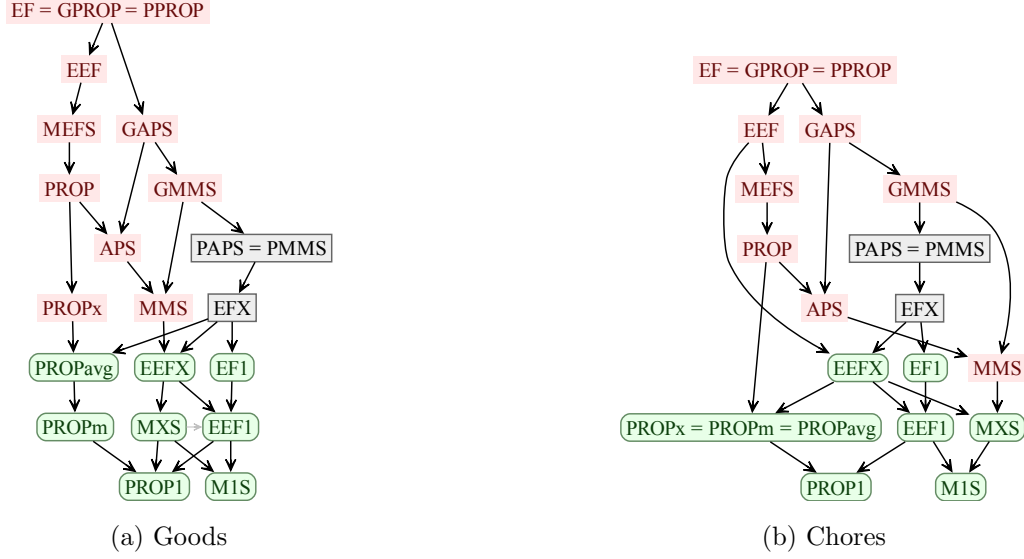


Figure 1: Implications between fairness notions for additive valuations over goods and over chores when agents have equal entitlements. There is a vertex for each fairness notion. Notion F_1 implies notion F_2 iff there is a path from F_1 to F_2 in the graph (except that it is not known whether MXS implies EEf1 for goods). Borderless vertices (red) are infeasible notions, vertices with rounded corners (green) are feasible notions, and the feasibility of the remaining vertices (gray) are open problems. Note that goods and chores have some key differences, e.g., for goods, $\text{PROP} \implies \text{MMS} \implies \text{EEf1} \implies \text{PROp1}$, but for chores, $\text{MMS} \not\Rightarrow \text{PROp1}$, and $\text{PROP} \not\Rightarrow \text{EEf1}$.

The literature on fair division covers a wide range of settings, including distinctions between goods, chores, and mixed manna, as well as varying entitlements (equal vs. unequal) and different classes of valuation functions. Special cases, such as identical valuations or fair division among only two agents, have also been explored. In this paper, we consider all combinations of these aspects of fair division. At first, this leads to a combinatorial explosion of possible settings. However, we address this challenge by encoding our results on implications and non-implications in a machine-readable format, and by implementing an *inference engine* that uses these results to automatically deduce new implications and non-implications.

For instance, if we query the inference engine with “Does epistemic envy-freeness (EEf) imply maximin share (MMS) for goods with additive valuations?”, it would give an affirmative answer based on these three results we show in the paper:

1. EEf implies minimum-EF-share fairness (MEFS).
2. MEFS implies PROP under subadditive valuations (Lemma 27 in Appendix D).
3. PROP implies MMS for superadditive valuations (Lemma 45 in Appendix D).

In Section 5, we give two more examples of inference that are much more involved, highlighting the engine’s usefulness.

1.1 Our Contributions

We establish several implications and counterexamples between fairness notions, leading to near-complete pictures of implications among fairness notions. For example, in Fig. 1a (additive valua-

tions over goods), there are $22 \times 21 = 462$ ordered pairs of notions, but our work leaves only one problem ($\text{MXS} \implies \text{EEF1}$) unresolved. We obtain such near-complete pictures for 76 different fair division settings, including important settings such as additive valuations over goods, chores, and mixed manna, and submodular and general valuations over goods. Section 4.2 precisely enumerates these 76 settings.

Figure 1 and Figs. 3–8 (pages 18–20) depict the near-complete pictures for 13 important settings. Similar figures for other settings can be generated using the companion web application [Sha24].

To achieve these tight results, we first curated a long list of implications and counterexamples, which we list in Tables 1–4 (pages 14–17). These tables list a total of 113 results, of which 46 are either trivial or already known, and the remaining 67 are new results proved in this paper. Moreover, the new counterexamples in this paper were carefully selected to be the simplest possible, so as to clearly illustrate the distinctions between fairness notions. Given the large number of results we prove, this was a challenging task.

Next, we developed a computer program, called the *inference engine*. It takes Tables 1–4 as input, and chains subsets of those results together to generate several additional results. The hundreds of results unearthed this way give us near-complete pictures of implications among fairness notions for many different settings. Our inference engine is implemented as a web application in JavaScript [Sha24]. The engine can be extended beyond fair division (see Section 5), and may have broader applications of independent interest.

We emphasize that the results in Tables 1–4 were proved manually, without using the inference engine.

We also wrote a Python program to automate the verification of counterexamples [Sha26]. It contains functions to check whether an allocation satisfies a fairness notion, and a representation of this paper’s counterexamples (Tables 2 and 4) as Python objects. Thus, to verify the correctness of our counterexamples, instead of manually verifying each counterexample, one can just run the code and verify the correctness of the fairness-checking functions.

Some fairness notions were originally defined for very specific settings; for example, EFX was introduced in [CKM⁺19] only for additive goods. We extend all fairness notions to the most general fair division setting we consider: mixed manna with non-additive valuations and unequal entitlements. In some cases, selecting an appropriate definition proved non-trivial, and we explain the insights that motivated our choices.

1.2 Related Work

[AAB⁺23] survey recent progress and open problems in fair division, where many different fair division settings and fairness notions are considered. [Suk25] gives a similar survey for unequal entitlements.

The most common setting in fair division is equally-entitled agents having additive valuations over goods. For this setting, [BL16] studied implications among 5 fairness notions: CEEI, EF, PROP, MMS, and min-max-share (also called minimum EF share). [ABM18] study implications between multiplicative approximations of fairness notions. [ACTW21] consider implications between EF, PROP, EF1, and PROP1 for mixed manna instead. [CSHS24] study implications for the weighted setting among EF1, PROP1, APS, MNW, and other notions. Over time, as new fairness notions were proposed [CGR⁺23, BEF23, BBKN18, ABC⁺18], their connections with other well-established notions were studied. However, the above works only consider a limited number of fairness notions and fair division settings. Our work, on the other hand, aims to be more exhaustive and thus have broader applicability.

1.3 Structure of the Paper

In Section 2, we formally define the fair division problem and describe different fair division settings. In Section 3, we describe the fairness notions we consider in this paper. In Section 4, we present a summary of our results. In Section 5, we describe our inference engine. Section 6 contains concluding remarks and open problems.

Appendices A and C contain details on fair division settings and fairness notions. Appendices D, E, F contain proofs of our implication and non-implication results. Appendix G contains results on (in)feasibility of fairness notions.

2 Preliminaries

In the fair division problem, there is a finite set M of items that must be distributed among a finite set N of agents fairly. Formally, we are given as input a *fair division instance* $\mathcal{I} := (N, M, V, w)$. Here $w := (w_i)_{i \in N}$ is a collection of positive numbers that sum to 1, and $V := (v_i)_{i \in N}$ is a collection of functions, where $v_i : 2^M \rightarrow \mathbb{R}$ and $v_i(\emptyset) = 0$ for each $i \in N$. v_i is called agent i 's *valuation function*, and w_i is called agent i 's *entitlement*. Our task is to find a fair allocation. An *allocation* $A := (A_i)_{i \in N}$ is a collection of pairwise-disjoint subsets of M such that $\bigcup_{i \in N} A_i = M$. The set A_i is called agent i 's *bundle* in A .

For any integer $k \geq 0$, define $[k] := \{1, 2, \dots, k\}$. We generally assume without loss of generality that $N = [n]$ and $M = [m]$. For an agent i and item j , we often write $v_i(j)$ instead of $v_i(\{j\})$ for notational convenience. We denote a fair division instance by $(N, M, V, -)$ when entitlements are equal. For any function $u : 2^M \rightarrow \mathbb{R}$ and sets $S, T \subseteq M$, the *marginal value* of S over T is defined as $u(S \mid T) := u(S \cup T) - u(T)$.

2.1 Fairness Notions

A *fairness notion* F is a function that takes as input a fair division instance $\mathcal{I}$, an allocation A , and an agent i , and outputs either true or false. When $F(\mathcal{I}, A, i)$ is true, we say that allocation A is F -fair to agent i , or that agent i is F -satisfied by allocation A . Allocation A is said to be F -fair if it is F -fair to every agent.

A fairness notion F is said to be *feasible* if for every fair division instance, there exists an F -fair allocation. We say that a notion F_1 of fairness *implies* another notion F_2 of fairness if every F_1 -fair allocation is also an F_2 -fair allocation. An allocation A is $(F_1 + F_2)$ -fair to an agent i if it is both F_1 -fair and F_2 -fair to agent i .

2.2 Fair Division Settings

We study many fair division settings in this paper. A setting is given by multiple *features*. By picking different values of these features, we get many different settings. We consider 5 features in this paper: (i) whether entitlements are equal, (ii) whether there are only two agents, (iii) whether agents have identical valuations, (iv) valuation function type, (v) marginal values. The first three are self-explanatory. We give an overview of the last two, and defer the details to Appendix A.

Valuation Function Type: This feature indicates how values of different sets of items are related to each other. We consider many popular function types like additive, subadditive, super-additive, submodular, supermodular, and general functions.

Marginal values: This feature indicates the possible marginal values items can have. Agent i 's marginal value for item j over set S is given by $v_i(j \mid S) := v_i(S \cup \{j\}) - v_i(S)$. We consider

several marginal value types, e.g., non-negative (goods), non-positive (chores), bivalued ($\{a, b\}$), binary ($\{0, 1\}$), negative binary ($\{0, -1\}$).

3 Fairness Notions

We now list all the fairness notions we consider in this paper. Due to the large number of notions we consider, we describe them briefly; details, motivation, and examples can be found in the papers we cite for each notion, or in the survey papers cited in Section 1.2.

3.1 Envy-Based Notions

Definition 1 (EF). Let $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$ be a fair division instance. In an allocation A , an agent $i \in [n]$ envies another agent $j \in [n] \setminus \{i\}$ if $v_i(A_i)/w_i < v_i(A_j)/w_j$. Agent i is envy-free in A (or A is EF-fair to i) if she does not envy any other agent in A .

For unequal entitlements, most papers use the term WEF (weighted EF), but we use the term EF instead to emphasize that unequal entitlements is a property of the setting, not the notion. It is easy to see that EF allocations may not exist, so several relaxations have been studied. Two of the most popular relaxations of EF are EF1 (envy-free up to one item) [Bud11, LMMS04], and EFX (envy-free up to any item) [CKM⁺19].

Definition 2 (EF1). Let $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$ be a fair division instance. In $\mathcal{I}$, an allocation A is EF1-fair to agent i if for every other agent j , either i does not envy j , or $v_i(A_i)/w_i \geq v_i(A_j \setminus \{g\})/w_j$ for some $g \in A_j$, or $v_i(A_i \setminus \{c\})/w_i \geq v_i(A_j)/w_j$ for some $c \in A_i$.

Equivalently, A is EF1-fair to i if for every other agent j and some $S \subseteq A_i \cup A_j$ such that $|S| \leq 1$, we have $v_i(A_i \setminus S)/w_i \geq v_i(A_j \setminus S)/w_j$.

EFX was originally defined by [CKM⁺19] for additive valuations over goods: an agent i strongly envies an agent j if, even after removing a good of non-zero value from j 's bundle, i still envies j . An allocation is EFX-fair to agent i if she does not strongly envy anyone. We extend this definition to non-additive valuations over mixed manna.

Definition 3 (EFX). Let $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$ be a fair division instance. In $\mathcal{I}$, an allocation A is EFX-fair to agent i if for each $j \in [n] \setminus \{i\}$, either i does not envy j , or both of the following hold:

1. Removing any positively-valued subset from j 's bundle eliminates i 's envy towards j :

$$\frac{v_i(A_i)}{w_i} \geq \frac{\max(\{v_i(A_j \setminus S) : S \subseteq A_j \text{ and } v_i(S \mid A_i) > 0\})}{w_j}.$$

2. Removing any negatively-valued subset from i 's bundle eliminates i 's envy towards j :

$$\frac{\min(\{v_i(A_i \setminus S) : S \subseteq A_i \text{ and } v_i(S \mid A_i \setminus S) < 0\})}{w_i} \geq \frac{v_i(A_j)}{w_j}.$$

Definition 3 looks different from the original definition by [CKM⁺19], and also differs from EFX₀, an alternative definition of EFX studied by many papers [PR20, CKMS21, CGM24]. However, we show in Appendix C.1 that Definition 3 is equivalent to the original definition of EFX for submodular valuations over goods and submodular valuations over chores, and equivalent to EFX₀ when marginals are (strictly) positive or negative. Thus, Definition 3 is a generalization of known definitions. In Appendix C.1, we give more details on our motivation for defining EFX this way.

3.2 Proportionality-Based Notions

Definition 4 (PROP). *Let $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$ be a fair division instance. For $\mathcal{I}$, agent i 's proportional share is $w_i \cdot v_i([m])$. Allocation A is proportional (PROP) to agent i if $v_i(A_i) \geq w_i \cdot v_i([m])$.*

A popular relaxation of PROP is PROP1 (proportional up to one item), where each agent believes her bundle is better than the proportional share after taking some good or giving away some chore.

Definition 5 (PROP1, [CFS17]). *Let $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$ be a fair division instance. In $\mathcal{I}$, an allocation A is PROP1-fair to agent i if either $v_i(A_i) \geq w_i \cdot v_i([m])$, or $v_i(A_i \cup \{g\}) > w_i \cdot v_i([m])$ for some $g \in [m] \setminus A_i$, or $v_i(A_i \setminus \{c\}) > w_i \cdot v_i([m])$ for some $c \in A_i$.*

Note that Definition 5 uses strict inequalities, whereas most papers do not. This definition is from [Fei25], and it makes PROP1 a slightly stronger notion without altering its fundamental properties. See Appendix C.4 for details.

PROPx (PROP up to any item) [AMS20, LLW22] is another relaxation of PROP. It was originally defined for additive valuations over goods: an agent i is PROPx-satisfied if transferring any good to her bundle makes her proportionally-satisfied. We extend this definition to non-additive valuations over mixed manna.

Definition 6 (PROPx). *For a fair division instance $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$, an allocation A is said to be PROPx-fair to agent i iff either $v_i(A_i) \geq w_i \cdot v_i([m])$ or both of these conditions hold:*

1. $v_i(A_i \cup S) > w_i \cdot v_i([m])$ for every $S \subseteq [m] \setminus A_i$ such that $v_i(S \mid A_i) > 0$.
2. $v_i(A_i \setminus S) > w_i \cdot v_i([m])$ for every $S \subseteq A_i$ such that $v_i(S \mid A_i \setminus S) < 0$.

Definition 6 differs from [AMS20] and [LLW22] in two ways. First, we use strict inequalities in Definition 6, similar to our definition of PROP1. Second, we replaced ‘transferring a good’ by ‘transferring a positive-valued subset’, and ‘transferring out a chore’ by ‘transferring out a negative-valued subset’, similar to our definition of EFX. We prove in Appendix C.5 that, just like with EFX, our definition of PROPx simplifies for submodular valuations over goods, for submodular valuations over chores, and for (strictly) positive or negative marginals.

PROPAvg (PROP up to the min-avg item) [KM25], and PROPM (PROP up to the minimax item) [BGS21a] are strengthenings of PROP1 and relaxations of PROPx. We formally describe them in Appendix C.6. Just like PROPx, our formal definitions for PROPM and PROPAvg differ from the original definitions, and our definitions simplify for special cases. See Appendix C.6 for details.

3.3 Maximin Share and AnyPrice Share

For equal entitlements, an agent’s maximin share (MMS) [Bud11] is the maximum value she can obtain by partitioning the goods into n bundles and picking the worst one. An allocation is MMS-fair to her if her bundle’s value is at least her maximin share.

Definition 7 (MMS, [Bud11]). *Let $([n], [m], (v_i)_{i=1}^n, -)$ be a fair division instance. Let $\Pi_n([m])$ be the set of all n -partitions of $[m]$. Define agent i 's maximin share to be*

$$\text{MMS}_i := \max_{P \in \Pi_n([m])} \min_{j=1}^n v_i(P_j).$$

An allocation A is MMS-fair to agent i if $v_i(A_i) \geq \text{MMS}_i$.

Weighted MMS (WMMS) [FGH⁺19] and pessimistic share (pessShare) [BEF23] are well-known extensions of MMS to the unequal entitlements setting. See Appendix C.2 for details.

AnyPrice Share (APS) [BEF23] is another fairness notion that is inspired by MMS.

Definition 8 (APS, [BEF23]). *For a fair division instance $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$, agent i 's AnyPrice Share (APS) is defined as*

$$\text{APS}_i := \min_{p \in \mathbb{R}^m} \max_{S \subseteq [m]: p(S) \leq w_i \cdot p([m])} v_i(S).$$

Here p is called the price vector. An allocation A is APS-fair to agent i if $v_i(A_i) \geq \text{APS}_i$.

[BEF23] define APS for goods and for chores, whereas Definition 8 works for mixed manna. They also give an alternative definition of APS, called the *dual definition*, which does not involve prices at all. See Appendix C.3 for a comparison of these definitions.

3.4 Derived Notions

New fairness notions can be obtained by systematically modifying existing notions. We start with two related concepts, epistemic fairness [ABC⁺18] and minimum fair share [CGR⁺23].

Definition 9 (epistemic fairness). *Let F be a fairness notion. An allocation A is epistemic- F -fair to an agent i if there exists another allocation B that is F -fair to agent i and $B_i = A_i$. B is called agent i 's epistemic- F -certificate for A .*

Definition 10 (minimum fair share). *For a fair division instance $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$ and fairness notion F , let $\mathcal{A}(\mathcal{I}, F, i)$ be the set of allocations that are F -fair to agent i . Then A is minimum- F -share-fair to agent i if there exists an allocation $B \in \mathcal{A}(\mathcal{I}, F, i)$ such that $v_i(A_i) \geq v_i(B_i)$. Here B is called agent i 's minimum- F -share-certificate for A . Equivalently, an allocation A is minimum- F -share-fair to agent i if $v_i(A_i)$ is at least her minimum- F -share, defined as*

$$\text{minFS}(\mathcal{I}, F, i) := \min_{X \in \mathcal{A}(\mathcal{I}, F, i)} v_i(X_i).$$

We now describe pairwise [CKM⁺19] and groupwise [BBKN18] fairness.

Definition 11 (restricting, pairwise and groupwise fairness). *Let $\mathcal{I} := (N, M, (v_i)_{i \in N}, w)$ be a fair division instance and A be an allocation. For a subset $S \subseteq N$ of agents, where $|S| \geq 2$, let $\text{restrict}(\mathcal{I}, A, S)$ be the pair $(\mathcal{I}^{(S)}, A^{(S)})$, where allocation $A^{(S)} := (A_j)_{j \in S}$, instance $\mathcal{I}^{(S)} := (S, \bigcup_{j \in S} A_j, (v_j)_{j \in S}, \hat{w})$, and weights $\hat{w}_j := w_j / \sum_{k \in S} w_k$.*

A is pairwise- F -fair to agent i if for all $j \in N \setminus \{i\}$, $A^{\{i,j\}}$ is F -fair to i in the instance $\mathcal{I}^{\{i,j\}}$. A is groupwise- F -fair to agent i if for all non-empty $S \subseteq N \setminus \{i\}$, $A^{\{i\} \cup S}$ is F -fair to i in the instance $\mathcal{I}^{\{i\} \cup S}$.

In this paper, we consider these derived notions: Epistemic envy-freeness (EEF), epistemic EFX (EEFX), epistemic EF1 (EEF1), minimum EF share (MEFS), minimum EFX share (MXS), minimum EF1 share (M1S), pairwise proportionality (PPROP), pairwise MMS (PMMS), pairwise APS (PAPS), groupwise proportionality (GPROP), groupwise MMS (GMMS), groupwise APS (GAPS).

4 Summary of Results

4.1 Manually-Proved Results

We first prove several implications and non-implications among fairness notions manually, i.e., without using the inference engine. We summarize implications in Tables 1 and 3 (pages 14 and 16), and defer the proofs to Appendix D. We state several counterexamples in Tables 2 and 4 (pages 15 and 17), and defer the proofs to Appendices E and F, respectively. All of these counterexamples, except Lemmas 123 and 124, can also be found in the checker program [Sha26], where they are written as Python objects. (Lemmas 123 and 124 are very large instances where agents have subadditive valuations, so standard fairness-checking algorithms are computationally intractable for them.)

In Appendix D.9, we prove additional implications for unit-demand valuations. We list results regarding the feasibility and infeasibility of fairness notions in Appendix G.

4.2 Automatically-Inferred Results

After proving implications and counterexamples manually, we feed them into our inference engine, which uses them to infer many more results. These results give us a near-complete picture of implications, similar to Fig. 1, for many different fair division settings.

In particular, for each of the following 76 settings, we get at most 3 unresolved implications, i.e., there are at most three pairs (F_1, F_2) where it is not known whether F_1 implies F_2 (whereas there are $22 \times 21 = 462$ ordered pairs of notions):

- $$\begin{array}{ccccccc} \text{entitlements} & & \text{no. of agents} & & \text{identical valuations} & & \text{valuation type} \\ \{ \text{equal, unequal} \} \times \{ n \geq 2, n = 2 \} \times \{ \text{ident, non-ident} \} \times \{ \text{additive} \} \times \\ \text{marginal values} & & & & & & \\ \{ \text{goods, chores, mixed, } \{0, 1\}, \{-1, 0\} \} \end{array}$$
- $$\begin{array}{ccccccc} \text{entitlements} & & \text{no. of agents} & & \text{identical valuations} & & \text{valuation type} \\ \{ \text{equal} \} \times \{ n \geq 2, n = 2 \} \times \{ \text{ident, non-ident} \} \times \{ \text{submod, subadditive, general} \} \times \\ \text{marginal values} & & & & & & \\ \{ \text{goods, } \mathbb{R}_{>0}, \{0, 1\} \} \end{array}$$

The almost-complete resolution of the above settings can be verified by running the `countOpenProblems.js` script included with the inference engine [Sha24].

We present implication diagrams for some important settings in Figs. 3–8 (pages 18–20). Dashed lines represent unresolved implications. Diagrams for other settings can be viewed in the inference engine web app [Sha24].

5 Inference Engine

We wrote a computer program, called the *inference engine*. We initialize it with a list of implications and counterexamples (Section 4.1), and then we repeatedly query it with a fair division setting. For each query, it infers additional implications and counterexamples using a method similar to transitive closure. Although the inference procedure is simple, the engine can still make non-trivial inferences. We present two such examples:

1. EEFX does not imply PROPavg (for equal entitlements over additive goods). This follows from a chain of (non-)implications: EEFX is implied by APS (Lemmas 48 and 41), APS does not imply PROPm (Example 92), and PROPm is implied by PROPavg. Finding such a chain

can be difficult if one is not intimately aware of all implications and counterexamples. The engine helps uncover many such non-obvious insights.

2. APS does not imply EEF1 for unequal entitlements (over additive goods). To prove this, one cannot find a chain as in the previous example, since it does not exist. However, for the simpler case of binary valuations among two agents, APS is equivalent to PROP1 (Lemmas 56 and 55), EEF1 is equivalent to M1S (Lemmas 51 and 22), and PROP1 does not imply M1S for this simpler case (Example 96).

For each fair division setting that we query the engine with, it summarizes its inferred results as a directed acyclic graph (DAG). See Fig. 1 and Figs. 3–8 (pages 18–20) for examples of the program’s output for various important fair division settings. The program is implemented as a web application; see Fig. 2 for the program’s screenshot. Its source code is available on Github [Sha24].

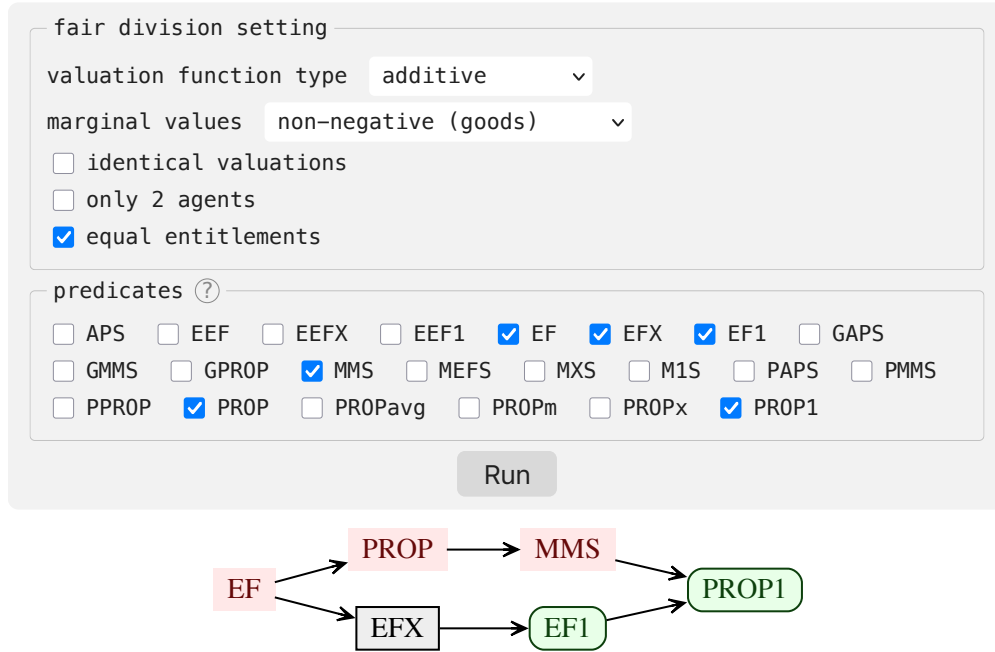


Figure 2: Screenshot from the inference engine’s web interface for fair division.

Our program is not limited to just fair division. It can be used more broadly for *conditional predicate implications*. A *predicate* is a function whose co-domain is $\mathbb{B} := \{\text{true}, \text{false}\}$. Given two predicates $\phi_1, \phi_2 : \Omega \rightarrow \mathbb{B}$, we say that ϕ_1 *implies* ϕ_2 conditioned on $S \subseteq \Omega$, denoted as $\phi_1 \implies_S \phi_2$, if $\phi_1(x) \implies \phi_2(x)$ for all $x \in S$. In fair division, Ω is the set of all pairs $(\mathcal{I}, A)$, where $\mathcal{I}$ is a fair division instance and A is an allocation for $\mathcal{I}$. A fair division setting is a subset of Ω , and fairness notions are predicates.

The inference engine’s input is a tuple $(\mathcal{F}, \Phi, I, C)$. $\mathcal{F}$ is a set family over a ground set Ω . In fair division, each set in $\mathcal{F}$ represents a setting. Since Ω can be uncountable, we represent sets in $\mathcal{F}$ implicitly (see Appendix H). Moreover, given $S_1, S_2 \in \mathcal{F}$, we should be able to efficiently tell whether $S_1 \subseteq S_2$. Φ is a set of predicates over Ω . I is a set of *conditional implications*, i.e., a set of triples $(\phi_1, \phi_2, S) \in \Phi \times \Phi \times \mathcal{F}$ where $\phi_1 \implies_S \phi_2$. C is a set of *conditional counterexamples*, i.e., a set of triples $(\phi_1, \phi_2, S) \in \Phi \times \Phi \times \mathcal{F}$, where $\phi_1(x) \not\implies \phi_2(x)$ for some $x \in S$.

We repeatedly query the engine with a set $S \in \mathcal{F}$, and it outputs all implications and counterexamples conditioned on S , even those not explicitly present in I and C .

The inference engine works in two steps. In step 1, we find all implications conditioned on S . To do this, we simply select implications from I that are conditioned on supersets of S , and compute their transitive closure. In step 2, we find all counterexamples conditioned on S . To do this, for each $(\phi_1, \phi_2, T) \in C$, we first find all implications conditioned on T like in step 1. Next, if $\phi_1 \implies_T \phi'_1$ and $\phi'_2 \implies_T \phi_2$, then we can infer that $\phi'_1 \not\implies_T \phi'_2$, because otherwise, by transitivity, we get $\phi_1 \implies_T \phi_2$. Using this technique, we expand the set of all counterexamples. Then we select counterexamples conditioned on subsets of S .

We can extend the inference engine to also make inferences about feasibility and infeasibility of fairness notions using data from the tables in Section 4.1. Specifically, if $F_1 \implies_S F_2$ and F_1 is feasible for setting S , then F_2 is also feasible for S . Contrapositively, if $F_1 \implies_S F_2$ and F_2 is infeasible for setting S , then F_1 is infeasible for setting S .

6 Conclusion and Open Problems

We prove several (non-)implications between fairness notions, and for many settings, we give an almost complete picture of implications. We believe our results would help inform further research in fair division. This would be especially useful if one wants to extend a fair division result to a stronger notion or a more general setting, or study a weaker notion or a simpler setting for a hard problem.

Our framework can easily accommodate new fairness notions. One just needs to prove a few key implications and counterexamples, and the rest can be inferred. In fact, we originally started with only 18 fairness notions, and gradually expanded the list to 22 notions as we found out about them. When we added PROPavg [KM25], for example, we only had to add the following 5 results: $\text{PROP}_x \implies \text{PROPavg}$, $\text{PROPavg} \implies \text{PROPM}$, $\text{EFX} \implies \text{PROPavg}$, $\text{PROPavg} \not\implies \text{PROP}_x$, and $\text{PROPM} \not\implies \text{PROPavg}$. The inference engine inferred PROPavg’s relationship to the remaining notions.

In Fig. 1, the only notions having unresolved feasibility are EFX and PMMS. For mixed manna, even the existence of MXS allocations is open. For equally-entitled agents having additive valuations over goods or over chores, EF1 and Pareto optimal (PO) allocations are known to exist [CKM⁺19, BKV18, Mah26], but their efficient computation remains open. Relaxing the problem to EEF1+PO can be a helpful first step.

Here are four interesting open problems regarding implications that we could not resolve:

1. For additive goods (equal entitlements), does MXS imply EEF1? Note that the implication holds for the special case of two agents (Lemma 22 in Appendix D).
2. For additive goods (unequal entitlements), does APS imply PROP1? This is open even for two agents.
3. For submodular goods (equal entitlements), does MXS imply PROP1? This is true for binary marginals (Lemma 68 in Appendix D).
4. For submodular goods with binary marginals (equal entitlements), does pairwise-MMS imply MMS?

For less-studied settings, like non-additive valuations over chores, many implications are still open.

Another interesting direction is to study implications of the form $F_1 + \text{PO} \implies F_2 + \text{PO}$. For additive valuations, we have already resolved most questions of this form. This is because if $F_1 \implies F_2$, then $F_1 + \text{PO} \implies F_2 + \text{PO}$. On the other hand, most of our counterexamples use identical valuations, where every allocation is trivially PO.

We omitted some fairness notions from our work because they are fundamentally different from the notions we consider. Equitability (EQ) [BT96], and its relaxations like EQ1 and EQX

[AAB⁺23, GMT14], compare different utility functions with each other. Notions like CEEI [Var74] and maximum Nash welfare [CKM⁺19] include an aspect of efficiency in addition to fairness. We also didn't study fair division of divisible items [Ste49, Str80, Var74], or a mix of divisible and indivisible items [LLSW24]. Nevertheless, we believe that these notions and settings offer an interesting line of research, and our inference engine (Section 5) can be readily adapted for them.

Some fairness notions and settings don't fit our model (Section 2), so it is unclear how to systematically represent results about them and extend the inference engine (Section 5) for them. Examples of such settings include constrained fair division [GMT14, BCE⁺17, BB18, EGK⁺24], notions based on social graphs [ABC⁺18], parametrized notions like WEF(x, y) [CSHS24], and multiplicative approximations of fairness notions [ABM18]. Specifically, studying multiplicative approximations presents the following challenges:

1. For all $\alpha \in (0, 1]$, α -EF1 implies $\alpha/(1 + \alpha)$ -PMMS, and α -PMMS implies $\alpha/(2 - \alpha)$ -EF1 (Propositions 3.8 and 4.6 in [ABM18]). It's unclear how to properly visually depict such results, since we can have infinite chains like $\text{EF1} \implies 1/2\text{-PMMS} \implies 1/3\text{-EF1} \implies 1/4\text{-PMMS} \dots$.
2. To infer new implications using existing ones, we need to be able to represent, compose, and evaluate arbitrary functions of approximation ratios. For some results, such representation is non-trivial. For example, $\text{MMS} \implies \text{EEFX}$ (Lemma 41 in Appendix D.6), but $(1 - \varepsilon)\text{-MMS}$ does not imply $\varepsilon\text{-EEFX}$ for $n = 2$ and any $\varepsilon > 0$ (Proposition 4.8 in [ABM18]).

Extending our techniques to these settings and notions would be an interesting line of research.

Table 1: Implications among fairness notions. For conciseness, we write $\text{ep-}F$ instead of $\text{epistemic-}F$, $\text{min-}F\text{-sh}$ instead of $\text{minimum-}F\text{-share}$, $\text{g-}F$ instead of $\text{groupwise-}F$, and $\text{p-}F$ instead of $\text{pairwise-}F$. For known results, citations can be found in the corresponding lemma statements.

		valuation	marginals	identical	n	entitlements		
1	$F \Rightarrow \text{ep-}F \Rightarrow \text{min-}F\text{-sh}$	–	–	–	–	–	Remark 17	trivial
2	$\text{g-}F \Rightarrow F + \text{p-}F$	–	–	–	–	–	Remark 18	trivial
3	$\text{ep-}F \Rightarrow F$	–	–	–	$n = 2$	–	Remark 17	trivial
4	$(F \text{ or } \text{p-}F) \Rightarrow \text{g-}F$	–	–	–	$n = 2$	–	Remark 18	trivial
5	$\text{EF} \Rightarrow \text{EFX} + \text{EF1}$	–	–	–	–	–	Remark 20	trivial
6	$\text{EEF} \Rightarrow \text{EEFX} + \text{EEF1}$	–	–	–	–	–	Remark 20	trivial
7	$\text{MEFS} \Rightarrow \text{MXS} + \text{M1S}$	–	–	–	–	–	Remark 20	trivial
8	$\text{EFX} \Rightarrow \text{EF1}^*$	additive	–	–	–	–	Lemma 21	trivial
9	$\text{EEFX} \Rightarrow \text{EEF1}^*$	additive	–	–	–	–	Lemma 21	trivial
10	$\text{MXS} \Rightarrow \text{M1S}^*$	additive	–	–	–	–	Lemma 21	trivial
11	$\text{MXS} \Rightarrow \text{M1S}$	–	≤ 0	–	–	–	Lemma 24	new
12	$\text{MXS} \Rightarrow \text{M1S}$	–	$\text{dblMono}^\dagger$	–	$n = 2$	–	Lemma 24	new
13	$\text{MXS} \Rightarrow \text{EF1}$	additive	–	–	$n = 2$	–	Lemma 22	new
14	$\text{PROP} \Rightarrow \text{PROP}_x$	–	–	–	–	–	–	trivial
15	$\text{PROP} \Rightarrow \text{PROP1}$	–	–	–	–	–	–	trivial
16	$\text{PROP}_x \Rightarrow \text{PROP}_{\text{avg}}$	–	–	–	–	–	Lemma 25	trivial
17	$\text{PROP}_{\text{avg}} \Rightarrow \text{PROP}_m$	–	–	–	–	–	–	trivial
18	$\text{PROP}_m \Rightarrow \text{PROP}_x$	–	–	–	$n = 2$	–	–	trivial
19	$\text{PROP}_m \Rightarrow \text{PROP}_x$	–	chores	–	–	–	–	trivial
20	$\text{PROP}_m \Rightarrow \text{PROP1}$	submodular	–	–	–	–	Lemma 26	folklore
21	$\text{PROP}_m \Rightarrow \text{PROP1}$	–	$> 0, < 0$	–	–	–	Lemma 26	folklore
22	$\text{MEFS} \Rightarrow \text{PROP}$	subadditive	–	–	–	–	Lemma 27	known
23	$\text{EF} \Rightarrow \text{GPROP}$	subadditive	–	–	–	–	Lemma 28	known
24	$\text{PROP} \Rightarrow \text{EF}$	superadditive	–	yes	–	–	Lemma 29	folklore
25	$\text{PPROP} \Rightarrow \text{EF}$	superadditive	–	–	–	–	Lemma 30	folklore
26	$\text{PPROP} \Rightarrow \text{GPROP}$	submodular	–	–	–	–	Lemma 31	new
27	$\text{EEF1} \Rightarrow \text{PROP1}$	submodular	–	–	–	equal	Lemma 34	new [§]
28	$\text{EEF1} \Rightarrow \text{PROP1}$	subadditive	chores	–	–	–	Lemma 33	new
29	$\text{EF1} \Rightarrow \text{PROP1}$	subadditive	–	–	$n = 2$	–	Lemma 35	new [¶]
30	$\text{EEFX} \Rightarrow \text{PROP}_x$	subadditive	chores	–	–	–	Lemma 36	known
31	$\text{EFX} \Rightarrow \text{PROP}_{\text{avg}}$	submodular	goods	–	–	equal	Lemma 37	new
32	$\text{EFX} \Rightarrow \text{PROP}_x$	subadditive	–	–	$n = 2$	–	Lemma 38	new
33	$\text{MXS} \Rightarrow \text{PROP1}$	additive	goods	–	–	equal	Lemma 32	known
34	$\text{PMMS} \Rightarrow \text{EFX}$	additive	–	–	–	equal	Lemma 40	folklore
35	$\text{PWMMS} \Rightarrow \text{EFX}$	–	goods	–	–	–	Lemma 40	folklore
36	$\text{WMMS} \Rightarrow \text{EEFX}$	–	goods	–	–	–	Lemma 41	known
37	$\text{MMS} \Rightarrow \text{MXS}$	additive	–	–	–	equal	Lemma 42	new
38	$\text{MMS} \Rightarrow \text{MXS} + \text{M1S}$	–	goods	–	–	equal	Lemma 43	new
39	$\text{PROP} \Rightarrow \text{APS}^\ddagger$	additive	–	–	–	–	Lemma 44	known
40	$\text{PROP} \Rightarrow \text{WMMS}^\ddagger$	superadditive	–	–	–	–	Lemma 45	folklore
41	$\text{APS} \Rightarrow \text{MMS}^\ddagger$	–	–	–	–	equal	Lemma 48	known
42	$\text{PWMMS} \Rightarrow \text{PAPS}$	additive	–	–	–	–	Lemma 50	known

* These results hold for additional settings. See Lemma 21 in Appendix D.2 for details.

† A function $v : 2^M \rightarrow \mathbb{R}$ is *doubly monotone* if there is a partition (G, C) of M such that $v(g \mid \cdot) \geq 0 \forall g \in G$ and $v(c \mid \cdot) \leq 0 \forall c \in C$.

‡ In addition to $F_1 \Rightarrow F_2$, we also get $\text{p-}F_1 \Rightarrow \text{p-}F_2$ and $\text{g-}F_1 \Rightarrow \text{g-}F_2$.

§ [ACIW21] proved this for additive valuations. Recently, [ACIS26] proved this independently for submodular valuations.

¶ Recently, [ACIS26] proved this independently.

Table 2: Non-implications among fairness notions (additive valuations).

		valuation	marginals	identical	n	entitlements		
1	$\text{APS} + \text{PROP}_x \not\Rightarrow \text{PROP}$	$m = 1$	$1, -1$	yes	any	equal	Example 74	trivial
2	$\text{APS} + \text{PROP}_x \not\Rightarrow \text{EF1}$	additive	1	yes	$n \geq 3$	equal	Example 75	folklore
3	$\text{APS} + \text{EEFX} \not\Rightarrow \text{EF1}$	additive	-1	yes	$n \geq 3$	equal	Lemma 76	folklore
4	$\text{EEF} \not\Rightarrow \text{EF1}$	additive	bival*	no	$n = 3$	equal	Lemma 77	new
5	$\text{PROP} \not\Rightarrow \text{MEFS}$	additive	> 0	no	$n = 3$	equal	Example 78	new
6	$\text{PROP} \not\Rightarrow \text{MEFS}$	additive	< 0	no	$n = 3$	equal	Example 79	new
7	$\text{MEFS} \not\Rightarrow \text{EEF}$	additive	> 0	no	$n = 3$	equal	Lemma 80	new
8	$\text{MEFS} \not\Rightarrow \text{EEF1}$	additive	< 0 bival	no	$n = 3$	equal	Lemma 82	new
9	$\text{EFX} \not\Rightarrow \text{MMS}$	additive	bival*	yes	$n = 2$	equal	Example 83	folklore
10	$\text{EF1} \not\Rightarrow \text{MXS or PROP}_x$	additive	bival*	yes	$n = 2$	equal	Example 84	new
11	$\text{PROP}_x \not\Rightarrow \text{M1S}$	additive	bival*	yes	$n = 2$	equal	Lemma 85	new
12	$\text{MXS} \not\Rightarrow \text{PROP}_x$	additive	bival*	yes	$n = 2$	equal	Example 86	known
13	$\text{M1S} \not\Rightarrow \text{PROP1}$	additive	bival*	yes	$n = 2$	equal	Lemma 87	new
14	$\text{GAPS} \not\Rightarrow \text{PROP}_x$	additive	> 0 bival	yes	$n = 3$	equal	Example 88	new
15	$\text{GMMS} \not\Rightarrow \text{APS}$	additive	$> 0, < 0$	yes	$n = 3$	equal	Example 90	known
16	$\text{PMMS} \not\Rightarrow \text{MMS}$	additive	$> 0, < 0$	yes	$n = 3$	equal	Example 91	known
17	$\text{APS} \not\Rightarrow \text{PROP}_m$	additive	> 0	yes	$n = 3$	equal	Example 92	new
18	$\text{APS} \not\Rightarrow \text{PROP1}$	additive	< 0 bival	yes	$n = 3$	equal	Example 93	new
19	$\text{GAPS} \not\Rightarrow \text{PROP}_m$	additive	mixed bival	yes	$n = 3$	equal	Lemma 94	new
20	$\text{PROP}_m \not\Rightarrow \text{PROP}_{\text{avg}}$	additive	> 0 bival	yes	$n = 3$	equal	Example 95	new
21	$\text{PROP1} \not\Rightarrow \text{M1S}$	additive	$-1, 1$	yes	$n = 2$	unequal	Example 96	new
22	$\text{GAPS} \not\Rightarrow \text{PROP}_x$	additive	bival*	yes	$n = 2$	unequal	Lemma 97	new
23	$\text{EF1} \not\Rightarrow \text{EFX}$	additive	$\{-1, 1\}$	yes	$n = 2$	unequal	Example 98	new
24	$\text{WMMS} \not\Rightarrow \text{M1S}$	additive	-1	yes	$n = 2$	unequal	Example 99	new
25	$\text{EFX} \not\Rightarrow \text{WMMS}$	additive	-1	yes	$n = 2$	unequal	Example 99	new
26	$\text{GWMMS} \not\Rightarrow \text{PROP1}$	additive	1	yes	$n = 3$	unequal	Lemma 100	new
27	$\text{GWMMS} \not\Rightarrow \text{PROP1}$	additive	-1	yes	$n = 3$	unequal	Lemma 101	new
28	$\text{PROP} \not\Rightarrow \text{M1S}$	additive	≤ 0 bival	no	$n = 3$	unequal	Lemma 102	new
29	$\text{PROP} \not\Rightarrow \text{MEFS}$	additive	$\{0, 1\}$	no	$n = 3$	unequal	Example 103	new

* Result holds for both positive bivalued marginals and negative bivalued marginals.

Table 3: Implications among fairness notions when marginals belong to the set $\{-1, 0, 1\}$.

		valuations	marginals	identical	$n = 2$	entitlements		
1	$EF1 \Rightarrow EFX^*$	additive	$\{-1, 0, 1\}$	–	–	equal	Lemma 51	trivial
2	$EF1 \Rightarrow EFX^*$	additive	$\{0, \pm 1\}$	–	–	–	Lemma 51	trivial
3	$PROP1 \Rightarrow PROP_x$	–	$\{-1, 0, 1\}$	–	–	–	Lemma 52	trivial
4	$PROP \Rightarrow EEF$	additive	$\{-1, 0, 1\}$	–	–	equal	Lemma 54	new
5	$APS \Rightarrow PROP_x$	additive	$\{-1, 0, 1\}$	–	–	–	Lemmas 55 and 56	new
6	$PROP1 \Rightarrow APS$	additive	$\{-1, 0, 1\}$	–	–	–	Lemmas 55 and 56	new
7	$M1S \Rightarrow APS$	additive	$\{-1, 0, 1\}$	–	–	equal	Lemmas 56 and 57	new
8	$MMS \Rightarrow EEFX$	additive	$\{-1, 0, 1\}$	–	–	equal	Lemma 59	new
9	$EF1 \Rightarrow GAPS$	additive	$\{-1, 0, 1\}$	–	–	equal	Lemma 60	new
10	$EF1 \Rightarrow PAPS$	additive	$\{-1, 0, 1\}$	–	–	–	Lemma 60	new
11	$EF1 \Rightarrow GAPS$	additive	$\{-1, 0\}$	–	–	–	Lemma 60	new
12	$EF1 \Rightarrow GWMS$	additive	$\{0, 1\}$	–	–	–	Lemma 61	new
13	$M1S \Rightarrow WMS$	additive	$\{0, 1\}$	–	–	–	Lemma 61	new
14	$MEFS \Rightarrow EEF$	additive	$\{-1, 0, 1\}$	–	–	–	Lemma 64	new
15	$MXS \Rightarrow EEFX$	additive	$\{0, \pm 1\}$	–	–	–	Lemma 64	new
16	$M1S \Rightarrow EEF1$	additive	$\{0, \pm 1\}$	–	–	–	Lemma 64	new
17	$MMS \Rightarrow APS^\dagger$	submodular	$\{0, 1\}$	–	–	equal	Lemma 65	known
18	$PROPM \Rightarrow PROP_{avg}$	–	$\{-1, 0, 1\}$	–	–	–	Lemma 67	new
19	$PROPM \Rightarrow PROP_x$	subadditive	$\{-1, 0, 1\}$	–	–	equal	Lemma 67	new
20	$M1S \Rightarrow PROP_x$	subadditive	$\{0, 1\}$	–	–	equal	Lemma 68	new
21	$M1S \Rightarrow PROP_x$	subadditive	$\{-1, 0, 1\}$	–	$n = 2$	–	Lemma 68	new
22	$M1S \Rightarrow PROP_x$	subadditive	$\{-1, 0\}$	–	–	–	Lemma 68	new

* $EEF1 \Rightarrow EEFX$ and $M1S \Rightarrow MXS$ also hold for the same conditions.

† $PMMS \Rightarrow PAPS$ and $GMMS \Rightarrow GAPS$ also hold for the same conditions.

Table 4: Non-implications among fairness notions (non-additive valuations).

		valuation	marginals	identical	n	entitlements		
1	$\text{EF} + \text{GAPS} \not\Rightarrow \text{PROP1 or PROPm}$	supermod	≥ 0 bival [†]	yes	$n \geq 2$	equal	Lemma 104	trivial
2	$\text{PPROP} \not\Rightarrow \text{PROP1}$	supermod	≥ 0 bival [†]	yes	$n \geq 3$	equal	Lemma 104	trivial
3	$\text{PROP} \not\Rightarrow \text{MIS}^*$	unit-dem	≥ 0	yes	$n = 2$	equal	Lemma 106	new
4	$\text{PROP1} \not\Rightarrow \text{PROPm}^*$	unit-dem	≥ 0	yes	$n = 2$	equal	Example 107	new
5	$\text{PROP} \not\Rightarrow \text{PPROP}^*$	unit-dem	≥ 0	yes	$n = 3$	equal	Example 108	new
6	$\text{PROPm} \not\Rightarrow \text{PROPavg}^*$	unit-dem	≥ 0	yes	$n = 3$	equal	Example 110	new
7	$\text{GAPS} \not\Rightarrow \text{PROPx}$	unit-dem	≥ 0	yes	$n = 3$	equal	Example 88	new
8	$\text{MMS} \not\Rightarrow \text{EF1 or PROPm}$	unit-dem	≥ 0	yes	$n = 4$	equal	Example 109	new
9	$\text{EF} \not\Rightarrow \text{MMS}$	submod	≥ 0 bival [†]	yes	$n = 2$	equal	Example 112	folklore
10	$\text{EEF} \not\Rightarrow \text{EF1 or PPROP}$	submod	≥ 0 bival [†]	yes	$n = 3$	equal	Example 115	new
11	$\text{MEFS} \not\Rightarrow \text{EF1}$	submod	≥ 0 bival [†]	yes	$n = 2$	equal	Example 114	new
12	$\text{PROP} \not\Rightarrow \text{MIS}$	submod	≥ 0 bival [†]	yes	$n = 2$	equal	Example 116	new
13	$\text{MMS} \not\Rightarrow \text{APS}$	submod	> 0	yes	$n = 2$	equal	Lemma 117	known
14	$\text{GAPS} + \text{PPROP} \not\Rightarrow \text{PROP1}$	subadd	≥ 0 bival [†]	yes	$n = 3$	equal	Example 122	new
15	$\text{EF1} \not\Rightarrow \text{MXS}$	submod	$\{0, 1\}$	yes	$n = 2$	equal	Example 113	new
16	$\text{GAPS} \not\Rightarrow \text{PROP1 or EF1}$	subadd	$\{0, 1\}$	yes	$n = 2$	equal	Example 119	new
17	$\text{EF1} \not\Rightarrow \text{PROP1}$	subadd	$\{0, 1\}$	yes	$n = 3$	equal	Example 120	new
18	$\text{PAPS} + \text{PPROP} \not\Rightarrow \text{PROPm or MIS}$	subadd	$\{0, 1\}$	yes	$n = 3$	equal	Lemma 123	new
19	$\text{GMMS} \not\Rightarrow \text{APS}$	subadd	$\{0, 1\}$	yes	$n = 3$	equal	Lemma 124	new
20	$\text{PROPavg} \not\Rightarrow \text{PROPx}$	supermod	$\{0, 1\}$	yes	$n \geq 3$	equal	Lemma 104	new

* Result also holds for submodular cancelable valuations with positive marginals.

† Result holds for both $\{0, 1\}$ marginals and positive bivalued marginals.

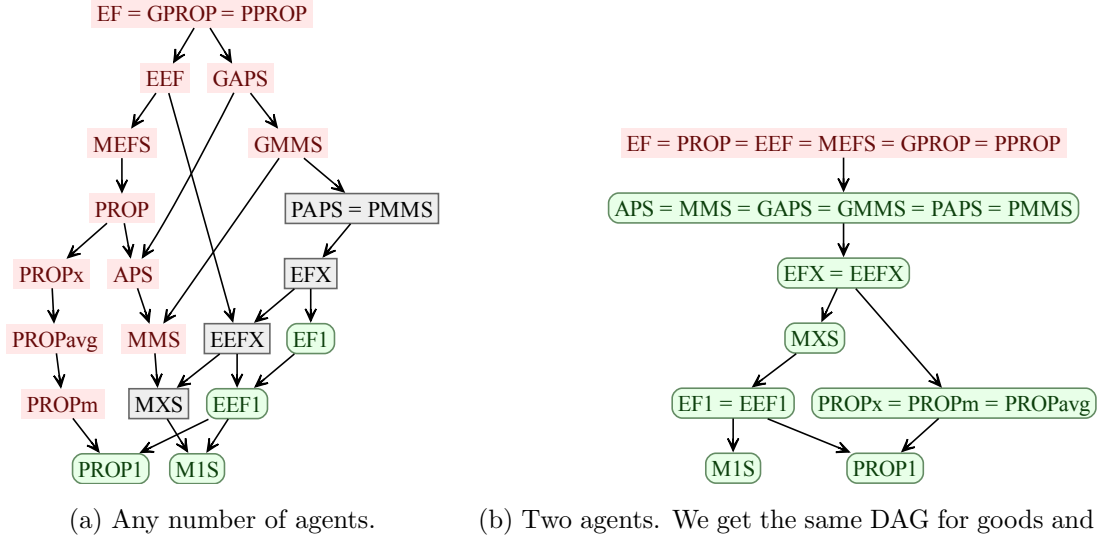


Figure 3: Additive valuations, mixed manna, equal entitlements.

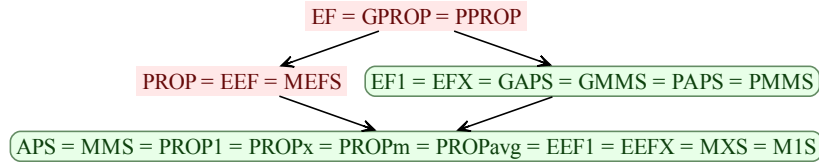


Figure 4: Additive valuations, marginals in $\{-1, 0, 1\}$, equal entitlements. We get the same DAG when marginals are in $\{0, -1\}$ or $\{0, 1\}$.

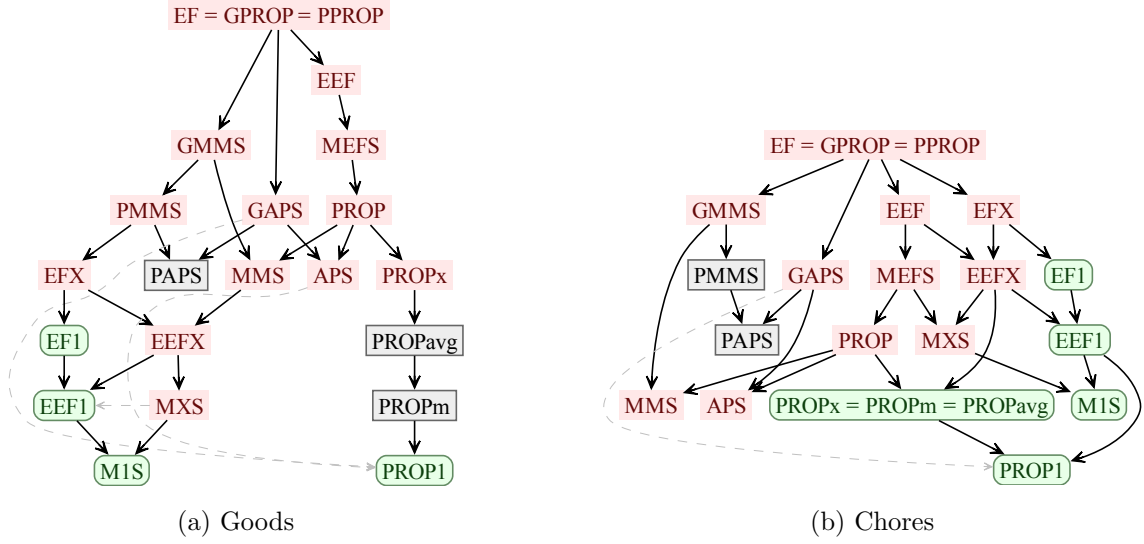


Figure 5: Additive valuations, unequal entitlements.

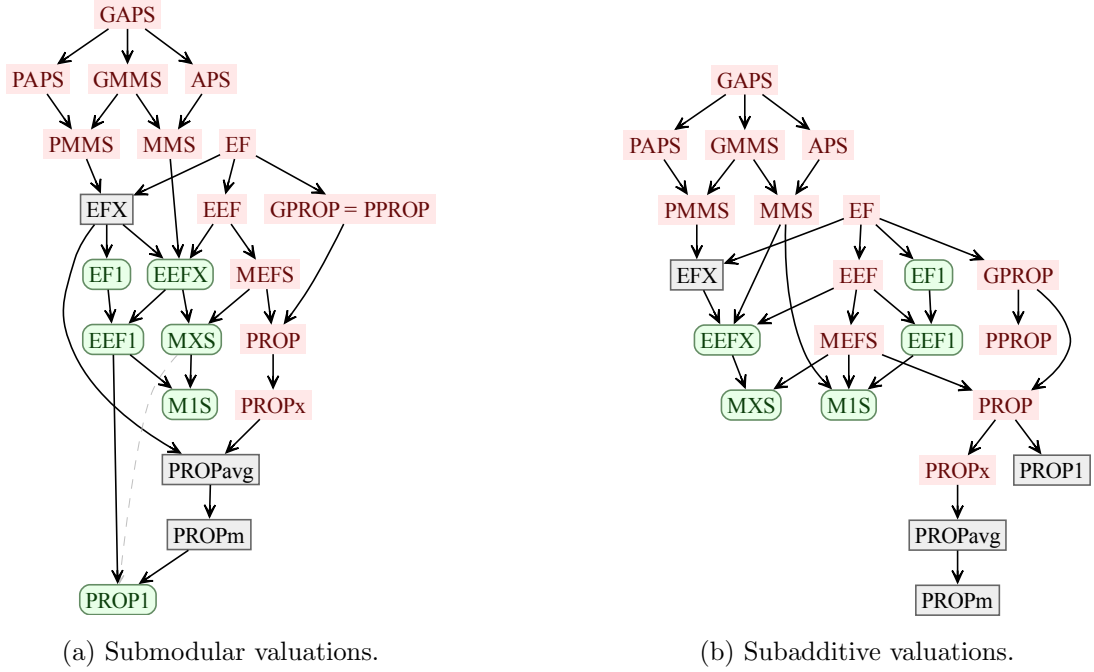


Figure 6: Submodular and subadditive valuations over goods with equal entitlements.

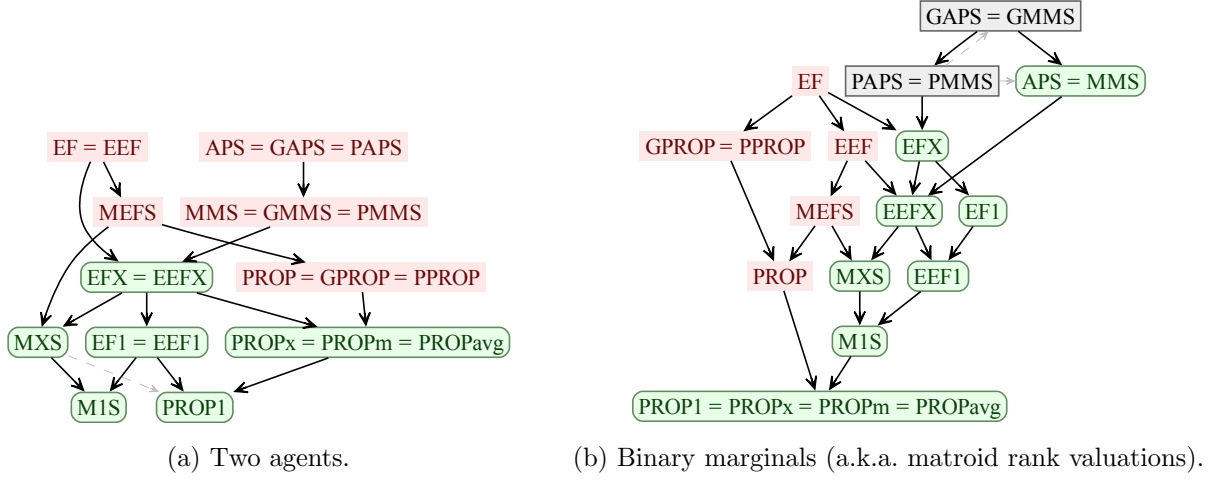


Figure 7: Special cases of submodular valuations over goods with equal entitlements.

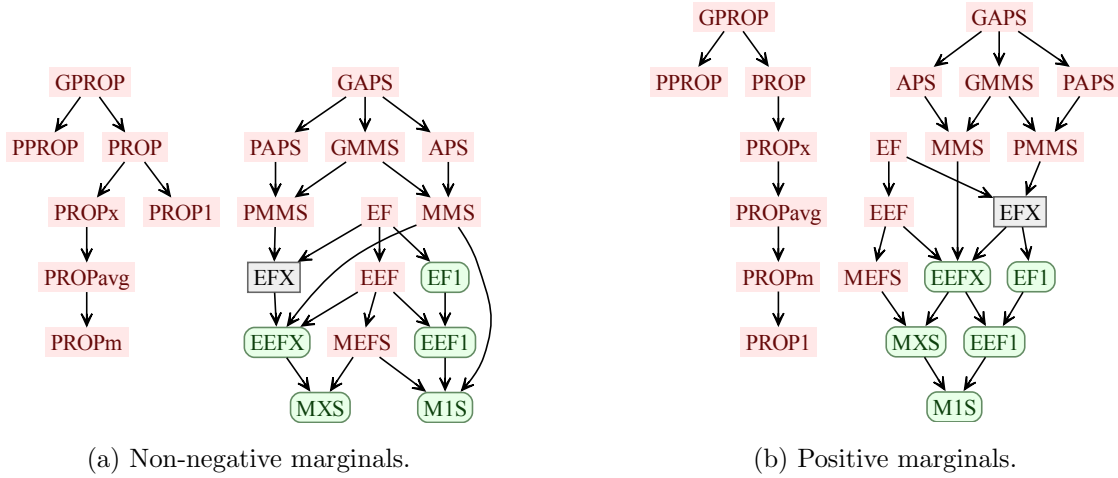


Figure 8: General valuations for goods with equal entitlements.

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A Details on Fair Division Settings

A.1 Valuation Function Type

A function $u : 2^M \rightarrow \mathbb{R}$ is

1. *additive* if for any two disjoint sets $S, T \subseteq M$, we have $u(S \cup T) = u(S) + u(T)$. Equivalently, for every set $S \subseteq M$, we have $u(S) = \sum_{j \in S} u(\{j\})$.
2. *subadditive* if for any two disjoint sets $S, T \subseteq M$, we have $u(S \cup T) \leq u(S) + u(T)$.
3. *superadditive* if for any two disjoint sets $S, T \subseteq M$, we have $u(S \cup T) \geq u(S) + u(T)$.
4. *submodular* if for any $S, T \subseteq M$, we have $u(S \cup T) + u(S \cap T) \leq u(S) + u(T)$.
5. *supermodular* if for any $S, T \subseteq M$, we have $u(S \cup T) + u(S \cap T) \geq u(S) + u(T)$.
6. *cancelable* [BCFF22] if for any $T \subseteq M$ and $S_1, S_2 \subseteq M \setminus T$, we have $u(S_1 \cup T) > u(S_2 \cup T) \implies u(S_1) > u(S_2)$.
7. *unit-demand* if $u(\emptyset) = 0$, and for any $\emptyset \neq S \subseteq M$, we have $u(S) \geq 0$ and $u(S) = \max_{j \in S} u(\{j\})$.
8. *XOS* (aka *fractionally subadditive*) [Nis00, Fei09] if there exist additive functions $a_1, \dots, a_k$, such that $u(S) = \max_{j=1}^k a_j(S)$ for all $S \subseteq M$.

When $|M| = 1$, u belongs to all of these classes. See Fig. 9a for relations between valuation function types.

A.2 Marginal Values

Marginal values for instance $(N, M, (v_i)_{i \in N}, w)$:

1. *goods*: $v_i(j \mid S) \geq 0 \forall S \subseteq M, j \in M \setminus S, i \in N$.
2. *chores*: $v_i(j \mid S) \leq 0 \forall S \subseteq M, j \in M \setminus S, i \in N$.
3. *positive*: $v_i(j \mid S) > 0 \forall S \subseteq M, j \in M \setminus S, i \in N$.
4. *negative*: $v_i(j \mid S) < 0 \forall S \subseteq M, j \in M \setminus S, i \in N$.
5. *bivalued*: $\exists a, b \in \mathbb{R}$ such that $v_i(j \mid S) \in \{a, b\} \forall S \subseteq M, j \in M \setminus S, i \in N$.
6. *binary*: $v_i(j \mid S) \in \{0, 1\} \forall S \subseteq M, j \in M \setminus S, i \in N$.
7. *negative binary*: $v_i(j \mid S) \in \{0, -1\} \forall S \subseteq M, j \in M \setminus S, i \in N$.
8. *doubly monotone*: $\forall i \in N$, there is a partition (G, C) of M such that $\forall S \subseteq M$, we have $v_i(g \mid S) \geq 0 \forall g \in G \setminus S$ and $v_i(c \mid S) \leq 0 \forall c \in C \setminus S$.
9. *doubly strictly monotone*: $\forall i \in N$, there is a partition (G, C) of M such that $\forall S \subseteq M$, we have $v_i(g \mid S) > 0 \forall g \in G \setminus S$ and $v_i(c \mid S) < 0 \forall c \in C \setminus S$.

We can break up the class of bivalued instances into positive bivalued, negative bivalued, binary, negative binary, and mixed bivalued (mixed means that exactly one of a and b is positive and the other is negative). See Fig. 9b for relations between marginal values.

B Properties of Submodular Functions

In this section, we describe some properties of submodular functions that would be useful to understand results in subsequent sections.

We begin by stating multiple equivalent definitions of submodularity. Recall that $u(X \mid Y) := u(X \cup Y) - u(Y)$ is called the *marginal* value of X given Y .

Definition 12 (submodularity, [Sch03]). *The following are equivalent definitions for a function $u : 2^M \rightarrow \mathbb{R}$ being submodular:*

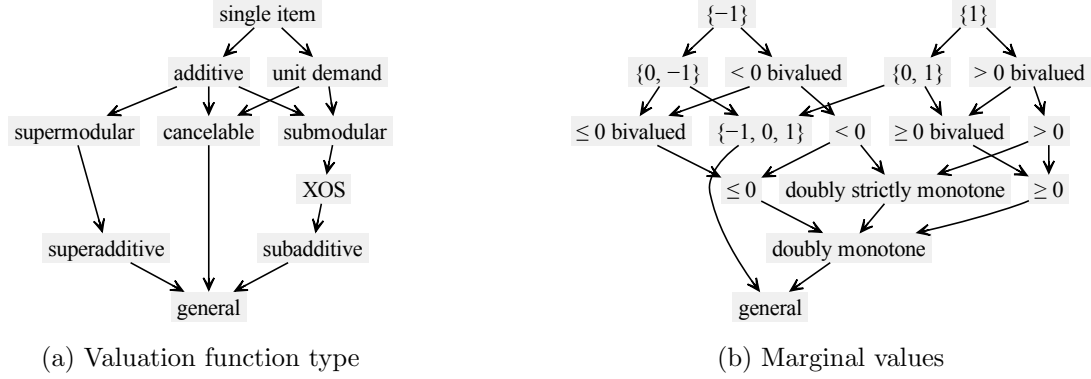


Figure 9: Valuation function type and marginal values represented as digraphs. There is a path from u to v iff u is a subclass of v .

1. for any $S, T \subseteq M$, we have $u(S \cup T) + u(S \cap T) \leq u(S) + u(T)$.
2. for any $S \subseteq T \subseteq M$ and $g \in M \setminus T$, we have $u(g | S) \geq u(g | T)$.
3. for any $S \subseteq T \subseteq M$ and $X \subseteq M \setminus T$, we have $u(X | S) \geq u(X | T)$.

Next, we analyze properties of functions defined using marginal values of submodular functions.

Lemma 1 (marginal gains are subadditive). *Let $u : 2^M \rightarrow \mathbb{R}$ be a submodular function and let $T \subseteq M$. For any $X \subseteq M \setminus T$, define $f(X) := u(X | T)$. Then f is subadditive, i.e., for any disjoint sets $X, Y \subseteq M \setminus T$, we have $f(X \cup Y) \leq f(X) + f(Y)$.*

Proof. $f(X \cup Y) = u(X \cup Y | T) = u(X | T) + u(Y | X \cup T) \leq u(X | T) + u(Y | T) = f(X) + f(Y)$. $\square$

Lemma 2 (marginal losses are superadditive). *Let $u : 2^M \rightarrow \mathbb{R}$ be a submodular function and let $T \subseteq M$. For any $X \subseteq T$, define $f(X) := u(X | T \setminus X)$. Then f is superadditive, i.e., for any disjoint sets $X, Y \subseteq T$, we have $f(X \cup Y) \geq f(X) + f(Y)$.*

Proof. $f(X \cup Y) = u(X \cup Y | T \setminus (X \cup Y))$
 $= u(X | T \setminus (X \cup Y)) + u(Y | T \setminus Y)$
 $\geq u(X | T \setminus X) + u(Y | T \setminus Y) = f(X) + f(Y)$. $\square$

C Details on Fairness Notions

Although we do not define any new fairness notions in this paper, we extend some of them to more general settings than they were originally defined for. For some notions, this extension is not obvious and is based on careful deliberation. Here we show how we arrived at these extensions and why they make sense.

C.1 EFX

Defining EFX in the fully general setting (non-additive valuations, mixed manna) is tricky. So, we start with the definition of EFX for additive goods, and gradually build up to the general definition of EFX from there. Some of these ideas also appear in [CGR⁺22].

There are actually two competing definitions of EFX for additive goods. According to the original definition by [CKM⁺19], an allocation A is EFX-fair to agent i if for every other agent j , removing any positively-valued good from j 's bundle eliminates i 's envy. Formally,

$$\frac{v_i(A_i)}{w_i} \geq \max_{g \in A_j: v_i(g) > 0} \frac{v_i(A_j \setminus \{g\})}{w_j}.$$

A different definition, often called EFX₀, does not require the good g to have a positive value to i [PR20]. See Definition 13 for a formal definition, and Definition 14 for the chores analogue.

Definition 13 (EFX₀ for goods). *In a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ over goods, an allocation A is EFX₀-fair to agent i if for every other agent j , and every $g \in A_j$,*

$$\frac{v_i(A_i)}{w_i} \geq \frac{v_i(A_j \setminus \{g\})}{w_j}.$$

Definition 14 (EFX₀ for chores). *In a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ over chores, an allocation A is EFX₀-fair to agent i if for every other agent j , and every $c \in A_j$,*

$$\frac{-v_i(A_i \setminus \{c\})}{w_i} \leq \frac{-v_i(A_j)}{w_j}.$$

EFX₀ is known to be incompatible with Pareto optimality (PO) based on the following simple example [PR20], whereas it is not known whether EFX is compatible with PO for additive valuations over goods.

Example 3. *Consider the fair division instance $([2], \{g_1, g_2, g_3\}, (v_i)_{i=1}^2, -)$, where v_1 and v_2 are additive, and*

$$\begin{array}{lll} v_1(g_1) = 1, & v_1(g_2) = 0, & v_1(g_3) = 10, \\ v_2(g_1) = 0, & v_2(g_2) = 1, & v_2(g_3) = 10. \end{array}$$

In Example 3, for any PO allocation A , we have $g_1 \in A_1$ and $g_2 \in A_2$ (otherwise we can transfer g_1 to agent 1 or g_2 to agent 2 to Pareto-dominate A). The agent who did not get g_3 in A is not EFX₀-satisfied, although she is EFX-satisfied.

While EFX₀ is trivial to extend to non-additive valuations, EFX is not. This is because there exist examples where every good in j 's bundle has zero value to agent i . (Additionally, the good's marginal value over A_j and A_i may also be zero.) Before we try to address this issue, let us instead jump to the setting of additive mixed manna.

One way to define EFX for mixed manna is this: agent i is EFX-satisfied by allocation A if for every other agent j , either agent i does not envy j , or i 's envy towards j vanishes after either removing the least valuable positively-valued item from j or after removing the most valuable negatively-valued item from i . However, we argue that this is not sufficient.

Example 4. *Consider a fair division instance $\mathcal{I}$ having 2 agents with equal entitlements, identical additive valuations, two goods of values 10 each, and two chores of values -9 each. Consider an allocation A where all the 4 items are allocated to agent 2. Then agent 1 would be EFX-satisfied by A , even though allocation B , where each agent gets one good and one chore, is intuitively fairer.*

For goods, EFX is considered one of the strongest fairness notions, so we would like it to be a very strong notion for mixed manna too. The key observation in Example 4 is that in allocation

A , we can transfer a set of items (containing one good and one chore) from agent 2 to agent 1 and get a fairer allocation. This suggests that instead of (hypothetically) removing a single good from j or a single chore from i in the definition of EFX, we should remove a positively-valued subset of A_j or a negatively-valued subset of A_i .

Hence, for additive mixed manna, we say that an allocation A is EFX-fair to agent i if for every other agent j , i does not envy j , or both of the following hold:

1. $\frac{v_i(A_i)}{w_i} \geq \frac{\max(\{v_i(A_j \setminus S) : S \subseteq A_j \text{ and } v_i(S) > 0\})}{w_j}$.
2. $\frac{\min(\{v_i(A_i \setminus S) : S \subseteq A_i \text{ and } v_i(S) < 0\})}{w_i} \geq \frac{v_i(A_j)}{w_j}$.

This definition also hints towards how to handle goods with non-additive valuations. Even if every good in j 's bundle has (marginal) value zero to agent i , some subset of j 's bundle must have positive (marginal) value. We replace $v(S) > 0$ by $v_i(S \mid A_i) > 0$ to ensure that transferring S from j to i leads to an increase in i 's valuation, otherwise we lose compatibility with PO, even with identical valuations, as the following example demonstrates.

Example 5. Consider a fair division instance with two equally-entitled agents having identical valuations. There are 2 red goods and 4 blue goods. The value of any bundle is $\max(k_r, k_b)$, where k_r and k_b are the numbers of red and blue goods in the bundle, respectively.

For Example 5, in any allocation, some agent gets at most 2 blue goods, and that agent's value for her own bundle can be at most 2. Also, the maximum value that any bundle can have is 4. Hence, an allocation is PO iff one agent gets all the blue goods and the other agent gets all the red goods. To ensure the compatibility of EFX and PO, we must define EFX such that this allocation is EFX.

Hence, the definition of EFX we converge on is Definition 3. We now identify special cases where this definition is equivalent to other well-known definitions of EFX.

Lemma 6. In the fair division instance $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$, if all marginals are positive for agent i (i.e., $v_i(g \mid R) > 0$ for all $R \subseteq [m]$ and $g \in [m] \setminus R$), then A is EFX-fair to agent i iff A is EFX₀-fair to i .

Proof sketch. $v_i(A_j \setminus S) = v_i(A_j) - v_i(S \mid A_j \setminus S)$. Thus, in Definition 3, we need to find $S \subseteq A_j$ such that $v_i(S \mid A_i) > 0$ and $v_i(S \mid A_j \setminus S)$ is minimized. Since marginals are positive, it suffices to only check sets of cardinality 1. $\square$

Lemma 7. In the fair division instance $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$, if all marginals are negative for agent i (i.e., $v_i(c \mid R) < 0$ for all $R \subseteq [m]$ and $c \in [m] \setminus R$), then A is EFX-fair to agent i iff A is EFX₀-fair to agent i .

Proof sketch. $v_i(A_i \setminus S) = v_i(A_i) - v_i(S \mid A_i \setminus S)$. Thus, in Definition 3, we need to find $S \subseteq A_i$ such that $-v_i(S \mid A_i \setminus S) > 0$ and $-v_i(S \mid A_i \setminus S)$ is minimized. Since marginals are negative, it suffices to only check sets of cardinality 1. $\square$

Lemma 8. In the fair division instance $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$, if v_i is submodular and all marginals are non-negative for agent i (i.e., $v_i(g \mid R) \geq 0$ for all $R \subseteq [m]$ and $g \in [m] \setminus R$), then A is EFX-fair to agent i iff for all $j \in [n] \setminus \{i\}$, we have

$$\frac{v_i(A_i)}{w_i} \geq \frac{\max(\{v_i(A_j \setminus \{g\}) : g \in A_j \text{ and } v_i(g \mid A_i) > 0\})}{w_j}.$$

Proof sketch. In Definition 3, we need to find $S \subseteq A_j$ such that $v_i(S \mid A_i) > 0$ and $v_i(S \mid A_j \setminus S)$ is minimized. By the subadditivity of v_i 's marginal gains (Lemma 1), we get $v_i(S \mid A_i) \leq \sum_{g \in S} v_i(g \mid A_i)$. Since $v_i(S \mid A_i) > 0$, there exists $\hat{g} \in S$ such that $v_i(\hat{g} \mid A_i) > 0$.

Moreover, $v_i(S \mid A_j \setminus S) \geq v_i(\hat{g} \mid A_j \setminus \{\hat{g}\})$ since $v_i(S \mid A_j \setminus S) = v_i(S \setminus \{\hat{g}\} \mid A_j \setminus S) + v_i(\hat{g} \mid A_j \setminus \{\hat{g}\})$. Thus, we can replace S by $\{\hat{g}\}$, so we can assume without loss of generality that $|S| = 1$. $\square$

Lemma 9. *In the fair division instance $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$, if v_i is submodular and all marginals are non-positive for agent i (i.e., $v_i(c \mid R) \leq 0$ for all $R \subseteq [m]$ and $c \in [m] \setminus R$), then A is EFX-fair to agent i iff for all $j \in [n] \setminus \{i\}$, we have*

$$\frac{\max(\{-v_i(A_i \setminus \{c\}) : c \in A_i \text{ and } v_i(c \mid A_i \setminus \{c\}) < 0\})}{w_i} \leq \frac{-v_i(A_j)}{w_j}.$$

Proof sketch. Let $d_i := -v_i$. In Definition 3, we need to find $S \subseteq A_i$ such that $d_i(S \mid A_i \setminus S) > 0$ and $d_i(S \mid A_i \setminus S)$ is minimized. By the superadditivity of v_i 's marginal losses (Lemma 2), we get $d_i(S \mid A_i \setminus S) \leq \sum_{c \in S} d_i(c \mid A_i \setminus \{c\})$. Since $d_i(S \mid A_i \setminus S) > 0$, there exists $\hat{c} \in A_i$ such that $d_i(\hat{c} \mid A_i \setminus \{\hat{c}\}) > 0$.

Moreover, $d_i(S \mid A_i \setminus S) \geq d_i(\hat{c} \mid A_i \setminus \{\hat{c}\})$ since $d_i(S \mid A_i \setminus S) = d_i(S \setminus \{\hat{c}\} \mid A_i \setminus S) + d_i(\hat{c} \mid A_i \setminus \{\hat{c}\})$. Thus, we can replace S by $\{\hat{c}\}$, so we can assume without loss of generality that $|S| = 1$. $\square$

C.2 MMS

Definition 15 (WMMS, [FGH⁺19]). *Let $\mathcal{I} := (N, M, (v_i)_{i \in N}, w)$ be a fair division instance. Let $\mathcal{A}$ be the set of all allocations for $\mathcal{I}$. Then agent i 's weighted maximin share (WMMS) is*

$$\text{WMMS}_i := w_i \max_{X \in \mathcal{A}} \min_{j \in N} \frac{v_i(X_j)}{w_j}.$$

An allocation A is WMMS-fair to agent i if $v_i(A_i) \geq \text{WMMS}_i$. An allocation X that maximizes $w_i \min_{j \in N} \frac{v_i(X_j)}{w_j}$ is called agent i 's WMMS partition.

For equal entitlements, we get $\text{WMMS}_i = \text{MMS}_i$.

Definition 16 (pessShare, [BEF23]). *Let $1 \leq \ell \leq d$. Let $\Pi_d(M)$ be the set of all d -partitions of M . Then agent i 's ℓ -out-of- d share is defined as*

$$\ell\text{-out-of-}d\text{-share}_i := \max_{\substack{X \in \Pi_d(M): \\ v_i(X_j) \leq v_i(X_{j+1}) \forall j \in [d-1]}} \sum_{j=1}^{\ell} v_i(X_j).$$

Then agent i 's pessimistic share is defined as

$$\text{pessShare}_i := \sup_{1 \leq \ell \leq d: \ell/d \leq w_i} \ell\text{-out-of-}d\text{-share}_i.$$

Allocation A is pessShare-fair to agent i if $v_i(A_i) \geq \text{pessShare}_i$.

Lemma 10. *For any fair division instance with equal entitlements, the pessShare of any agent is the same as her maximin share.*

Proof. Let $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$ be a fair division instance, where $w_i = 1/n$ for all $i \in [n]$. Any agent i 's 1-out-of- n share is the same as MMS_i , so $\text{pessShare}_i \geq \text{MMS}_i$.

Let $\ell/d \leq n$ and let $X \in \Pi_d(M)$ such that $v_i(X_j) \leq v_i(X_{j+1})$ for all $j \in [d-1]$. Now let Y_1 be the union of the first ℓ bundles of X , let Y_2 be the union of the next ℓ bundles of X , and so on. Add any remaining bundles of X to Y_n . Formally, $Y \in \Pi_n(M)$ where $Y_k := \bigcup_{j=(k-1)\ell+1}^{k\ell} X_j$ for $k \in [n-1]$ and $Y_n := M \setminus \bigcup_{j=1}^{n-1} Y_j$. Then Y_n contains at least ℓ bundles of X , since $d \geq \ell n$. Hence, $v_i(Y_1) \leq v_i(Y_2) \leq \dots \leq v_i(Y_n)$, and $v_i(Y_1)$ is agent i 's ℓ -out-of- d share.

By definition of MMS, $v_i(Y_1) \leq \text{MMS}_i$. Hence, for any ℓ and d such that $\ell/d \leq n$, agent i 's ℓ -out-of- d share is at most her MMS. Hence, $\text{pessShare} \leq \text{MMS}_i$. $\square$

C.3 APS

For any $x \in \mathbb{R}^m$ and $S \subseteq [m]$, define $x(S) := \sum_{j \in S} x_j$.

Definition 17 (APS, [BEF23]). For a fair division instance $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$, agent i 's AnyPrice Share (APS) is defined as

$$\text{APS}_i := \min_{p \in \mathbb{R}^m} \max_{S \subseteq [m]: p(S) \leq w_i p([m])} v_i(S).$$

Here p is called the price vector. A vector $p^* \in \mathbb{R}^m$ is called an optimal price vector for agent i if

$$p^* \in \operatorname{argmin}_{p \in \mathbb{R}^m} \max_{S \subseteq [m]: p(S) \leq w_i p([m])} v_i(S).$$

Definition 17 is slightly different from the original definition given in [BEF23]. However, they assume that all items are goods, and for that special case, we can prove that their definition is equivalent to Definition 17. We begin by proving that, without loss of generality, we can assume that goods have a non-negative price and chores have a non-positive price.

Lemma 11. Let $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$ be a fair division instance. For agent i , let G be the set of goods and C be the set of chores, i.e., $G := \{g \in [m] : v_i(g \mid R) \geq 0 \ \forall R \subseteq [m] \setminus \{g\}\}$ and $C := \{c \in [m] : v_i(c \mid R) \leq 0 \ \forall R \subseteq [m] \setminus \{c\}\}$. Then for some optimal price vector $\hat{p} \in \mathbb{R}^m$, we have $\hat{p}_g \geq 0$ for all $g \in G$ and $\hat{p}_c \leq 0$ for all $c \in C$.

Proof. Let $p^* \in \mathbb{R}^m$ be an optimal price vector. Let $\hat{G} := \{g \in G : p_g^* < 0\}$ and $\hat{C} := \{c \in C : p_c^* > 0\}$. The high-level idea is that if we change the price of $\hat{G} \cup \hat{C}$ to 0, we get potentially better prices. Let $\hat{p}_j := 0$ if $j \in \hat{G} \cup \hat{C}$ and let $\hat{p}_j := p_j^*$ otherwise. Let

$$\hat{S} \in \operatorname{argmax}_{S \subseteq [m]: \hat{p}(S) \leq w_i \hat{p}([m])} v_i(S).$$

Since $\hat{p}(\hat{S} \cup \hat{G} \setminus \hat{C}) = \hat{p}(\hat{S})$ and $v_i(\hat{S} \cup \hat{G} \setminus \hat{C}) \geq v_i(\hat{S})$, we assume without loss of generality that $\hat{G} \subseteq \hat{S}$ and $\hat{C} \cap \hat{S} = \emptyset$. Moreover, $p^*(\hat{S}) - w_i p^*([m]) = (\hat{p}(\hat{S}) - w_i \hat{p}([m])) - w_i p^*(\hat{C}) - (1 - w_i)(-p^*(\hat{G})) \leq 0$. Hence,

$$\max_{\substack{S \subseteq [m]: \\ p^*(S) \leq w_i p^*([m])}} v_i(S) \geq v_i(\hat{S}) = \max_{\substack{S \subseteq [m]: \\ \hat{p}(S) \leq w_i \hat{p}([m])}} v_i(S),$$

so $\hat{p}$ is also an optimal price vector. $\square$

For any positive integer m , define $\Delta_m := \{x \in \mathbb{R}_{\geq 0}^m : \sum_{j=1}^m x_j = 1\}$. When all items are goods, by Lemma 11, we can assume without loss of generality that p is non-negative and $p([m]) = 1$. Hence,

$$\text{APS}_i = \min_{p \in \Delta_m} \max_{S \subseteq [m]: p(S) \leq w_i} v_i(S).$$

When all items are chores, by Lemma 11, we can assume that p is non-positive and $p([m]) = -1$. Hence,

$$-\text{APS}_i = \max_{q \in \Delta_m} \min_{S \subseteq [m]: q(S) \geq w_i} (-v_i(S)).$$

[BEF23] give another equivalent definition of APS, called the *dual* definition.

Definition 18 (APS (dual)). *Let $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$ be a fair division instance. For an agent i and any $z \in \mathbb{R}$, let $\mathcal{S}_z := \{S \subseteq [m] : v_i(S) \geq z\}$. Then agent i 's AnyPrice share, denoted by APS_i , is the largest value z such that*

$$\exists x \in \mathbb{R}_{\geq 0}^{\mathcal{S}_z}, \sum_{S \in \mathcal{S}_z} x_S = 1 \text{ and } \left(\forall j \in [m], \sum_{S \in \mathcal{S}_z: j \in S} x_S = w_i \right).$$

Definition 18 can be interpreted as a linear programming relaxation of MMS. Formally, when entitlements are equal, adding the integrality constraints $nx_S \in \mathbb{Z}_{\geq 0}$ for all $S \in \mathcal{S}_z$ gives us an alternate definition of MMS.

[BEF23] show that the primal and dual definitions of APS are equivalent. They prove this only for goods, but their proof can be easily adapted to the case of mixed manna.

Lemma 12. *Definitions 17 and 18 are equivalent.*

Proof. Let pAPS_i and dAPS_i be agent i 's AnyPrice shares given by the primal and dual definitions, respectively. We will show that for any $z \in \mathbb{R}$, $\text{pAPS}_i \geq z$ iff $\text{dAPS}_i \geq z$. This would prove that $\text{pAPS}_i = \text{dAPS}_i$.

$\text{dAPS}_i \geq z$ iff the following LP has a feasible solution:

$$\min_{x \in \mathbb{R}_{\geq 0}^{\mathcal{S}_z}} 0 \text{ where } \sum_{S \in \mathcal{S}_z} x_S = 1 \text{ and } \left(\forall j \in [m], \sum_{S \in \mathcal{S}_z: j \in S} x_S = w_i \right).$$

Its dual is

$$\max_{p \in \mathbb{R}^m, r \in \mathbb{R}} r - w_i p([m]) \text{ where } p(S) \geq r \text{ for all } S \in \mathcal{S}_z.$$

The dual LP is feasible since $(0, 0)$ is a solution. Furthermore, if (p, r) is feasible for the dual LP, then $(\alpha p, \alpha r)$ is also feasible, for any $\alpha \geq 0$. Hence, by strong duality of LPs, the primal LP is feasible iff all feasible solutions to the dual have objective value at most 0.

For a given p , the optimal r is $\min_{S \in \mathcal{S}_z} p(S)$. Hence, the dual LP is bounded iff for all $p \in \mathbb{R}^m$,

$$\min_{S \in \mathcal{S}_z} p(S) \leq w_i p([m]).$$

Furthermore,

$$\forall p \in \mathbb{R}^m, \min_{S \in \mathcal{S}_z} p(S) \leq w_i p([m])$$

$$\begin{aligned}
&\iff \forall p \in \mathbb{R}^m, \exists S \subseteq [m] \text{ such that } p(S) \leq w_i p([m]) \text{ and } v_i(S) \geq z \\
&\iff \left(\min_{p \in \mathbb{R}^m} \max_{S \subseteq [m]: p(S) \leq w_i p([m])} v_i(S) \right) \geq z \\
&\iff \text{pAPS}_i \geq z.
\end{aligned}$$

Hence, $\text{dAPS}_i \geq z \iff \text{pAPS}_i \geq z$. $\square$

C.4 PROP1

We use the definition of PROP1 from [Fei25]. This definition (c.f. Definition 5) uses strict inequalities, i.e., $v_i(A_i \cup \{g\}) > w_i v_i([m])$ instead of $v_i(A_i \cup \{g\}) \geq w_i v_i([m])$, and $v_i(A_i \setminus \{c\}) > w_i v_i([m])$ instead of $v_i(A_i \setminus \{c\}) \geq w_i v_i([m])$. This is different from the convention followed by most fair division papers.

We define it this way to make it a slightly stronger notion. In particular, when the items are identical, the stricter notion aligns with the intuitive idea that each agent must get at least $\lfloor m/n \rfloor$ goods or at most $\lceil m/n \rceil$ chores. However, outside of the identical items case, this nuance does not affect our results much. Almost all implications, counterexamples, and (in)feasibility results hold for both the stronger and weaker definitions, and we explicitly mention when they don't. We alter PROP_x , PROP_m , and PROP_{avg} to also use strict inequalities.

C.5 PROP_x

We identify special cases where our definition of PROP_x (Definition 6) is equivalent to well-known definitions of PROP_x [AMS20, LLW22].

Lemma 13. *In the fair division instance $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$, if all marginals are positive for agent i (i.e., $v_i(g \mid R) > 0$ for all $R \subseteq [m]$ and $g \in [m] \setminus R$), then A is PROP_x -fair to agent i iff $v_i(A_i) \geq w_i v_i([m])$ or $v_i(A_i \cup \{g\}) > w_i v_i([m])$ for some $g \in [m] \setminus A_i$.*

Proof sketch. In Definition 6, we need to find $S \subseteq [m] \setminus A_i$ such that $v_i(S \mid A_i) > 0$ and $v_i(S \mid A_i)$ is minimized. Since marginals are positive, we require $|S| = 1$. $\square$

Lemma 14. *In the fair division instance $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$, if all marginals are negative for agent i (i.e., $v_i(c \mid R) < 0$ for all $R \subseteq [m]$ and $c \in [m] \setminus R$), then A is PROP_x -fair to agent i iff $-v_i(A_i) \leq w_i \cdot (-v_i([m]))$ or $-v_i(A_i \setminus \{c\}) < w_i \cdot (-v_i([m]))$ for some $c \in A_i$.*

Proof sketch. In Definition 6, we need to find $S \subseteq A_i$ such that $-v_i(S \mid A_i \setminus S) > 0$ and $-v_i(S \mid A_i \setminus S)$ is minimized. Since marginals are negative, we require $|S| = 1$. $\square$

Lemma 15. *In the fair division instance $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$, if v_i is submodular and all marginals are non-negative for agent i (i.e., $v_i(g \mid R) \geq 0$ for all $R \subseteq [m]$ and $g \in [m] \setminus R$), then A is PROP_x -fair to agent i iff $v_i(A_i) \geq w_i v_i([m])$ or $v_i(A_i \cup \{g\}) > w_i v_i([m])$ for some $g \in [m] \setminus A_i$ such that $v_i(g \mid A_i) > 0$.*

Proof sketch. In Definition 6, we need to find $S \subseteq [m] \setminus A_i$ such that $v_i(S \mid A_i) > 0$ and $v_i(S \mid A_i)$ is minimized. By the subadditivity of v_i 's marginal gains (Lemma 1), $v_i(S \mid A_i) > 0$ tells us that $\exists \hat{g} \in S$ such that $v_i(\hat{g} \mid A_i) > 0$. Since $v_i(S \mid A_i) \geq v_i(\hat{g} \mid A_i)$, we can replace S by $\{\hat{g}\}$, so we can assume without loss of generality that $|S| = 1$. $\square$

Lemma 16. *In the fair division instance $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, w)$, if v_i is submodular and all marginals are non-positive for agent i (i.e., $v_i(c \mid R) \leq 0$ for all $R \subseteq [m]$ and $c \in [m] \setminus R$), then A is PROPx-fair to agent i iff $-v_i(A_i) \leq w_i \cdot (-v_i([m]))$ or $-v_i(A_i \setminus \{c\}) < w_i \cdot (-v_i([m]))$ for some $c \in A_i$ such that $v_i(c \mid A_i \setminus \{c\}) < 0$.*

Proof sketch. In Definition 6, we need to find $S \subseteq A_i$ such that $-v_i(S \mid A_i \setminus S) > 0$ and $-v_i(S \mid A_i \setminus S)$ is minimized. By the superadditivity of v_i 's marginal losses (Lemma 2), $-v_i(S \mid A_i \setminus S) > 0$ tells us that $\exists \hat{c} \in S$ such that $-v_i(\hat{c} \mid A_i \setminus \{\hat{c}\}) > 0$. Since $-v_i(S \mid A_i \setminus S) \geq -v_i(\hat{c} \mid A_i \setminus \{\hat{c}\})$, we can replace S by $\{\hat{c}\}$, so we can assume without loss of generality that $|S| = 1$. $\square$

C.6 PROPM and PROPavg

Definition 19 (PROPM and PROPavg). *Let A be an allocation for the fair division instance $([n], [m], (v_i)_{i=1}^n, w)$. Fix an agent i . For every $j \in [n] \setminus \{i\}$, let $K_j := \{v_i(S \mid A_i) : S \subseteq A_j \text{ and } v_i(S \mid A_i) > 0\}$. Let $\tau_j := 0$ if $K_j = \emptyset$ and $\tau_j := \min(K_j)$ otherwise. Let $T := \{\tau_j : j \in [n] \setminus \{i\} \text{ and } \tau_j > 0\}$. Define conditions C1, C2, C3, and C4 as follows:*

- (C1) $v_i(A_i) \geq w_i \cdot v_i([m])$.
- (C2) $v_i(A_i \setminus S) > w_i \cdot v_i([m])$ for every $S \subseteq A_i$ such that $v_i(S \mid A_i \setminus S) < 0$.
- (C3) $T = \emptyset$, or $v_i(A_i) + \max(T) > w_i \cdot v_i([m])$.
- (C4) $T = \emptyset$, or $v_i(A_i) + \frac{1}{|T|} \sum_{\tau_j \in T} \tau_j > w_i \cdot v_i([m])$.

A is PROPM-fair to i if either C1 holds or both C2 and C3 hold. A is PROPavg-fair to i if either C1 holds or both C2 and C4 hold.

PROPM was originally defined by [BGGS21a], and PROPavg was originally defined in [KM25], both for the additive goods setting. Note that PROPM, PROPavg, and PROPx are all equivalent to each other for chores.

The definitions of PROPM and PROPavg simplify under submodular valuations over goods and over chores. The proof is similar to that of PROPx (Lemmas 15 and 16) and is based on Lemmas 1 and 2, so we omit the details here.

The original definitions of PROPM and PROPavg have minor errors, which our definitions fix. We now explain these errors and motivate why we decided to define them the way we have done.

C.6.1 Defining PROPM

PROPM was first defined in [BGGS21a] for equal entitlements and goods with additive valuations. Moreover, they claimed that PROPx implies PROPM and PROPM implies PROP1.

According to [BGGS21a], when dividing a set $[m]$ of goods among n agents, A is PROPM-fair to agent i if $v_i(A_i) + \max_{j \neq i} m_i(A_j) \geq v_i([m])/n$, where $m_i(S) := \min_{g \in S} v_i(g)$. However, [BGGS21a] do not explicitly state what $m_i(\emptyset)$ is. The well-known convention of $\min(\emptyset) = \infty$ leads to the strange phenomenon where every agent is PROPM-satisfied if two agents receive no goods (whereas PROP1 is not guaranteed). One way to fix this is to only consider agents with non-empty bundles, i.e., we say that i is PROPM-satisfied by A if $v_i(A_i) + \max_{j \in [n] \setminus \{i\}: A_j \neq \emptyset} m_i(A_j) \geq v_i([m])/n$. This does not give a satisfactory definition when $A_j = \emptyset$ for all $j \in N \setminus \{i\}$ (assuming $\max(\emptyset) = -\infty$), so we define A to be PROPM-fair to i for that edge case.

The above idea, along with extending the definition to mixed manna in the same way as PROPx (Definition 6), gives us our definition of PROPM (Definition 19). Moreover, the following two example instances (having 3 equally-entitled agents with identical additive valuations) guided our definition of PROPM.

1. Consider three goods of values 100, 10, and 1. Intuitively, each agent should get 1 good each, and that should be considered fair.
2. Consider 5 items of values -100 , -100 , -100 , 10, 1. Intuitively, allocation $(\{-100, 10, 1\}, \{-100\}, \{-100\})$ should not be fair, and allocation $(\{-100, 10\}, \{-100, 1\}, \{-100\})$ should be fair. In both allocations, removing a chore makes an agent PROP-satisfied, so just satisfying this condition is not enough. We also need to look at the goods.

For mixed manna, [LMM22] define a notion called PropMX, but that definition is weak: when all items are goods, every allocation is trivially PropMX.

[BGS21b] showed that for equal entitlements and goods with additive valuations, a PROPm allocation always exists and can be computed in polynomial time. It can be verified that their result also works for our definition of PROPm (Definition 19).

C.6.2 Defining PROPavg

PROPavg was first defined by [KM25] for additive goods. Moreover, they claimed that PROPx implies PROPavg and PROPavg implies PROPm. However, their former claim is incorrect for their definition.

Their definition of PROPavg is equivalent to ours for equal entitlements and additive goods, except that in condition C4, they use $\text{sum}(T)/(n-1)$ instead of $\text{sum}(T)/|T|$. Consider a fair division instance with three equally-entitled agents and two identical goods. Let A be an allocation where agent 1 gets all goods and the other agents get nothing. This allocation is PROPx and PROPm, but is not PROPavg by [KM25]’s definition.

We believe that finding a feasible fairness notion in between PROPx and PROPm is well-motivated, so we slightly change [KM25]’s definition to ensure that PROPx implies PROPavg. Also, for additive goods, when no agent gets an empty bundle in an allocation, then $|T| = n - 1$, so both definitions are the same.

D Proofs of Implications

D.1 Among Derived Notions

Remark 17. *For any fairness notion F , if an allocation is F -fair to an agent i , then it is also epistemic- F -fair to agent i . If an allocation is epistemic- F -fair to an agent i , then it is also minimum- F -share-fair to agent i . If there are only two agents, then an allocation is epistemic- F -fair to an agent i iff it is F -fair to agent i .*

Remark 18. *For any fairness notion F , if an allocation is groupwise- F -fair to an agent i , then it is also pairwise- F -fair to agent i and F -fair to agent i . When there are only two agents, all three of these notions are equivalent.*

Lemma 19. *Let Ω be a set containing pairs of the form $(\mathcal{I}, A)$, where $\mathcal{I}$ is a fair division instance and A is an allocation for $\mathcal{I}$. For any two fairness notions F_1 and F_2 , let $\phi(F_1, F_2)$ be the proposition “ $\forall (\mathcal{I}, A) \in \Omega$, for every agent i in $\mathcal{I}$, A is F_2 -fair to i whenever A is F_1 -fair to i ”. Then $\phi(F_1, F_2)$ implies $\phi(\text{epistemic-}F_1, \text{epistemic-}F_2)$ and $\phi(\text{min-}F_1\text{-share}, \text{min-}F_2\text{-share})$.*

Proof sketch. If $\phi(F_1, F_2)$, then an epistemic- F_1 -certificate is also an epistemic- F_2 -certificate, and a min- F_1 -share-certificate is also a min- F_2 -share-certificate. $\square$

D.2 Among EF, EFX, EF1

Here we look at implications among EF, EFX, EF1, and their epistemic variants.

Remark 20 ($\text{EF} \implies \text{EFX} + \text{EF1}$). *If an allocation is EF-fair to agent i , then it is also EFX-fair and EF1-fair to i .*

Because of how we define EFX (Definition 3), EFX does not always imply EF1 (see Example 119 for a counterexample with subadditive valuations). However, $\text{EFX} \implies \text{EF1}$ for many common settings, as the following lemma shows.

Lemma 21 ($\text{EFX} \implies \text{EF1}$). *For the fair division instance $([n], [m], (v_i)_{i=1}^n, w)$, let allocation A be EFX-fair to agent i . Then agent i is EF1-satisfied in these scenarios:*

1. v_i is additive.
2. v_i is doubly strictly monotone ($\exists$ a partition (G, C) of $[m]$ s.t. $v_i(g|\cdot) > 0 \ \forall g \in G$, and $v_i(c|\cdot) < 0 \ \forall c \in C$).
3. Agents have equal entitlements, v_i is submodular, and all items are goods for agent i .
4. v_i is submodular and all items are chores for agent i .

Proof. Suppose i is EFX-satisfied but EF1-envies agent j .

Since i EF1-envies j , for all $t \in A_j$, we have

$$\frac{v_i(A_i)}{w_i} < \frac{v_i(A_j \setminus \{t\})}{w_j}.$$

If $v_i(t | A_i) > 0$ for some $t \in A_j$, then i would EFX-envy j , so we get $v_i(t | A_i) \leq 0$ for all $t \in A_j$.

Since i EF1-envies j , for all $t \in A_i$, we have

$$\frac{v_i(A_i \setminus \{t\})}{w_i} < \frac{v_i(A_j)}{w_j}.$$

If $v_i(t | A_i \setminus \{t\}) < 0$ for some $t \in A_i$, then i would EFX-envy j , so we get $v_i(t | A_i \setminus \{t\}) \geq 0$ for all $t \in A_i$.

If v_i is additive, we get $v_i(A_i) \geq 0 \geq v_i(A_j)$, which is a contradiction. If v_i is doubly strictly monotone, then all items in A_j are chores and all items in A_i are goods. Hence, $v_i(A_i) \geq 0 \geq v_i(A_j)$, which is a contradiction.

Suppose agents have equal entitlements, all items are goods for i , and v_i is submodular. By the subadditivity of marginal gains (Lemma 1), we get

$$v_i(A_j | A_i) \leq \sum_{g \in A_j} v_i(g | A_i) \leq 0.$$

Hence, $v_i(A_j) \leq v_i(A_i \cup A_j) = v_i(A_i) + v_i(A_j | A_i) \leq v_i(A_i)$, which is a contradiction.

Suppose all items are chores for i and v_i is submodular. By the superadditivity of marginal losses (Lemma 2), we get

$$v_i(A_i) \geq \sum_{c \in A_i} v_i(c | A_i \setminus \{c\}) \geq 0.$$

Hence, $v_i(A_i) \geq 0 \geq v_i(A_j)$, which is a contradiction.

A contradiction implies that it is impossible for agent i to be EFX-satisfied but not EF1-satisfied. $\square$

Lemma 22 (MXS $\implies$ EF1 for $n = 2$). *Let $([2], [m], (v_1, v_2), w)$ be a fair division instance with indivisible items. If v_1 is additive and agent 1 is MXS-satisfied by allocation A , then agent 1 is also EF1-satisfied by A .*

Proof. Suppose A is MXS-fair to agent 1 but not EF1-fair to her. Then agent 1 envies agent 2 in A , so $v_1(A_1) < v_1(A_2)$. Let B be agent 1's MXS-certificate for A . Then $v_1(B_1) \leq v_1(A_1)$. Moreover, $v_1(A_2) = v_1([m]) - v_1(A_1) \leq v_1([m]) - v_1(B_1) = v_1(B_2)$. Hence, we get $v_1(B_1) \leq v_1(A_1) < v_1(A_2) \leq v_1(B_2)$.

Let $G := \{g \in [m] : v_1(g) > 0\}$ and $C := \{c \in [m] : v_1(c) < 0\}$. Let $\max(\emptyset) := -\infty$ and $\min(\emptyset) := \infty$.

Since agent 1 is EFX-satisfied by B and not EF1-satisfied by A , for every $\hat{g} \in A_2$, we get

$$\begin{aligned} \frac{v_1(A_2) - v_1(\hat{g})}{w_2} &> \frac{v_1(A_1)}{w_1} \geq \frac{v_1(B_1)}{w_1} \\ &\geq \frac{1}{w_2} \left(v_1(B_2) - \min_{g \in B_2 \cap G} v_1(g) \right) \\ &\geq \frac{1}{w_2} \left(v_1(A_2) - \min_{g \in B_2 \cap G} v_1(g) \right). \end{aligned}$$

Hence, for every $\hat{g} \in A_2$, we get $v_1(\hat{g}) < \min_{g \in B_2 \cap G} v_1(g)$. Hence, $A_2 \cap G$ and $B_2 \cap G$ are disjoint, so $A_2 \cap G \subseteq B_1 \cap G$. Let $d_i := -v_i$ for all i . Then for every $\hat{c} \in A_1$, we get

$$\begin{aligned} \frac{d_1(A_1) - d_1(\hat{c})}{w_1} &> \frac{d_1(A_2)}{w_2} \geq \frac{d_1(B_2)}{w_2} \\ &\geq \frac{1}{w_1} \left(d_1(B_1) - \min_{c \in B_1 \cap C} d_1(c) \right) \\ &\geq \frac{1}{w_1} \left(d_1(A_1) - \min_{c \in B_1 \cap C} d_1(c) \right). \end{aligned}$$

Hence, for every $\hat{c} \in A_1$, we have $d_1(\hat{c}) < \min_{c \in B_1 \cap C} d_1(c)$. Hence, $A_1 \cap C$ and $B_1 \cap C$ are disjoint, so $B_1 \cap C \subseteq A_2 \cap C$. Hence,

$$\begin{aligned} v_1(A_2) &= v_1(A_2 \cap G) - d_1(A_2 \cap C) \\ &\leq v_1(B_1 \cap G) - d_1(B_1 \cap C) = v_1(B_1), \end{aligned}$$

which is a contradiction. Hence, it's not possible for A to be MXS-fair to agent 1 but not EF1-fair to her. $\square$

We will now show that every MXS allocation is also M1S when marginal values are *doubly monotone*. We first state a simple observation on how EFX and EF1 are related. Then we use that as the basis for an iterative algorithm for computing an agent's M1S certificate using their MXS certificate.

Observation 23. *Let $([n], [m], (v_i)_{i=1}^n, w)$ be a fair division instance where v_i is (G, C) -doubly-monotone for some agent i (i.e., $[m] = G \cup C$, $v_i(g \mid \cdot) \geq 0 \ \forall g \in G$, and $v_i(c \mid \cdot) \leq 0 \ \forall c \in C$). If all of the following hold for an allocation A , then A is EF1-fair to agent i .*

1. A is EFX-fair to agent i .
2. $v_i(g \mid A_i) > 0$ for all $g \in G \setminus A_i$.
3. $v_i(c \mid A_i \setminus \{c\}) < 0$ for all $c \in A_i \cap C$.

Algorithm 1 $\text{improve}_i(Z)$: Here Z is an allocation for the fair division instance $([n], G \cup C, (v_j)_{j=1}^n, w)$.

```

1: while true do
2:   if  $\exists j \in [n] \setminus \{i\}$  and  $\exists g \in G \cap Z_j$  s.t.  $v_i(g \mid Z_i) = 0$  then
3:     Set  $Z_i = Z_i \cup \{g\}$  and  $Z_j = Z_j \setminus \{g\}$ .
4:   else if  $\exists c \in Z_i \cap C$  s.t.  $v_i(c \mid Z_i \setminus \{c\}) = 0$  then
5:     Set  $Z_i = Z_i \setminus \{c\}$ .
6:     Pick any  $j \in [n] \setminus \{i\}$ . Set  $Z_j = Z_j \cup \{c\}$ .
7:   else
8:     return  $Z$ .
9:   end if
10: end while

```

Lemma 24. Let $([n], [m], (v_i)_{i=1}^n, w)$ be a fair division instance where v_i is (G, C) -doubly-monotone for some agent i (i.e., $[m] = G \cup C$, $v_i(g \mid \cdot) \geq 0 \ \forall g \in G$, and $v_i(c \mid \cdot) \leq 0 \ \forall c \in C$). If an allocation A is MXS-fair to agent i , then it is also M1S-fair to agent i if $n = 2$ or $G = \emptyset$.

Proof. Without loss of generality, let $i = 1$. Let B be agent 1's MXS-certificate for A . Then B is EFX-fair to agent 1 and $v_1(B_1) \leq v_1(A_1)$. Define $\widehat{B} := \text{improve}_1(B)$, where improve_1 is defined in Algorithm 1.

In any iteration of the while loop in Algorithm 1, let X be the allocation at the beginning of the iteration. Either X is returned at line 8, or the allocation gets modified to Y , and (as we will prove soon) the following invariants hold:

1. If X is EFX-fair to agent 1, then Y is also EFX-fair to agent 1.
2. $|Y_1 \cap G| - |Y_1 \cap C| > |X_1 \cap G| - |X_1 \cap C|$.
3. $v_1(Y_1) = v_1(X_1)$.

The first invariant ensures that $\widehat{B}$ is also EFX-fair to agent 1. The second invariant ensures that improve_1 terminates. Then using ?? 23, we get that $\widehat{B}$ is EF1-fair to agent 1. The third invariant ensures that $v_1(\widehat{B}_1) = v_1(B_1) \leq v_1(A_1)$, which makes $\widehat{B}$ an M1S-certificate for allocation A and agent 1. The second and third invariants are trivial to show. We now prove the first invariant if $n = 2$ or $G = \emptyset$.

Suppose $Y_1 = X_1 \cup \{g\}$ and $Y_j = X_j \setminus \{g\}$, where $g \in G \cap X_j$ such that $v_1(g \mid X_1) = 0$. Since $G \neq \emptyset$, we have $n = j = 2$. Pick any $S \subseteq Y_2$ such that $v_1(S \mid Y_1) > 0$. Then

$$\begin{aligned}
& v_1(S \cup \{g\} \mid X_1) \\
&= v_1(S \cup \{g\} \cup X_1) - v_1(X_1) \\
&= (v_1(S \cup Y_1) - v_1(Y_1)) + (v_1(Y_1) - v_1(X_1)) \\
&= v_1(S \mid Y_1) > 0.
\end{aligned}$$

Since X is EFX-fair to agent 1, we get

$$\frac{v_1(Y_1)}{w_1} = \frac{v_1(X_1)}{w_1} \geq \frac{v_1(X_2 \setminus (S \cup \{g\}))}{w_2} = \frac{v_1(Y_2 \setminus S)}{w_j},$$

so 1 doesn't EFX-envy agent j . Hence, Y is EFX-fair to 1.

Suppose $Y_1 = X_1 \setminus \{c\}$ and $Y_j = X_j \cup \{c\}$, where $c \in X_1 \cap C$ such that $v_1(c \mid X_1 \setminus \{c\}) = 0$. Pick any $S \subseteq Y_1$ such that $v_1(S \mid Y_1 \setminus S) < 0$. Then

$$\begin{aligned} & v_1(S \cup \{c\} \mid X_1 \setminus (S \cup \{c\})) \\ &= v_1(X_1) - v_1(X_1 \setminus (S \cup \{c\})) \\ &= (v_1(X_1) - v_1(Y_1)) + (v_1(Y_1) - v_1(Y_1 \setminus S)) \\ &= v_1(S \mid Y_1 \setminus S) < 0. \end{aligned}$$

Since X is EFX-fair to agent 1, we get that for any other agent $k \in [n] \setminus \{1\}$,

$$\frac{v_1(Y_1 \setminus S)}{w_1} = \frac{v_1(X_1 \setminus (S \cup \{c\}))}{w_1} \geq \frac{v_1(X_k)}{w_k} \geq \frac{v_1(Y_k)}{w_k}.$$

Hence, agent 1 is EFX-satisfied with Y .

Hence, invariant 1 holds, so $\hat{B}$ is agent 1's M1S-certificate for allocation A . $\square$

D.3 Among PROP-Based Notions

Lemma 25 ($\text{PROPx} \implies \text{PROPavg}$). *In a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$, if an allocation is PROPx -fair to agent i , then it is also PROPavg -fair to agent i .*

Proof. Assume (for the sake of contradiction) that there is an allocation A where agent i is PROPx -satisfied but not PROPavg -satisfied. Since i is not PROPavg -satisfied, we get $v_i(A_i) \leq w_i v_i([m])$. Since i is PROPx -satisfied, we get

- $v_i(A_i \setminus S) > w_i v_i([m])$ for all $S \subseteq A_i$ such that $v_i(S \mid A_i \setminus S) < 0$.
- $v_i(A_i \cup S) > w_i v_i([m])$ for all $S \subseteq [m] \setminus A_i$ such that $v_i(S \mid A_i) > 0$.

Since i is not PROPavg -satisfied, we get that $T \neq \emptyset$ and $v_i(A_i) + \text{sum}(T)/|T| \leq w_i v_i([m])$ (T is defined in Definition 19). For each $\tau_j \in T$, we have $0 < \tau_j = v_i(S_j \mid A_i)$ for some $S_j \subseteq A_j$. Since A is PROPx -fair to i , we have $v_i(A_i) + \tau_j > w_i v_i([m])$ for all $\tau_j \in T$. On averaging these inequalities, we get $v_i(A_i) + \text{sum}(T)/|T| > w_i v_i([m])$, which implies that A is PROPavg -satisfied, which is a contradiction. Hence, if i is PROPx -satisfied by A , then she is also PROPavg -satisfied by A . $\square$

Lemma 26 ($\text{PROPM} \implies \text{PROP1}$). *For a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$, if an allocation A is PROPM -fair to agent i , then it is also PROP1 -fair to agent i if at least one of these conditions holds:*

1. v_i is submodular.
2. v_i is doubly strictly monotone ($\exists$ a partition (G, C) of $[m]$ s.t. $v_i(g \mid \cdot) > 0 \ \forall g \in G$, and $v_i(c \mid \cdot) < 0 \ \forall c \in C$).

Proof. Suppose allocation A is PROPM -fair to i but not PROP1 -fair to i . Then

1. $v_i(A_i) < w_i v_i([m])$ (not PROP1).
2. $v_i(A_i \setminus \{c\}) \leq w_i v_i([m])$ for all $c \in A_i$ (not PROP1).
3. $v_i(A_i \cup \{g\}) \leq w_i v_i([m])$ for all $g \notin A_i$ (not PROP1).
4. $v_i(A_i \setminus \{c\}) > w_i v_i([m])$ for all $c \in A_i$ such that $v_i(c \mid A_i \setminus \{c\}) < 0$ (by PROPM fairness).
5. $T = \emptyset$ or $v_i(A_i) + \max(T) > w_i v_i([m])$ (by PROPM fairness; c.f. Definition 19 for the definition of T).

By 2 and 4, we get $v_i(c \mid A_i \setminus \{c\}) \geq 0$ for all $c \in A_i$. If v_i is doubly strictly monotone, then A_i only contains goods, so $v_i(A_i) \geq 0$. If v_i is submodular, then by superadditivity of v_i 's marginal loss (Lemma 2), we get

$$v_i(A_i) \geq \sum_{c \in A_i} v_i(c \mid A_i \setminus \{c\}) \geq 0.$$

Suppose $T = \emptyset$. Then $\tau_j = 0$ for all $j \in [n] \setminus \{i\}$. Hence, for all $j \in [n] \setminus \{i\}$, we have $v_i(A_j \mid A_i) \leq 0$. If v_i is doubly strictly monotone, then $[m] \setminus A_i$ contains only chores, so $v_i([m] \setminus A_i \mid A_i) \leq 0$. If v_i is submodular, then $v_i(\cdot \mid A_i)$ is subadditive by Lemma 1, so

$$v_i([m] \setminus A_i \mid A_i) \leq \sum_{j \in [n] \setminus \{i\}} v_i(A_j \mid A_i) \leq 0.$$

Hence, $v_i(A_i) \geq v_i([m])$. If $v_i([m]) \leq 0$, then $v_i(A_i) \geq 0 \geq w_i v_i([m])$, and if $v_i([m]) \geq 0$, then $v_i(A_i) \geq v_i([m]) \geq w_i v_i([m])$. This contradicts 1, so $T \neq \emptyset$.

Let $\max(T) = \tau_{\hat{j}} > 0$. By definition of $\tau_{\hat{j}}$, we get

$$\begin{aligned} 0 < \tau_{\hat{j}} &= \min(\{v_i(S \mid A_i) \mid S \subseteq A_{\hat{j}} \text{ and } v_i(S \mid A_i) > 0\}) \\ &\leq \min(\{v_i(g \mid A_i) \mid g \in A_{\hat{j}} \text{ and } v_i(g \mid A_i) > 0\}). \end{aligned}$$

Case 1: $v_i(g \mid A_i) \leq 0$ for all $g \in A_{\hat{j}}$.

If v_i is doubly strictly monotone, then $A_{\hat{j}}$ only has chores, so $v_i(S \mid A_i) \leq 0$ for all $S \subseteq A_{\hat{j}}$. This contradicts $\tau_{\hat{j}} > 0$. Now let v_i be submodular. Since $v_i(\cdot \mid A_i)$ is subadditive by Lemma 1, for any $S \subseteq A_{\hat{j}}$, we get $v_i(S \mid A_i) \leq \sum_{c \in S} v_i(c \mid A_i) \leq 0$. This contradicts $\tau_{\hat{j}} > 0$.

Case 2: $v_i(\hat{g} \mid A_i) > 0$ for some $\hat{g} \in A_{\hat{j}}$.

Then $\max(T) = \tau_{\hat{j}} \leq v_i(\hat{g} \mid A_i)$. By 5, we get $w_i v_i([m]) < v_i(A_i) + \max(T) \leq v_i(A_i \cup \{\hat{g}\})$. But this contradicts 3.

Hence, it cannot happen that i is PROPm-satisfied by A but not PROP1-satisfied. $\square$

D.4 EF vs PROP

Here we look at implications between EF (and its epistemic variants) and PROP.

Lemma 27 (MEFS $\implies$ PROP, [BL16]). *For a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$, if v_i is subadditive and an allocation A is MEFS-fair to i , then A is also PROP-fair to i .*

Proof. Let B be agent i 's MEFS-certificate for A . Then for all $j \in [n]$, we have $v_i(B_i)/w_i \geq v_i(B_j)/w_j$. Sum these inequalities over all $j \in [n]$, weighting each by w_j , to get $v_i(B_i)/w_i \geq \sum_{j=1}^n v_i(B_j)$. Since v_i is subadditive, we get $v_i([m]) \leq \sum_{j=1}^n v_i(B_j)$. Hence,

$$\frac{v_i(A_i)}{w_i} \geq \frac{v_i(B_i)}{w_i} \geq \sum_{j=1}^n v_i(B_j) \geq v_i([m]). \quad \square$$

Lemma 28 (EF $\implies$ GPROP, [BL16]). *For a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$, if v_i is subadditive and agent i is envy-free in A , then A is groupwise-PROP-fair to i .*

Proof. Let S be a subset of agents containing i . Let $\hat{A}$ be the allocation obtained by restricting A to S (c.f. Definition 11). Then i is also envy-free in $\hat{A}$. $\hat{A}$ is also MEFS-fair to i , since $\hat{A}$ is its own MEFS-certificate. By Lemma 27, agent i is PROP-satisfied by $\hat{A}$. Since this is true for all S containing i , we get that A is groupwise-PROP-fair to agent i . $\square$

Lemma 29 (PROP $\implies$ EF for idval). *In a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ with identical superadditive valuations, a PROP allocation is also an EF allocation.*

Proof. Let v be the common valuation function. Let A be a PROP allocation. Then $v(A_i) \geq w_i v([m])$ for each agent i . Suppose $v(A_k) > w_k v([m])$ for some agent k . Sum these inequalities to get $\sum_{i=1}^n v(A_i) > v([m])$. This contradicts superadditivity of v , so $v(A_i) = w_i v([m])$ for each agent i . Hence, $v(A_i)/w_i = v([m])$ for all i , so A is EF. $\square$

Lemma 30 (PROP $\implies$ EF for $n = 2$). *In the fair division instance $([2], [m], (v_1, v_2), w)$, for some agent i , if v_i is superadditive and agent i is PROP-satisfied by allocation A , then she is also envy-free in A .*

Proof. Assume $i = 1$ without loss of generality. Then

$$\begin{aligned} \frac{v_1(A_1)}{w_1} &\geq v_1([m]) \geq v_1(A_1) + v_1(A_2) \\ \implies v_1(A_2) &\leq v_1(A_1) \left(\frac{1}{w_1} - 1 \right) = w_2 \frac{v_1(A_1)}{w_1}. \end{aligned}$$

Hence, agent 1 does not envy agent 2. $\square$

Lemma 31 (PPROP $\implies$ GPROP). *In instance $([n], [m], (v_i)_{i=1}^n, w)$, if v_i is submodular and allocation A is PPROP-fair to agent i , then A is also GPROP-fair to i .*

Proof. Since A is PPROP-fair to i , for all $j \in [n] \setminus \{i\}$,

$$\begin{aligned} v_i(A_i) &\geq \frac{w_i}{w_i + w_j} v_i(A_i \cup A_j) \\ \implies (w_i + w_j) v_i(A_i) &\geq w_i (v_i(A_j \mid A_i) + v_i(A_i)) \\ \implies w_j v_i(A_i) &\geq w_i v_i(A_j \mid A_i). \end{aligned}$$

Sum these inequalities over all $j \in [n] \setminus \{i\}$ to get

$$(1 - w_i) v_i(A_i) \geq w_i \sum_{j \neq i} v_i(A_j \mid A_i).$$

By Lemma 1, $v_i(\cdot \mid A_i)$ is subadditive. Hence,

$$\begin{aligned} (1 - w_i) v_i(A_i) &\geq w_i \sum_{j \neq i} v_i(A_j \mid A_i) \geq w_i v_i([m] \setminus A_i \mid A_i) \\ \implies v_i(A_i) &\geq w_i v_i([m]). \end{aligned}$$

So, A is PROP to i . This also works if we restrict the allocation to a subset of agents, so A is also GPROP to i . $\square$

D.5 EFX, EF1 vs PROP_x, PROP_m, PROP1

Lemma 32 (MXS $\implies$ PROP1, Theorem 3 of [CGR⁺23]). *In a fair division instance with equal entitlements, if v_i is additive for some agent i , $v_i(g) \geq 0$ for every item g , and an allocation A is MXS-fair to agent i , then A is also PROP1-fair to agent i .*

Lemma 33 (EEF1 $\implies$ PROP1 for chores). *For a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ where all items are chores to agent i and v_i is subadditive, if an allocation X is EEF1-fair to agent i , then X is also PROP1-fair to i .*

Proof. Let A be agent i 's EEF1-certificate for X . Suppose A is not PROP1-fair to i . We get $v_i(A_i) < w_i v_i([m])$ and $v_i(A_i \setminus \{c\}) \leq w_i v_i([m])$ for all $c \in A_i$.

Since v_i is subadditive, there exists $j \in [n] \setminus \{i\}$ such that $v_i(A_j) > w_j v_i([m])$ (otherwise $v_i([m]) \leq \sum_{j=1}^n v_i(A_j) < \sum_{j=1}^n w_j v_j([m]) = v_i([m])$). Hence,

$$\frac{v_i(A_i)}{w_i} < v_i([m]) < \frac{v_i(A_j)}{w_j}.$$

Thus, i envies j . But A is EF1-fair to i , so $\exists c \in A_i$ such that

$$\frac{v_i(A_i \setminus \{c\})}{w_i} \geq \frac{v_i(A_j)}{w_j}.$$

Since i is PROP1-unsatisfied and $v_i(A_j) > w_j v_i([m])$,

$$\frac{v_i(A_j)}{w_j} \leq \frac{v_i(A_i \setminus \{c\})}{w_i} \leq v_i([m]) < \frac{v_i(A_j)}{w_j},$$

which is a contradiction. Hence, A is PROP1-fair to i , and so, X is PROP1-fair to agent i . $\square$

Lemma 34 (EEF1 $\implies$ PROP1). *Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ where v_i is submodular for some agent i and $w_i \leq w_j$ for all $j \in [n]$. If an allocation X is EEF1-fair to agent i , then X is also PROP1-fair to i .*

Proof. Let A be agent i 's EEF1-certificate for X . Then A is EF1-fair to i . We will show that A is PROP1-fair to i (which would prove that X is also PROP1-fair to i). This is trivial if A is PROP-fair to i , so assume $v_i(A_i) < w_i v_i([m])$.

By Lemma 1, we get that $v_i(\cdot \mid A_i)$ is subadditive. Hence,

$$v_i([m]) = v_i(A_i) + v_i([m] \setminus A_i \mid A_i) \leq v_i(A_i) + \sum_{j \neq i} v_i(A_j \mid A_i).$$

Hence, for some $j \in [n] \setminus \{i\}$, we get $v_i(A_j \mid A_i) > w_j v_i([m])$. Hence,

$$\frac{v_i(A_i)}{w_i} < v_i([m]) < \frac{v_i(A_j \mid A_i)}{w_j} \leq \frac{v_i(A_j)}{w_j}.$$

So, i envies j in A . Since A is EF1-fair to i , we get two cases:

Case 1: $\exists c \in A_i$ such that $\frac{v_i(A_i \setminus \{c\})}{w_i} \geq \frac{v_i(A_j)}{w_j}$.

Then $v_i(A_i \setminus \{c\}) > w_i v_i([m])$, so A is PROP1-fair.

Case 2: $\exists g \in A_j$ such that $\frac{v_i(A_i)}{w_i} \geq \frac{v_i(A_j \setminus \{g\})}{w_j}$.

$$\begin{aligned} \frac{v_i(A_i \cup \{g\})}{w_i} &= \frac{v_i(A_i)}{w_i} + \frac{v_i(g \mid A_i)}{w_i} \\ &\geq \frac{v_i(A_j \setminus \{g\})}{w_j} + \frac{v_i(g \mid A_i)}{w_j} \end{aligned} \quad (w_i \leq w_j)$$

$$\begin{aligned}
&\geq \frac{v_i(A_j \setminus \{g\} \mid A_i) + v_i(g \mid A_i)}{w_j} && (v_i \text{ is submodular}) \\
&\geq \frac{v_i(A_j \mid A_i)}{w_j} && (v_i(\cdot \mid A_i) \text{ is subadditive}) \\
&> v_i([m]).
\end{aligned}$$

In both cases, we get that A is PROP1-fair to i . Hence, X is also PROP1-fair to i . $\square$

Lemma 35 (EF1 $\implies$ PROP1 for $n = 2$). *Consider a fair division instance $([2], [m], (v_1, v_2), w)$ where v_1 is subadditive. If an allocation A is EF1-fair to agent 1, then it is also PROP1-fair to agent 1.*

Proof. If A is PROP-fair to agent 1, then we are done, so assume otherwise. By subadditivity of v_1 , we get $v_1([m]) \leq v_1(A_1) + v_1(A_2)$. Since $v_1(A_1) < w_1 v_1([m])$, we get $v_1(A_2) > w_2 v_1([m])$. Hence, $v_1(A_1)/w_1 < v_1([m]) < v_1(A_2)/w_2$, so agent 1 envies agent 2. Since A is EF1-fair to agent 1, there are two cases:

Case 1: $\exists c \in A_1$ such that $\frac{v_1(A_1 \setminus \{c\})}{w_1} \geq \frac{v_1(A_2)}{w_2}$.

Then $v_1(A_1 \setminus \{c\}) > w_1 v_1([m])$, so A is PROP1-fair to 1.

Case 2: $\exists \hat{g} \in A_2$ such that $\frac{v_1(A_1)}{w_1} \geq \frac{v_1(A_2 \setminus \{\hat{g}\})}{w_2}$. By subadditivity of v_1 and PROP-unfairness of A , we get

$$\begin{aligned}
v_1([m]) &\leq v_1(A_1 \cup \{\hat{g}\}) + v_1(A_2 \setminus \{\hat{g}\}) \leq v_1(A_1 \cup \{\hat{g}\}) + \frac{w_2}{w_1} v_1(A_1) < v_1(A_1 \cup \{\hat{g}\}) + w_2 v_1([m]) \\
&\implies w_1 v_1([m]) < v_1(A_1 \cup \{\hat{g}\}),
\end{aligned}$$

so A is PROP1-fair to agent 1. $\square$

Lemma 36 (EEFX $\implies$ PROP_x, Lemma 2.1 of [LLW22]). *Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ where the items are chores to agent i and v_i is subadditive. If an allocation A is EEFX-fair to agent i , then it is also PROP_x-fair to agent i .*

Proof. Let B be agent i 's EEFX-certificate for A .

Since i is EFX-satisfied with B , for all $j \in [n] \setminus \{i\}$,

$$\min_{S \in \mathcal{S}} \frac{v_i(B_i \setminus S)}{w_i} \geq \frac{v_i(B_j)}{w_j},$$

where $\mathcal{S} := \{S \subseteq B_i : v_i(S \mid B_i \setminus S) < 0\}$. Add these inequalities for all j , weighting each by w_j , to get

$$\min_{S \in \mathcal{S}} \frac{v_i(B_i \setminus S)}{w_i} > \sum_{j=1}^n v_i(B_j) \geq v_i([m]).$$

Thus, B is PROP_x-fair to i , so A is also PROP_x-fair to i . $\square$

Lemma 37 (EFX $\implies$ PROP_{avg}). *Let $([n], [m], (v_i)_{i=1}^n, -)$ be a fair division instance where v_i is submodular for some agent i and $v_i(A_i) \geq 0$. If an allocation A is EFX-fair to agent i , then it is also PROP_{avg}-fair to agent i .*

Proof. If A is PROP-fair to agent i , we are done, so assume otherwise. By Lemma 1, $v_i(\cdot \mid A_i)$ is subadditive. Hence, $v_i([m]) = v_i(A_i) + v_i([m] \mid A_i) \leq v_i(A_i) + \sum_{j \neq i} v_i(A_j \mid A_i)$. Then for some agent $\hat{j} \in [n] \setminus \{i\}$, we have

$$v_i(A_i) < \frac{v_i([m])}{n} < v_i(A_{\hat{j}} \mid A_i) \leq v_i(A_{\hat{j}}).$$

Since i doesn't EFX-envy $\hat{j}$, for all $S \subseteq A_i$ such that $v_i(S \mid A_i \setminus S) < 0$, we get

$$v_i(A_i \setminus S) \geq v_i(A_{\hat{j}}) > \frac{v_i([m])}{n}.$$

Thus, condition C2 of PROPavg holds (c.f. Definition 19).

For any $j \in [n] \setminus \{i\}$, recall that $\tau_j := 0$ if $v_i(S \mid A_i) \leq 0 \ \forall S \subseteq A_j$, and $\tau_j := \min(\{v_i(S \mid A_i) : S \subseteq A_j \text{ and } v_i(S \mid A_i) > 0\})$ otherwise. If $\tau_j = 0$, then $v_i(A_j \mid A_i) \leq 0 \leq v_i(A_i) + \tau_j$. Now let $\tau_j > 0$. Then $\tau_j = v_i(S \mid A_i) > 0$ for some $S \subseteq A_j$. Since A is EFX-fair to i ,

$$\begin{aligned} v_i(A_i) + \tau_j &= v_i(A_i) + v_i(S \mid A_i) \\ &\geq v_i(A_j \setminus S) + v_i(S \mid A_i) \\ &\geq v_i(A_j \setminus S \mid A_i) + v_i(S \mid A_i) && (v_i \text{ is submodular}) \\ &\geq v_i(A_j \mid A_i). && (v_i(\cdot \mid A_i) \text{ is subadditive}) \end{aligned}$$

Let $T := \{\tau_j : j \in [n] \setminus \{i\} \text{ and } \tau_j > 0\}$. On adding the above inequalities for all $j \in [n] \setminus \{i\}$, and using the subadditivity of $v_i(\cdot \mid A_i)$, we get

$$\begin{aligned} (n-1)v_i(A_i) + \sum_{j \neq i} \tau_j &\geq \sum_{j \neq i} v_i(A_j \mid A_i) \geq v_i([m] \setminus A_i \mid A_i) \\ \implies nv_i(A_i) + \text{sum}(T) &\geq v_i([m]). \end{aligned}$$

If $T = \emptyset$, condition C4 of PROPavg trivially holds. Else, $v_i(A_i) + \text{sum}(T)/(n-1) > v_i(A_i) + \text{sum}(T)/n \geq v_i([m])/n$. $\square$

Lemma 38 (EFX $\implies$ PROPx for $n = 2$). *Let $([2], [m], (v_1, v_2), w)$ be a fair division instance where v_i is subadditive for some agent i . If an allocation A is EFX-fair to agent i , then it is also PROPx-fair to agent i .*

Proof. Let the two agents be i and j . If A is PROP-fair to agent i , we are done, so assume $v_i(A_i) < w_i v_i([m])$. By subadditivity of v_i , we get $v_i([m]) \leq v_i(A_i) + v_i(A_j)$. Hence,

$$\frac{v_i(A_i)}{w_i} < v_i([m]) < \frac{v_i(A_j)}{w_j}.$$

Hence, i envies j . Let $S \subseteq A_i$ such that $v_i(S \mid A_i \setminus S) < 0$. Since A is EFX-fair to i , we get

$$\frac{v_i(A_i \setminus S)}{w_i} \geq \frac{v_i(A_j)}{w_j} > v_i([m]).$$

Let $S \subseteq A_j$ such that $v_i(S \mid A_i) > 0$. Then

$$\begin{aligned} v_i([m]) &\leq v_i(A_i \cup S) + v_i(A_j \setminus S) && (v_i \text{ is subadditive}) \\ &\leq v_i(A_i \cup S) + \frac{w_j}{w_i} v_i(A_i) && (A \text{ is EFX-fair to } i) \\ &< v_i(A_i \cup S) + \frac{w_j}{w_i} v_i(A_i \cup S) = \frac{v_i(A_i \cup S)}{w_i}. \end{aligned}$$

Hence, A is PROPx-fair to i . $\square$

D.6 MMS vs EFX

We prove some results connecting MMS, EFX, and related notions, using techniques from [PR20, CGR⁺23].

Lemma 39. *For a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$, suppose an allocation A is WMMS-fair to agent i and i envies every other agent. Then A is also EFX-fair to agent i if at least one of these conditions hold:*

1. *The items are goods to agent i .*
2. *v_i is additive and $w_i \leq w_j$ for all $j \in [n] \setminus \{i\}$.*

Proof. Suppose agent i is not EFX-satisfied by A , i.e., she EFX-envies some agent j . Then $\exists S \subseteq A_i \cup A_j$ where either

1. $S \subseteq A_j$, $v_i(S \mid A_i) > 0$, and $\frac{v_i(A_i)}{w_i} < \frac{v_i(A_j \setminus S)}{w_j}$.
2. $S \subseteq A_i$, $v_i(S \mid A_i \setminus S) < 0$, and $\frac{v_i(A_i \setminus S)}{w_i} < \frac{v_i(A_j)}{w_j}$.

If all items are goods, case 2 doesn't occur.

Case 1: $S \subseteq A_j$

Let B be the allocation obtained by transferring S from A_j to A_i . Formally, let $B_i := A_i \cup S$, $B_j := A_j \setminus S$, and $B_k := A_k$ for all $k \in [n] \setminus \{i, j\}$. Then

$$\begin{aligned} \frac{v_i(B_i)}{w_i} &= \frac{v_i(A_i) + v_i(S \mid A_i)}{w_i} > \frac{v_i(A_i)}{w_i}, \\ \frac{v_i(B_j)}{w_j} &= \frac{v_i(A_j \setminus S)}{w_j} > \frac{v_i(A_i)}{w_i}, \end{aligned}$$

and for any $k \in [n] \setminus \{i, j\}$, we get

$$\frac{v_i(B_k)}{w_k} = \frac{v_i(A_k)}{w_k} > \frac{v_i(A_i)}{w_i}.$$

Hence,

$$\min_{k=1}^n \frac{v_i(B_k)}{w_k} > \frac{v_i(A_i)}{w_i} \geq \frac{\text{WMMS}_i}{w_i},$$

which is a contradiction.

Case 2: $S \subseteq A_i$

Let v_i be additive and $w_i \leq w_j$. Let B be the allocation obtained by transferring S from A_i to A_j . Formally, let $B_i := A_i \setminus S$, $B_j := A_j \cup S$, and $B_k := A_k$ for all $k \in [n] \setminus \{i, j\}$. Then

$$\begin{aligned} \frac{v_i(B_i)}{w_i} &= \frac{v_i(A_i) - v_i(S)}{w_i} > \frac{v_i(A_i)}{w_i}, \\ \frac{v_i(B_j)}{w_j} &= \frac{v_i(A_j) + v_i(S)}{w_j} > \frac{v_i(A_i \setminus S)}{w_i} - \frac{(-v_i(S))}{w_j} \geq \frac{v_i(A_i)}{w_i}, \end{aligned}$$

and for any $k \in [n] \setminus \{i, j\}$, we get

$$\frac{v_i(B_k)}{w_k} = \frac{v_i(A_k)}{w_k} > \frac{v_i(A_i)}{w_i}.$$

Hence,

$$\min_{k=1}^n \frac{v_i(B_k)}{w_k} > \frac{v_i(A_i)}{w_i} \geq \frac{\text{WMMS}_i}{w_i},$$

which is a contradiction. $\square$

Lemma 40 (MMS $\implies$ EFX for $n = 2$). *For a fair division instance $([2], [m], (v_i)_{i=1}^2, w)$, suppose an allocation A is WMMS-fair to agent 1. Then A is also EFX-fair to agent 1 if at least one of these conditions hold:*

1. *The items are goods to agent 1.*
2. *v_1 is additive and $w_1 \leq w_2$.*

Proof. If agent 1 doesn't envy agent 2, she is EFX-satisfied. Otherwise, she is EFX-satisfied because of Lemma 39. $\square$

Theorem 2 in [CGR⁺23] proves that MMS $\implies$ EEFX for additive valuations over goods and equal entitlements. The proof can be easily adapted to non-additive valuations over goods and unequal entitlements. For the sake of completeness, we give a proof below.

Lemma 41 (MMS $\implies$ EEFX, [CGR⁺23]). *For a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$, if all items are goods to agent i and allocation A is WMMS-fair to agent i , then A is also EEFX-fair to i .*

Proof. In any allocation X , let E_X be the set of agents envied by i , S_X be the set of agents EFX-envied by i , W_X be the total number of items among the agents in S_X .

$E_X := \{t \in [n] \setminus \{i\} : i \text{ envies } t \text{ in } X\}$, $S_X := \{t \in [n] \setminus \{i\} : i \text{ EFX-envies } t \text{ in } X\}$, $W_X := \sum_{t \in S_X} |X_t|$, and $\phi(X) := (-|E_X|, W_X)$.

First, we show that for any allocation X where $|E_X| \leq n - 2$ and $S_X \neq \emptyset$, there exists a *better* allocation Y , i.e. $Y_i = X_i$ and $\phi(Y) < \phi(X)$ (tuples are compared lexicographically).

Let $j \in S_X$ and $k \in [n] \setminus \{i\} \setminus E_X$. Since i EFX-envies j , $\exists S \subseteq X_j$ such that $v_i(S \mid X_i) > 0$ and

$$\frac{v_i(X_i)}{w_i} < \frac{v_i(X_j \setminus S)}{w_j}.$$

Let $Y_k := X_k \cup S$, $Y_j := X_j \setminus S$, and $Y_t := X_t$ for all $t \in [n] \setminus \{j, k\}$.

Then i envies j in Y . Hence, $E_X \subseteq E_Y$. If $k \in E_Y$, then $|E_Y| > |E_X|$, so $\phi(Y) < \phi(X)$. If $k \notin E_Y$, then $W_Y < W_X$, so $\phi(Y) < \phi(X)$. Hence, Y is better than X .

Set $X = A$. As long as $|E_X| < n - 1$ and $S_X \neq \emptyset$, keep modifying X as per Lemma 1. This process will eventually end, since $\phi(X)$ keeps reducing, and there are a finite number of different values $\phi(X)$ can take. Let B be the final allocation thus obtained. Then $B_i = A_i$, and $|E_B| = n - 1$ or $S_B = \emptyset$.

By Lemma 39, $|E_B| = n - 1$ implies $S_B = \emptyset$. Hence, B is agent i 's EEFX-certificate for A . $\square$

Definition 20 (leximin partition). *Let R be a totally-ordered set. For any sequence $X = (x_i)_{i=1}^n$ from R , define $\text{sorted}(X)$ to be a permutation of X where entries occur in non-decreasing order. For any two sequences $X = (x_i)_{i=1}^n$ and $Y = (y_i)_{i=1}^n$ from R , we say that $X \leq Y$ if $\exists i \in [n]$ such that $x_i \leq y_i$ and $x_j = y_j$ for all $j \in [i - 1]$. (Note that this relation $\leq$ over sequences is a total ordering.)*

For any set $S \subseteq M$, let $\Pi_n(S)$ denote the set of all n -partitions of S . We say that $P \in \Pi_n([m])$ is a leximin n -partition of a function $f : 2^{[m]} \rightarrow R$ if

$$P \in \operatorname{argmax}_{X \in \Pi_n([m])} \text{sorted}((f(X_j))_{j=1}^n).$$

Lemma 42 (MMS $\implies$ MXS). *Let $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, -)$ be a fair division instance. Let allocation A be MMS-fair to agent i . Then A is also MXS-fair to i if either all items are goods to agent i or v_i is additive.*

Proof. We will show that agent i 's leximin n -partition is her MXS-certificate for A . Without loss of generality, let $i = 1$. Let P be a leximin n -partition of v_1 such that $v_1(P_1) \leq \dots \leq v_1(P_n)$. Then $v_1(P_1) = \text{MMS}_1 \leq v_1(A_1)$.

For any agent $j \geq 2$, the allocation (P_1, P_j) is leximin for the instance $\widehat{\mathcal{I}} := (\{i, j\}, P_1 \cup P_j, (v_1, v_j), -)$. Hence, agent 1 is MMS-satisfied by (P_1, P_j) in $\widehat{\mathcal{I}}$. Since either all items are goods to agent 1 or v_1 is additive, by Lemma 40, agent 1 is EFX-satisfied by (P_1, P_j) in $\widehat{\mathcal{I}}$. Thus, P is EFX-fair to agent 1 in $\mathcal{I}$, so P is agent 1's MXS-certificate for A . $\square$

Lemma 43 (MMS $\implies$ MXS₀). *Let $\mathcal{I} := ([n], [m], (v_i)_{i=1}^n, -)$ be a fair division instance over goods. If an allocation A is MMS-fair to agent i , then it is also MXS₀-fair to i . (EFX₀ is defined in Definition 13. MXS₀ is the minimum-EFX₀-share. EFX₀ implies EF1, so MXS₀ implies M1S.)*

Proof. Without loss of generality, let $i = 1$. Let $\widehat{v}(X) := (v_1(X), |X|)$. Then $\widehat{v}$'s co-domain is totally-ordered, assuming pairs are compared lexicographically. Let P be a leximin n -partition of $\widehat{v}$ (c.f. Definition 20) such that $\widehat{v}(P_1) \leq \dots \leq \widehat{v}(P_n)$. (P is also called the leximin++ partition of v_1 [PR20]). Then $v_1(P_1) = \min_{j=1}^n v_1(P_j)$. Hence, $v_1(P_1) = \text{MMS}_1 \leq v_1(A_1)$.

For any $j \geq 2$, we get that (P_1, P_j) is also a leximin 2-partition of $\widehat{v}$ (when $\widehat{v}$'s domain is restricted to $P_1 \cup P_j$). We will now show that agent 1 doesn't EFX₀-envy agent j , i.e., $v_1(P_1) \geq \max_{g \in P_j} v_1(P_j \setminus \{g\})$. (If $P_j = \emptyset$, this is trivially true.) Suppose $v_1(P_1) < v_1(P_j \setminus \{g\})$ for some $g \in P_j$. Let $Q_1 := P_1 \cup \{g\}$ and $Q_j := P_j \setminus \{g\}$. Then $\widehat{v}(Q_1) > \widehat{v}(P_1)$ and $\widehat{v}(Q_j) > \widehat{v}(P_1)$. Hence, (P_1, P_j) is not a leximin 2-partition of $\widehat{v}$, which is a contradiction. Hence, agent 1 doesn't EFX₀-envy agent j .

Thus, P is EFX₀-fair to agent 1 in $\mathcal{I}$, so P is agent 1's MXS₀-certificate for A . $\square$

D.7 Among PROP, APS, MMS

Lemma 44 (PROP $\implies$ APS, Proposition 4 of [BEF23]). *For any fair division instance $([n], [m], (v_i)_{i=1}^n, w)$, $\text{APS}_i \leq w_i v_i([m])$ for agent i if v_i is additive.*

Proof. Set the price $p(g)$ of each item g to $v_i(g)$. Then

$$\text{APS}_i \leq \max_{\substack{S \subseteq [m]: \\ p(S) \leq w_i p([m])}} v_i(S) = \max_{\substack{S \subseteq [m]: \\ v_i(S) \leq w_i v_i([m])}} v_i(S) \leq w_i v_i([m]). \quad \square$$

Lemma 45 (PROP $\implies$ WMMS). *In any instance $([n], [m], (v_i)_{i=1}^n, w)$, if v_i is superadditive, then $\text{WMMS}_i \leq w_i v_i([m])$.*

Proof. Let P be agent i 's WMMS partition. Then

$$v_i([m]) \geq \sum_{j=1}^n v_i(P_j) = \sum_{j=1}^n w_j \left(\frac{v_i(P_j)}{w_j} \right) \geq \sum_{j=1}^n w_j \frac{\text{WMMS}_i}{w_i} = \frac{\text{WMMS}_i}{w_i}. \quad \square$$

Lemma 46. *Let $a_1 \leq a_2 \leq \dots \leq a_n$ be n real numbers. Let $s_k := \sum_{i=1}^k a_i$ for any $0 \leq k \leq n$. Then $s_k \leq (k/n)s_n$.*

Proof.

$$s_n = s_k + \sum_{i=k+1}^n a_i \geq s_k + (n-k)a_k \geq s_k + (n-k)\frac{s_k}{k} = \frac{n}{k}s_k. \quad \square$$

Lemma 47 (PROP $\implies$ pessShare). *For any fair division instance $([n], [m], (v_i)_{i=1}^n, w)$, if v_i is superadditive, then $\text{pessShare}_i \leq w_i v_i([m])$.*

Proof. Let P be agent i 's ℓ -out-of- d -partition. Assume without loss of generality that $v_i(P_1) \leq v_i(P_2) \leq \dots \leq v_i(P_d)$. Then by Lemma 46 and superadditivity of v_i , we get

$$\ell\text{-out-of-}d\text{-share}_i = \sum_{j=1}^{\ell} v_i(P_j) \leq \frac{\ell}{d} \sum_{j=1}^d v_i(P_j) \leq \frac{\ell}{d} v_i([m]).$$

Hence,

$$\text{pessShare}_i := \sup_{1 \leq \ell \leq d: \ell/d \leq w_i} \ell\text{-out-of-}d\text{-share}_i \leq \sup_{1 \leq \ell \leq d: \ell/d \leq w_i} (\ell/d) v_i([m]) \leq w_i v_i([m]). \quad \square$$

Lemma 48 (APS $\implies$ pessShare, [BEF23]). *In any fair division instance, $\text{APS}_i \geq \text{pessShare}_i$ for each agent i .*

Proof. Proposition 2 in [BEF23] proves this for goods, but their proof works for mixed manna too. $\square$

Lemma 49. *For a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ and any agent i , define*

$$\beta_i := \max_{j=1}^n \max_{S \subseteq [m]: v_i(S) \leq w_j v_i([m])} \frac{v_i(S)}{w_j}.$$

If v_i is superadditive, then $\text{WMMS}_i \leq w_i \beta_i$. If v_i is additive and $n = 2$, then $\text{WMMS}_i = w_i \beta_i$.

Proof. $\beta_i = \max_{j=1}^n v_i(S_j)/w_j$, where

$$S_j := \operatorname{argmax}_{S \subseteq [m]: v_i(S) \leq w_j v_i([m])} \frac{v_i(S)}{w_j}.$$

Let $\Pi_n([m])$ be the set of all n -partitions of $[m]$. Let

$$P := \operatorname{argmax}_{P \in \Pi_n([m])} \min_{j=1}^n \frac{v_i(P_j)}{w_j}, \quad t := \operatorname{argmin}_{j=1}^n \frac{v_i(P_j)}{w_j}.$$

Then $\text{WMMS}_i/w_i := v_i(P_t)/w_t$.

By Lemma 45, we get $v_i(P_t)/w_t \leq v_i([m])$. Hence,

$$\frac{\text{WMMS}_i}{w_i} = \frac{v_i(P_t)}{w_t} \leq \frac{v_i(S_t)}{w_t} \leq \max_{j=1}^n \frac{v_i(S_j)}{w_j} = \beta_i.$$

Now let v_i be additive and $n = 2$. For any $j \in [2]$, let $Q^{(j)}$ be an allocation where $Q_j^{(j)} := S_j$ and $Q_k^{(j)} := [m] \setminus S_j$. Then $v_i(Q_j^{(j)}) = v_i(S_j) \leq w_j v_i([m])$ and $v_i(Q_k^{(j)}) = v_i([m]) - v_i(S_j) \geq v_i([m]) - w_j v_i([m]) = w_k v_i([m])$. Hence,

$$\frac{v_i(Q_j^{(j)})}{w_j} \leq v_i([m]) \leq \frac{v_i(Q_k^{(j)})}{w_k}.$$

Therefore,

$$\begin{aligned}
\frac{\text{WMMS}_i}{w_i} &= \min \left(\frac{v_i(P_1)}{w_1}, \frac{v_i(P_2)}{w_2} \right) \\
&\geq \max_{j=1}^n \min \left(\frac{v_i(Q_j^{(j)})}{w_j}, \frac{v_i(Q_k^{(j)})}{w_k} \right) && \text{(by definition of } P) \\
&= \max_{j=1}^n \frac{v_i(S_j)}{w_j} = \beta_i.
\end{aligned}$$

Hence, $\text{WMMS}_i/w_i = \beta_i$. $\square$

Lemma 50 (WMMS $\implies$ APS for $n = 2$, [BEF23]). *For a fair division instance $([2], [m], (v_i)_{i=1}^2, w)$, if v_i is additive for some i , then $\text{APS}_i \leq \text{WMMS}_i$. Moreover, when entitlements are equal, we get $\text{APS}_i = \text{MMS}_i$.*

Proof. By Lemma 49, we get

$$\frac{\text{WMMS}_i}{w_i} = \beta_i := \max_{j=1}^2 \frac{v_i(S_j)}{w_j}, \quad \text{where } S_j := \underset{\substack{S \subseteq [m]: \\ v_i(S) \leq w_j v_i([m])}}{\operatorname{argmax}} v_i(S).$$

Setting $p = v_i$ in the definition of APS gives us

$$\text{APS}_i \leq \max_{\substack{S \subseteq [m]: \\ p(S) \leq w_i p([m])}} v_i(S) = v_i(S_i) \leq w_i \beta_i = \text{WMMS}_i.$$

For equal entitlements, $\text{APS}_i \geq \text{MMS}_i$ by Lemma 48. $\square$

D.8 Binary and Negative-Binary Valuations

In this section, we study implications between fairness notions when agents' marginals are *triboolean*, i.e., for all $R \subseteq [m]$ and $t \in [m] \setminus R$, we have $v_i(t \mid R) \in \{-1, 0, 1\}$.

We begin by proving $\text{EF1} \implies \text{EFX}$ (Lemma 51) and $\text{PROP1} \implies \text{PROPx}$ (Lemma 52) for additive valuations.

Lemma 51 ($\text{EF1} \implies \text{EFX}$). *Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ where v_i is additive for some agent i . If an allocation A is EF1-fair to i , then it is also EFX-fair to i if at least one of these conditions is satisfied:*

1. *agents have equal entitlements and v_i 's marginals are triboolean (i.e., $v_i(t \mid \cdot) \in \{-1, 0, 1\} \forall t \in [m]$).*
2. *v_i 's marginals are binary (i.e., $v_i(t \mid \cdot) \in \{0, 1\} \forall t \in [m]$).*
3. *v_i 's marginals are negative binary (i.e., $v_i(t \mid \cdot) \in \{0, -1\} \forall t \in [m]$).*

Proof. Suppose agents have equal entitlements and v_i 's marginals are triboolean. Since i is EF1-satisfied, for any other agent j , either $v_i(A_i) \geq v_i(A_j)$, or $v_i(A_i) \geq v_i(A_j \setminus \{g\})$ for some $g \in A_j$, or $v_i(A_i \setminus \{c\}) \geq v_i(A_j)$ for some $c \in A_i$. Hence, we get $v_i(A_i) \geq v_i(A_j) - 1$. For any $S \subseteq A_j$ such that $v_i(S \mid A_i) > 0$, we have $v_i(A_j \setminus S) = v_i(A_j) - v_i(S) \leq v_i(A_j) - 1 \leq v_i(A_i)$. For any $S \subseteq A_i$ such that $v_i(S \mid A_i \setminus S) < 0$, we have $v_i(A_i \setminus S) = v_i(A_i) - v_i(S) \geq v_i(A_i) + 1 \geq v_i(A_j)$. Hence, i doesn't EFX-envy j . Since this holds for every j , A is EFX-fair to agent i .

Suppose v_i 's marginals are binary. Since i is EF1-satisfied, for any other agent j , either $v_i(A_i)/w_i \geq v_i(A_j)/w_j$, or for some $g \in A_j$, we have $v_i(A_i)/w_i \geq v_i(A_j \setminus \{g\})/w_j$, which implies $v_i(A_i)/w_i \geq (v_i(A_j) - 1)/w_j$. For any $S \subseteq A_j$ such that $v_i(S | A_i) > 0$, we have

$$\frac{v_i(A_j \setminus S)}{w_j} = \frac{v_i(A_j) - v_i(S)}{w_j} \leq \frac{v_i(A_j) - 1}{w_j} \leq \frac{v_i(A_i)}{w_i}.$$

Hence, i doesn't EFX-envy agent j . Since this holds for every j , A is EFX-fair to agent i .

Suppose v_i 's marginals are negative binary. Since i is EF1-satisfied, for any other agent j , either $v_i(A_i)/w_i \geq v_i(A_j)/w_j$, or for some $c \in A_i$, we have $v_i(A_i \setminus \{c\})/w_i \geq v_i(A_j)/w_j$, which implies $(v_i(A_i) + 1)/w_i \geq v_i(A_j)/w_j$. For any $S \subseteq A_i$ such that $v_i(S | A_i \setminus S) < 0$, we have

$$\frac{v_i(A_i \setminus S)}{w_i} = \frac{v_i(A_i) - v_i(S)}{w_i} \geq \frac{v_i(A_i) + 1}{w_i} \geq \frac{v_i(A_j)}{w_j}.$$

Hence, i doesn't EFX-envy agent j . Since this holds for every j , A is EFX-fair to agent i . $\square$

Lemma 52 (PROP1 $\implies$ PROPx). *Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ where marginals are triboolean for some agent i (i.e., $v_i(t | \cdot) \in \{-1, 0, 1\} \forall t \in [m]$). Then if an allocation A is PROP1-fair to agent i , then it is also PROPx-fair to i .*

Proof. If $v_i(A_i) \geq w_i v_i([m])$, then A is PROPx. Otherwise, we get $v_i(A_i) + 1 > w_i v_i([m])$ since A is PROP1 and marginals are triboolean. Removing any positive-disutility subset from A_i or adding any positive-utility subset to A_i will increase its value by at least 1. Hence, A is PROPx. $\square$

We now characterize different fairness notions. If agent i having entitlement w_i has an additive valuation function v_i with triboolean marginals, then for any allocation A ,

1. for equal entitlements, A is PROP-fair to $i \iff A$ is EEF-fair to $i \iff v_i(A_i) \geq \lceil v_i([m])/n \rceil$ (Lemma 54).
2. A is PROP1-fair to $i \iff A$ is PROPx-fair to $i \iff A$ is APS-fair to $i \iff v_i(A_i) \geq \lfloor w_i \cdot v_i([m]) \rfloor$ (Lemmas 55 and 56).
3. for equal entitlements, A is MMS-fair to $i \iff A$ is M1S-fair to $i \iff A$ is EEFX-fair to $i \iff v_i(A_i) \geq \lfloor v_i([m])/n \rfloor$ (Lemmas 57 and 59).

Lemma 53. *For an additive function $f : 2^{[m]} \rightarrow \{-1, 0, 1\}$, there exists an n -partition P such that $|f(P_i) - f(P_j)| \leq 1$ for all $i, j \in [n]$ and $\lfloor f([m])/n \rfloor \leq f(P_i) \leq \lceil f([m])/n \rceil$ for all $i \in [n]$.*

Proof. Partition $[m]$ into goods $M_+ := \{g \in [m] : v_i(g) > 0\}$, chores $M_- := \{c \in [m] : v_i(c) < 0\}$, and neutral items $M_0 := \{t \in [m] : v_i(t) = 0\}$. Fuse items M_0 , $\min(|M_+|, |M_-|)$ goods, and $\min(|M_+|, |M_-|)$ chores into a single item h . Then we are left with only goods and a neutral item, or only chores and a neutral item. Using round-robin, one can allocate items such that any two bundles differ by at most one item. $\square$

Lemma 54 (PROP $\implies$ EEF). *Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, -)$ where v_i is additive and $v_i(t) \in \{-1, 0, 1\}$ for all $t \in [m]$ for some agent i . If an allocation A is PROP-fair to i , then it is also EEF-fair to i .*

Proof. Since A is PROP-fair to i , and $v_i(S) \in \mathbb{Z}$ for all $S \subseteq [m]$, we get $v_i(A_i) \geq \lceil v_i([m])/n \rceil$.

Construct allocation B where $B_i = A_i$, and items $[m] \setminus A_i$ are allocated among agents $[n] \setminus \{i\}$ using Lemma 53 with $f = v_i$. We show that B is agent i 's EEF-certificate for A .

In B , for each agent $j \in [n] \setminus \{i\}$, we have

$$v_i(B_j) \leq \left\lceil \frac{v_i([m] \setminus A_i)}{n-1} \right\rceil \leq \left\lceil \frac{v_i([m]) - v_i([m])/n}{n-1} \right\rceil \leq \left\lceil \frac{v_i([m])}{n} \right\rceil \leq v_i(A_i).$$

Hence, i doesn't envy anyone in B . Hence, B is agent i 's EEF-certificate for A . $\square$

Lemma 55. Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ where v_i is additive and $v_i(t) \in \{-1, 0, 1\}$ for all $t \in [m]$ for some agent i . Then the following statements are equivalent:

1. Allocation A is PROP1-fair to i .
2. Allocation A is PROP x -fair to i .
3. $v_i(A_i) \geq \lfloor w_i v_i([m]) \rfloor$.

Proof. Partition $[m]$ into goods $M_+ := \{g \in [m] : v_i(g) > 0\}$, chores $M_- := \{c \in [m] : v_i(c) < 0\}$, and neutral items $M_0 := \{t \in [m] : v_i(t) = 0\}$.

Case 1: A_i has all goods and no chores.

Then $v_i(A_i) \geq \max(0, v_i([m]))$. If $v_i([m]) \geq 0$, then $v_i(A_i) \geq v_i([m]) \geq w_i v_i([m])$, else $v_i(A_i) \geq 0 \geq w_i v_i([m])$. Hence, $v_i(A_i) \geq w_i v_i([m])$ and A is PROP x +PROP1.

Case 2: A_i has a chore or some good is outside A_i .

Then adding a good to A_i or removing a chore from A_i makes its value more than $w_i v_i([m])$ iff $v_i(A_i) \geq \lfloor w_i v_i([m]) \rfloor$. So, A is PROP1 iff A is PROP x iff $v_i(A_i) \geq \lfloor w_i v_i([m]) \rfloor$. $\square$

Lemma 56. Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ where v_i is additive and $v_i(t) \in \{-1, 0, 1\}$ for all $t \in [m]$ for some agent i . Then $\text{APS}_i = \lfloor w_i v_i([m]) \rfloor$.

Proof. Partition $[m]$ into goods $M_+ := \{g \in [m] : v_i(g) > 0\}$, chores $M_- := \{c \in [m] : v_i(c) < 0\}$, and neutral items $M_0 := \{t \in [m] : v_i(t) = 0\}$.

First, set $p(t) = v_i(t)$ for each $t \in [m]$ to get $\text{APS}_i \leq w_i v_i([m])$. Since the APS is the value of some bundle, and bundle values are integers, we get $\text{APS}_i \leq \lfloor w_i v_i([m]) \rfloor$.

Pick an arbitrary price vector $p \in \Delta_m$. We will construct a set S such that $p(S) \leq w_i p([m])$ and $v_i(S) \geq \lfloor w_i v_i([m]) \rfloor$. Fuse items M_0 , $\min(|M_+|, |M_-|)$ goods, and $\min(|M_+|, |M_-|)$ chores into a single item h . Let M'_+ and M'_- be the remaining goods and chores, respectively. Then $M'_+ = \emptyset$ or $M'_- = \emptyset$. Using techniques from Lemma 11, we can assume without loss of generality that $p_g \geq 0$ for all $g \in M'_+$, $p_c \leq 0$ for all $c \in M'_-$, and $p_h = 0$.

Case 1: $M'_- = \emptyset$.

Let $m_i := |M'_+| = v_i([m])$. Let S be the cheapest $\lfloor w_i m_i \rfloor$ items in M'_+ . Then using Lemma 46, we get

$$p(S) \leq \frac{\lfloor w_i m_i \rfloor}{m_i} p(M_+) \leq w_i p([m]).$$

Then S is affordable, so $\text{APS}_i \geq \lfloor w_i m_i \rfloor$.

Case 2: $M'_+ = \emptyset$.

Let $m_i := |M'_-| = -v_i([m])$. Let S be the cheapest $\lceil w_i m_i \rceil$ items in $M'_- \cup \{h\}$. Then using Lemma 46, we get

$$-p(S) \geq \frac{\lceil w_i m_i \rceil}{m_i} (-p([m])) \geq w_i (-p([m])).$$

Then S is affordable and $-v_i(S) \leq \lceil w_i m_i \rceil = -\lfloor w_i v_i([m]) \rfloor$. Hence, $\text{APS}_i \geq \lfloor w_i v_i([m]) \rfloor$. $\square$

Lemma 57. Consider an instance $([n], [m], (v_i)_{i=1}^n, -)$ where v_i is additive and $v_i(t) \in \{-1, 0, 1\}$ for all $t \in [m]$ for some agent i . Then $\text{MMS}_i = \text{M1S}_i = \lfloor v_i([m])/n \rfloor$.

Proof. Allocate items $[m]$ among agents $[n]$ using Lemma 53 with $f = v_i$. Then any two bundles differ by a value of at most one. Hence, $\text{MMS}_i = \lfloor m'/n \rfloor$ and $\text{M1S}_i \leq \lfloor m'/n \rfloor$, where $m' := v_i([m])$.

Let X be an allocation where agent i is EF1-satisfied. Then any two bundles can differ by a value of at most one. Hence, the smallest value $v_i(X_i)$ can have is $\lfloor m'/n \rfloor$. Hence, $\text{M1S}_i \geq \lfloor m'/n \rfloor$. $\square$

Remark 58. For any $m \in \mathbb{Z}$ and $n \in \mathbb{Z}_{\geq 1}$, we have

$$\left\lfloor \frac{m}{n} \right\rfloor = \left\lceil \frac{m+1}{n} \right\rceil - 1.$$

Lemma 59. Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, -)$ where v_i is additive and $v_i(t) \in \{-1, 0, 1\}$ for all $t \in [m]$ for some agent i . Then an allocation A is EEFX-fair to i iff $v_i(A_i) \geq \lfloor v_i([m])/n \rfloor$.

Proof. Suppose $v_i(A_i) \geq \lfloor v_i([m])/n \rfloor$. Construct an allocation B where $B_i = A_i$, and items $[m] \setminus A_i$ are allocated among agents $[n] \setminus \{i\}$ using Lemma 53 with $f = v_i$. We will show that B is agent i 's EEFX-certificate for A .

Let $k := v_i([m])$. Suppose $v_i(A_i) \geq \lfloor k/n \rfloor = \lceil (k+1)/n \rceil - 1$ (c.f. Remark 58). Then for any other agent $j \in [n] \setminus \{i\}$, we get

$$v_i(B_j) \leq \left\lceil \frac{v_i([m] \setminus A_i)}{n-1} \right\rceil \leq \left\lceil \frac{k - (k+1)/n + 1}{n-1} \right\rceil = \left\lceil \frac{k+1}{n} \right\rceil \leq v_i(B_i) + 1.$$

If B_j contains no goods and B_i contains no chores, then $v_i(B_i) \geq 0 \geq v_i(B_j)$, so i doesn't envy j in B . Otherwise, removing a good from B_j or removing a chore from B_i eliminates i 's envy towards j . Hence, B is agent i 's EEFX-certificate for A , so A is EEFX-fair to i .

Now suppose $v_i(A_i) < \lfloor v_i([m])/n \rfloor$. Let B be any allocation where $B_i = A_i$. Then $\exists j \in [n] \setminus \{i\}$ such that $v_i(B_j) > v_i([m])/n$. If $v_i([m])/n$ is an integer, then $v_i(B_j) > v_i([m])/n > v_i(B_i)$, so $v_i(B_j) \geq v_i(B_i) + 2$. Otherwise, $v_i(B_j) \geq \lceil v_i([m])/n \rceil > \lfloor v_i([m])/n \rfloor > v_i(B_i)$, so $v_i(B_j) \geq v_i(B_i) + 2$. Thus, B_j has a good or B_i has a chore. Even after removing a good from B_j or removing a chore from B_i , agent i will still envy agent j , so B is not EFX-fair to agent i . Thus, A is not EEFX-fair to i . $\square$

We now prove more implications for additive valuations.

Lemma 60 (EF1 $\implies$ GAPS). Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ where v_i is additive and $v_i(t) \in \{-1, 0, 1\}$ for all $t \in [m]$ for some agent i . If an allocation A is EF1-fair to i , then it is also groupwise-APS-fair to i if at least one of these conditions hold:

1. $n = 2$
2. $w_i \leq w_j$ for all $j \in [n] \setminus \{i\}$.
3. $v_i(c) \in \{0, -1\}$ for all $c \in [m]$.

Proof. On restricting A to any subset S of agents, we get an allocation B that is EF1-fair to i . B is also PROP1-fair to i by Lemmas 33, 34, and 35, and APS-fair to i by Lemmas 55 and 56. Hence, A is groupwise-APS-fair to i . $\square$

Lemma 61 (EF1 $\implies$ GWMS, M1S $\implies$ WMMS). Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ where for some agent i , v_i is additive and $v_i(g) \in \{0, 1\}$ for all $g \in [m]$. If an allocation A is EF1-fair to i , then it is also GWMS-fair to i . If an allocation B is M1S-fair to i , then it is also WMMS-fair to i .

Proof. Suppose A is not WMMS-fair to i . By the definition of WMMS, we get

$$\frac{\text{WMMS}_i}{w_i} = \min_{k=1}^n \frac{v_i(X_k^*)}{w_k}, \quad \text{where } X^* \in \operatorname{argmax}_X \min_{k=1}^n \frac{v_i(X_k)}{w_k}.$$

Then $v_i(X_i^*) \geq \text{WMMS}_i > v_i(A_i)$. Now pick any $j \in [n] \setminus \{i\}$. Since A is EF1-fair to i , we get

$$\frac{v_i(X_j^*)}{w_j} \geq \frac{\text{WMMS}_i}{w_i} > \frac{v_i(A_i)}{w_i} \geq \frac{v_i(A_j) - 1}{w_j}.$$

Bundle values are integers, so $v_i(X_j^*) \geq v_i(A_j)$. Thus,

$$v_i([m]) = \sum_{k=1}^n v_i(X_k^*) > \sum_{k=1}^n v_i(A_k) = v_i([m]),$$

which is a contradiction. Thus, A is WMMS-fair to i .

Restricting an EF1 allocation to a subset of agents gives us another EF1 allocation, so A is also groupwise WMMS.

Since allocation B is M1S-fair to i , there is an allocation C that is EF1-fair to i and has $v_i(C_i) \leq v_i(B_i)$. Thus, C is also WMMS-fair to i , so B is WMMS-fair to i . $\square$

Lemma 62 (monotonicity of envy-freeness). *Let $Q \subseteq \{-1, 0, 1\}$. Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ where for some agent i , v_i is additive and $v_i(t) \in Q$ for all $t \in [m]$. Let $F \in \{\text{EF}, \text{EF1}, \text{EFX}\}$. Let allocation A be F -fair to agent i . Let B be an allocation such that $B_i = A_i$ and $v_i(B_j) \leq v_i(A_j)$ for some agent j . Then i does not F -envy j in B if $F = \text{EF}$, $Q = \{0, 1\}$, or $Q = \{-1, 0\}$.*

Proof. This holds trivially if $F = \text{EF}$, or if i does not envy j in A or B . So now assume F is EF1 or EFX and i envies j in A and B .

First, let us consider $F = \text{EFX}$. The chores condition of EFX continues to be satisfied in B . We need to verify the goods condition. The goods condition holds trivially if B_j has no goods, so we are done for EFX if $Q = \{-1, 0\}$. Now assume B_j has at least one good. We need to prove that $v_i(B_i)/w_i \geq (v_i(B_j) - 1)/w_j$. If $Q = \{0, 1\}$, then A_j always contains a good since $v_i(A_j)/w_j > v_i(A_i)/w_i \geq 0$. Since i does not F -envy j in A , we get

$$\frac{v_i(B_i)}{w_i} = \frac{v_i(A_i)}{w_i} \geq \frac{v_i(A_j) - 1}{w_j} \geq \frac{v_i(B_j) - 1}{w_j}.$$

Hence, i does not EFX-envy j if $Q = \{-1, 0\}$ or $Q = \{0, 1\}$.

Now let us consider $F = \text{EF1}$. We have two cases, depending on the value of Q .

1. If $Q = \{-1, 0\}$, then A_i contains a chore since i envies j in A . Since i does not EF1-envy j in A , we get $(v_i(A_i) + 1)/w_i \geq v_i(A_j)/w_j \geq v_i(B_j)/w_j$.
2. If $Q = \{0, 1\}$, then A_j contains a good and B_j contains a good since i envies j in A and B . Since i does not EF1-envy j in A , we get $v_i(A_i)/w_i \geq (v_i(A_j) - 1)/w_j \geq (v_i(B_j) - 1)/w_j$.

In both cases, i does not EF1-envy j in B . $\square$

Lemma 63. *Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ where for some agent i , v_i is additive and $v_i(t) \in \{-1, 0, 1\}$ for all $t \in [m]$. Let $F \in \{\text{EF}, \text{EF1}, \text{EFX}\}$. Let allocation A be F -fair to agent i .*

Let $j, k \in [n] \setminus \{i\}$ be agents such that $v_i(A_j) > 0$ and $v_i(A_k) < 0$. Obtain allocation B by moving a good from A_j to A_k , i.e., for some $g \in A_j$ such that $v_i(g) > 0$, let $B_j := A_j \setminus \{g\}$, $B_k := A_k \cup \{g\}$, and $B_\ell := A_\ell$ for all $\ell \in [n] \setminus \{j, k\}$. Then B is also F -fair to i .

Proof. Since the bundles of agents $[n] \setminus \{j, k\}$ didn't change, agent i doesn't F -envy them. Since A is F -fair to i , and $v_i(A_j) > 0$, we get $v_i(A_i) \geq 0$. Then i doesn't envy k because $v_i(B_k)/w_k \leq 0 \leq v_i(A_i)/w_i = v_i(B_i)/w_i$. Since $v_i(B_j) = v_i(A_j) - 1$, one can verify that i doesn't F -envy agent j . Thus, B is F -fair to agent i . $\square$

Lemma 64 (min- F -share $\implies$ epistemic- F). *Let $Q \subseteq \{-1, 0, 1\}$. Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, w)$ where for some agent i , v_i is additive and $v_i(t) \in Q$ for all $t \in [m]$. Let $F \in \{EF, EF1, EFX\}$. If allocation A is min- F -share-fair to agent i , then it is also epistemic- F -fair to i if $F = EF$, $Q = \{-1, 0\}$, or $Q = \{0, 1\}$.*

Proof. Since A is min- F -share-fair to i , there exists an allocation B such that $v_i(B_i) \leq v_i(A_i)$ and B is F -fair to i . By repeatedly applying the transformation of Lemma 63, we can assume without loss of generality that either $v_i(B_j) \geq 0$ for all j or $v_i(B_j) \leq 0$ for all j .

Since $v_i([m] \setminus A_i) \leq v_i([m] \setminus B_i)$, we can redistribute $[m] \setminus A_i$ to get an allocation C such that $C_i = A_i$ and $v_i(C_i) \leq v_i(B_i)$. This is because we can fuse items in $[m] \setminus A_i$ such that we either get rid of all goods or all chores. By Lemma 62, C is F -fair to agent i . Thus, C is agent i 's epistemic- F -certificate for A . $\square$

We now look at non-additive valuations.

Lemma 65 (MMS = APS, [KKM24]). *For a fair division instance $([n], [m], (v_i)_{i=1}^n, -)$, if v_i is submodular for some agent i , and v_i has binary marginals (i.e., $v_i(g \mid X) \in \{0, 1\}$ for some $X \subseteq [m]$ and $g \in [m] \setminus X$), then $\text{APS}_i = \text{MMS}_i$.*

Lemma 66. *Let $f : 2^{[m]} \rightarrow \mathbb{R}$ such that f has tribool marginals, i.e., for all $X \subseteq [m]$ and $j \in [m] \setminus X$, we have $f(j \mid X) \in \{-1, 0, 1\}$. For any disjoint sets $S, T \subseteq [m]$ such that $f(T \mid S) > 0$, there exists $T' \subseteq T$ such that $f(T' \mid S) = 1$.*

Proof. Let $T = \{g_1, \dots, g_k\}$. For any $0 \leq j \leq k$, define $T_j := \{g_1, \dots, g_j\}$ and $\alpha_j := f(T_j \mid S)$. Then $\alpha_0 = f(\emptyset \mid S) = 0$, and $\alpha_k = f(T \mid S) > 0$. Let t be the smallest value in $[k]$ such that $\alpha_t > 0$. Then $\alpha_t = f(T_t \mid S) = \alpha_{t-1} + f(g_t \mid S \cup T_{t-1})$. Since $\alpha_{t-1} \leq 0$, we get $\alpha_t \leq f(g_t \mid S \cup T_{t-1}) \leq 1$. Since marginals are in $\{-1, 0, 1\}$, we get $\alpha_t = 1$, so $f(T_t \mid S) = 1$. $\square$

Lemma 67 (PROPm vs PROPavg vs PROPx for binary subadd). *Let $([n], [m], (v_i)_{i=1}^n, w)$ be a fair division instance where v_i has tribool marginals for some agent i , i.e., for all $X \subseteq [m]$ and $j \in [m] \setminus X$, we have $v_i(j \mid X) \in \{-1, 0, 1\}$. Suppose agent i is PROPm-satisfied by allocation A . Then A is also PROPavg-fair to i . Moreover, if v_i is subadditive and $w_i = 1/n$, then A is also PROPx-fair to i .*

Proof. If A is PROP-fair to i , then we are done, so assume otherwise. For $j \neq i$, recall the definition of τ_j from Definition 19. We have two cases:

Case 1: $\tau_j = 0$ for all $j \neq i$.

Then A is PROPavg by definition. Let v_i be subadditive. Then for all $j \in [n] \setminus \{i\}$, we have

$$\tau_j = 0 \iff v_i(A_j \mid A_i) = 0 \implies v_i(A_j) \leq v_i(A_i \cup A_j) = v_i(A_i).$$

Hence,

$$v_i([m]) \leq \sum_{j=1}^n v_i(A_i) \leq n v_i(A_i).$$

Therefore, if $w_i = 1/n$, then A is PROP-fair to i .

Case 2: $\tau_j > 0$ for some $j \neq i$.

By Lemma 66, we get $\tau_j \in \{0, 1\}$ for all j . Since A is PROPm-fair to i , we get $v_i(A_i) + 1 > w_i v_i([m])$. The average of $\{\tau_j : j \neq i, \tau_j > 0\}$ is 1, so A is also PROPavg-fair to i . If $v_i([m] \setminus A_i \mid A_i) = 0$, then $v_i(A_i) = v_i([m]) > w_i v_i([m])$, so A is PROP-fair to i . Otherwise, for every $S \subseteq [m] \setminus A_i$ such that $v_i(S \mid A_i) > 0$, we get $v_i(A_i \cup S) \geq v_i(A_i) + 1 > w_i v_i([m])$, so A is PROPx-fair to i . $\square$

Lemma 68 (M1S $\implies$ PROPx for binary subadditive). *Let $([n], [m], (v_i)_{i=1}^n, w)$ be a fair division instance where v_i is subadditive and has $\{-1, 0, 1\}$ marginals, i.e., $v_i(g \mid X) \in \{-1, 0, 1\}$ for all $X \subseteq [m]$ and $g \in [m] \setminus X$. Let allocation A be M1S-fair to agent i . Then A is also PROPx-fair to agent i if at least one of the following hold:*

1. $n = 2$.
2. $w_i < 1/(n - 1)$ and v_i has $\{0, 1\}$ marginals.
3. v_i has $\{-1, 0\}$ marginals.

Proof. Let B be agent i 's M1S-certificate for A . Let $k := v_i(B_i)$. Then $v_i(A_i) \geq k$ and B is EF1-fair to i . If $v_i(A_i) \geq w_i v_i([m])$, then A is PROP-fair to i , and we are done. Now let $k \leq v_i(A_i) < w_i v_i([m])$. Define sets J_0 , J_G , and J_C :

$$\begin{aligned} J_0 &:= \left\{ j \in [n] \setminus \{i\} : \frac{v_i(B_i)}{w_i} \geq \frac{v_i(B_j)}{w_j} \right\}, \\ J_G &:= \left\{ j \in [n] \setminus \{i\} : \frac{v_i(B_j \setminus \{g\})}{w_j} \leq \frac{v_i(B_i)}{w_i} < \frac{v_i(B_j)}{w_j} \text{ for some } g \in B_j \right\}, \\ J_C &:= \left\{ j \in [n] \setminus \{i\} : \frac{v_i(B_i)}{w_i} < \frac{v_i(B_j)}{w_j} \leq \frac{v_i(B_i \setminus \{c\})}{w_i} \text{ for some } c \in B_i \right\}. \end{aligned}$$

Since B is EF1-fair to i , we have $J_0 \cup J_G \cup J_C = [n] \setminus \{i\}$. Also, J_0 and $J_G \cup J_C$ are disjoint.

If all items are goods, then $J_C = \emptyset$. If all items are chores, then $J_G = \emptyset$. If $n = 2$, then $J_G = \emptyset$ or $J_C = \emptyset$. Hence, if at least one of the three conditions of the lemma hold, then either $J_G = \emptyset$, or $J_C = \emptyset$ and $w_i < 1/(n - 1)$. We first show that, under these conditions, $w_i v_i([m]) < k + 1$.

Case 1: $J_C = \emptyset$ and $w_i < 1/(n - 1)$.

Pick $j \in [n] \setminus \{i\}$. If $j \in J_0$, then $v_i(B_j) \leq k w_j / w_i$. Otherwise, $\exists g \in B_j$ such that $v_i(B_j \setminus \{g\}) \leq k w_j / w_i$. So, $v_i(B_j) = v_i(B_j \setminus \{g\}) + v_i(g \mid B_j \setminus \{g\}) \leq k(w_j / w_i) + 1$. Hence,

$$v_i([m]) \leq \sum_{j=1}^n v_i(B_j) \leq (n - 1) + k / w_i.$$

So, $w_i v_i([m]) \leq k + w_i(n - 1) < k + 1$ (since $w_i < 1/(n - 1)$).

Case 2: $J_G = \emptyset$

Pick $j \in [n] \setminus \{i\}$. If $j \in J_0$, then $v_i(B_j) \leq k w_j / w_i$. Otherwise, $\exists c \in B_i$ such that $v_i(B_i \setminus \{c\}) / w_i \geq v_i(B_j) / w_j$. $v_i(B_i \setminus \{c\}) = v_i(B_i) - v_i(c \mid B_i \setminus \{c\}) \leq k + 1$. So, $v_i(B_j) \leq (k + 1) w_j / w_i$. Hence,

$$v_i([m]) \leq \sum_{j=1}^n v_i(B_j) \leq \frac{k + 1}{w_i} - 1.$$

Hence, $w_i v_i([m]) \leq k + 1 - w_i < k + 1$.

Now we will show that A is PROPx-fair to i . Note that $k \leq v_i(A_i) < w_i v_i([m]) < k + 1$, so $v_i(A_i) = k$. If $S \subseteq [m] \setminus A_i$ such that $v_i(S \mid A_i) > 0$, then $v_i(A_i \cup S) \geq k + 1 > w_i v_i([m])$. If $S \subseteq A_i$ such that $v_i(S \mid A_i \setminus S) < 0$, then $v_i(A_i \setminus S) = v_i(A_i) - v_i(S \mid A_i \setminus S) \geq k + 1 > w_i v_i([m])$. Hence, A is PROPx-fair to agent i . $\square$

D.9 Unit-Demand Valuations

A function $v : 2^{[m]} \rightarrow \mathbb{R}_{\geq 0}$ is called *unit-demand* if $v(\emptyset) = 0$, and for any non-empty set $X \subseteq M$, we have $v(X) = \max_{g \in X} v(\{g\})$. Such functions capture scenarios where agents want just one good, e.g., when the goods are houses. Unit-demand functions are known to be submodular and cancelable.

Table 5: Implications among fairness notions when valuations are unit-demand, marginals are non-negative, and entitlements are equal.

implication	proof
MEFS $\implies$ EF	Remark 69
MMS $\implies$ APS	Lemmas 71 and 72
PMMS $\implies$ PAPS	Lemmas 71 and 72
GMMS $\implies$ GAPS	Lemmas 71 and 72
M1S $\implies$ APS	Lemmas 70 and 72
EF1 $\implies$ GAPS	Lemmas 70 and 72

Remark 69. Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, -)$ over goods where v_i is unit-demand for some agent i . Without loss of generality, let $v_i(1) \geq \dots \geq v_i(m)$. Then agent i is envy-free in allocation A iff $v_i(A_i) = v_i(1)$.

Lemma 70. Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, -)$ over goods where v_i is unit-demand for some agent i . Without loss of generality, let $v_i(1) \geq \dots \geq v_i(m)$. Then allocation A is M1S-fair to agent i iff $v_i(A_i) \geq v_i(n)$.

Proof. Suppose A is M1S-fair to agent i . Let B be agent i 's M1S-certificate for A . Let $G := \{g \in [m] : v_i(g) > v_i(B_i)\}$. Then B is EF1-fair to agent i , so every agent $j \in [n] \setminus \{i\}$ has at most one good from G , so $|G| \leq n - 1$. If $v_i(B_i) < v_i(n)$, then $G \supseteq [n]$, which is a contradiction. Hence, $v_i(A_i) \geq v_i(B_i) \geq v_i(n)$.

Suppose $v_i(A_i) \geq v_i(n)$. Let B be an allocation where each agent in $[n] \setminus \{i\}$ gets exactly one good from $[n - 1]$, and agent i gets $[m] \setminus [n - 1]$. Then B is EF1-fair to i and $v_i(A_i) \geq v_i(n) = v_i(B_i)$. Hence, B is an M1S-certificate for agent i for allocation A . Hence, A is M1S-fair to agent i . $\square$

Lemma 71. Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, -)$ over goods where v_i is unit-demand for some agent i . Without loss of generality, let $v_i(1) \geq \dots \geq v_i(m)$. Then agent i 's maximin share is $v_i(n)$.

Proof. In every n -partition P of $[m]$, some bundle P_k does not contain a good from $[n - 1]$. Then $v_i(P_k) \leq v_i(n)$. Hence, $\text{MMS}_i \leq v_i(n)$. Let P be a partition where $P_j = \{j\}$ for $j \in [n - 1]$ and $P_n = [m] \setminus [n - 1]$. Then $\min_{j=1}^n v_i(P_j) = v_i(n)$. Hence, $\text{MMS}_i = v_i(n)$. $\square$

Lemma 72. Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, -)$ over goods where v_i is unit-demand for some agent i . Without loss of generality, let $v_i(1) \geq \dots \geq v_i(m)$. Then agent i 's anyprice share is $v_i(n)$.

Proof. Set the price of the first $n - 1$ goods to $1/(n - 1)$ each and price the remaining goods at 0. Then $\text{APS}_i \leq v_i([m] \setminus [n - 1]) = v_i(n)$. For any non-negative price vector, some item from $[n]$ is affordable, so $\text{APS}_i \geq v_i(n)$. $\square$

Lemma 73. Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, -)$ over goods where v_i is unit-demand for some agent i . Without loss of generality, let $v_i(1) \geq \dots \geq v_i(m)$. Then every allocation is PROP1-fair to agent i .

Proof. $v_i(A_i \cup \{1\}) = v_i([m]) > v_i([m])/n$. □

E Non-Implications (Additive Valuations)

E.1 Trivial Examples

Example 74 (single item). Consider a fair division instance with n agents and one item (which is either a good to everyone or a chore to everyone). Then every allocation is EFX, EF1, APS, MMS, PROP_x, GAPS, GMMS, PAPS, PMMS, EEFX, EEF1, MXS, M1S, PROP_m, and PROP1, but not EF or PROP or GPROP or PPROP or EEF or MEFS.

Lemma 75. Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, -)$ with $n \geq 3$, $m = 2n - 1$, and identical additive valuations, where each item has value 1 to each agent. Let A be an allocation where agent n gets n goods, and all other agents get 1 good each. Then this allocation is APS+MMS+EEFX+PROP_x, but not EF1.

Proof. By Lemmas 44 and 48, $1 = \text{MMS}_i \leq \text{APS}_i \leq v([m])/n = 2 - 1/n$. Hence, $\text{APS}_i = 1$, since the APS is the value of some bundle. Hence, A is APS+MMS+PROP_x. Agent n 's EEFX-certificate for A is A itself. For $i \neq n$, agent i 's EEFX-certificate is B , where $|B_i| = 1$ and $|B_j| = 2$ for $j \neq i$. A is not EF1: agents $[n - 1]$ EF1-envy agent n . □

Lemma 76. Consider a fair division instance $([n], [m], (v_i)_{i=1}^n, -)$ with $n \geq 3$, identical additive disutilities, and $m = n + 1$ chores, each of disutility 1. Let A be an allocation where agents 1 and 2 get 2 chores each, agent n gets 0 chores, and the remaining agents get 1 chore each. Then this allocation is APS+MMS+EEFX+PROP_x, but not EF1.

Proof. $-2 = \text{MMS}_i \leq \text{APS}_i \leq v([m])/n = -1 - 1/n$. Hence, $\text{APS}_i = -2$, since the APS is the value of some bundle. Hence, A is APS+MMS+PROP_x.

Agents $[n] \setminus [2]$ do not EFX-envy anyone in A . For $i \in \{1, 2\}$, agent i 's epistemic-EFX-certificate can be obtained by transferring a chore from agent $3 - i$ to agent n . Hence, A is epistemic EFX. A is not EF1 because agents 1 and 2 EF1-envy agent n . □

E.2 From EEF, MEFS, PROP

Lemma 77 (EEF $\not\Rightarrow$ EF1). Let $0 \leq 2a < b$. Let $f_1, f_2, f_3 : 2^{[12]} \rightarrow \mathbb{R}_{>0}$ be additive sets functions:

	1	2	3	4	5	6	7	8	9	10	11	12
f_1	a	a	b	b	b	b	b	b	a	a	a	a
f_2	a	a	a	a	a	a	b	b	b	b	b	b
f_3	b	b	b	b	a	a	a	a	a	a	b	b

Let $t \in \{-1, 1\}$ and let $\mathcal{I} := ([3], [12], (v_i)_{i=1}^3, -)$ be an instance where $v_i := tf_i$ for all $i \in [3]$. Then allocation $A := ([4], [8] \setminus [4], [12] \setminus [8])$ is EEF+PROP but not EF1.

Proof. For $t = 1$, agent 1 EF1-envies agent 2 in A , and for $t = -1$, agent 1 EF1-envies agent 3 in A . $B = ([4], \{5, 6, 9, 10\}, \{7, 8, 11, 12\})$ is agent 1's EEF-certificate. A similar argument holds for agents 2 and 3 too. □

Example 78 ($\text{PROP} \not\Rightarrow \text{MEFS}$). Consider a fair division instance with 3 equally-entitled agents having additive valuations over 3 goods:

	1	2	3
v_1	10	20	30
v_2	20	10	30
v_3	10	20	30

Then the allocation $(\{2\}, \{1\}, \{3\})$ is PROP, but no allocation is MEFS (every agent's min EF share is 30).

Example 79 ($\text{PROP} \not\Rightarrow \text{MEFS}$). Consider a fair division instance with 3 equally-entitled agents having additive disutilities over 3 chores:

	1	2	3
$-v_1$	30	20	10
$-v_2$	20	30	10
$-v_3$	30	20	10

Then the allocation $(\{2\}, \{1\}, \{3\})$ is PROP, but no allocation is MEFS (every agent's min EF share is -10).

Lemma 80 ($\text{MEFS} \not\Rightarrow \text{EEF}$). Consider a fair division instance with 3 equally-entitled agents having additive valuations over 6 goods:

	1	2	3	4	5	6
v_1	20	20	20	10	10	10
v_2, v_3	20	10	10	1	1	1

Then the allocation $A := (\{4, 5, 6\}, \{1\}, \{2, 3\})$ is MEFS, but no allocation is epistemic EF.

Proof. Agents 2 and 3 are envy-free in A . Agent 1 has $B := (\{1, 4\}, \{2, 5\}, \{3, 6\})$ as her MEFS-certificate for A . Hence, A is MEFS.

Suppose an epistemic EF allocation X exists. Let $Y^{(i)}$ be each agent i 's epistemic-EF-certificate. For agent 2 to be envy-free in $Y^{(2)}$, we require $Y_2^{(2)} \supseteq \{1\}$ or $Y_2^{(2)} \supseteq \{2, 3\}$. Similarly, $Y_3^{(3)} \supseteq \{1\}$ or $Y_3^{(3)} \supseteq \{2, 3\}$. Since $Y_i^{(i)} = X_i$ for all i , we get $X_2 \cup X_3 \supseteq \{1, 2, 3\}$. Hence, $X_1 \subseteq \{4, 5, 6\}$. But then no epistemic-EF-certificate exists for agent 1 for X , contradicting our assumption that X is epistemic EF. Hence, no epistemic EF allocation exists. $\square$

Example 81 ($\text{MEFS} \not\Rightarrow \text{EEF}$). Consider a fair division instance with 3 equally-entitled agents having additive disutilities over 6 chores:

	1	2	3	4	5	6
$-v_1$	20	20	20	10	10	10
$-v_2, -v_3$	20	10	10	10	10	10

Then the allocation $A := (\{4, 5, 6\}, \{1\}, \{2, 3\})$ is MEFS (agents 2 and 3 are EF, agent 1's MEFS-certificate is $(\{1, 4\}, \{2, 5\}, \{3, 6\})$). Agent 1 is not EEF-satisfied by A .

Lemma 82 ($\text{MEFS} \not\Rightarrow \text{EEF1}$). Consider an instance with 3 equally-entitled agents having additive disutilities over 12 chores. $v_1(1) = v_1(2) = v_1(3) = 70$ and $v_1(c) = 10$ for all $c \in [12] \setminus [3]$. Agents 2 and 3 have disutility 10 for each chore. Then $A := ([12] \setminus [3], [2], \{3\})$ is a MEFS+PROP allocation where agent 1 is not EEF1-satisfied.

Proof. $\text{PROP}_1 = -100$ and $\text{PROP}_2 = \text{PROP}_3 = -40$. $\text{MEFS}_1 \leq -100$ because of the allocation $(\{1, 4, 5, 6\}, \{2, 7, 8, 9\}, \{3, 10, 11, 12\})$. $\text{MEFS}_i \leq -40$ for $i \in \{2, 3\}$ because of the allocation $([4], [8] \setminus [4], [12] \setminus [8])$. Agent 1 has disutility 90 in A , so A is MEFS-fair and PROP-fair to agent 1. Agents 2 and 3 have disutility at most 20 in A , so A is MEFS-fair and PROP-fair to them.

Agent 1 is not EEF1-satisfied by A , since in any EEF1-certificate B , some agent $j \in \{2, 3\}$ receives at most one chore of value 70, and agent 1 would EF1-envy j . $\square$

E.3 Two Equally-Entitled Agents

Example 83 (EFX $\not\Rightarrow$ MMS). *Let $t \in \{-1, 1\}$. Consider a fair division instance with 2 equally-entitled agents having an identical additive valuation function v over 5 items. $v(1) = v(2) = 3t$ and $v(3) = v(4) = v(5) = 2t$. Then allocation $A := (\{1, 3\}, \{2, 4, 5\})$ is EFX. The MMS is $6t$, since $P = (\{3t, 3t\}, \{2t, 2t, 2t\})$ is an MMS partition. But in A , some agent doesn't get her MMS.*

Example 84 (EF1 $\not\Rightarrow$ PROPX or MXS). *Let $t \in \{-1, 1\}$. Consider an instance with 2 equally-entitled agents having an identical additive valuation function v over 5 items: $v(1) = v(2) = 4t$ and $v(3) = v(4) = v(5) = t$. Allocation $(\{1\}, [5] \setminus \{1\})$ is EF1 but not PROPX and not MXS.*

Lemma 85 (PROPX $\not\Rightarrow$ M1S). *Let $t \in \{-1, 1\}$ and $0 < \varepsilon < 1/2$. Consider a fair division instance with 2 equally-entitled agents having an identical additive valuation function v over 4 items. Let $v(4) = (1 + 2\varepsilon)t$ and $v(j) = t$ for $j \in [3]$. Then allocation $A := (\{4\}, [3])$ is PROPX but not M1S.*

Proof. $v([m])/2 = (2 + \varepsilon)t$, so A is PROPX. For $t = 1$, in any allocation B where agent 1 doesn't EF1-envy agent 2, she must have at least 2 goods. But $v(A_1) = 1 + 2\varepsilon$, so agent 1 doesn't have an M1S-certificate for A . Hence, A is not M1S. For $t = -1$, in any allocation B where agent 2 doesn't EF1-envy agent 1, she must have at most 2 chores. But $v(A_2) = -3$, so agent 1 doesn't have an M1S-certificate for A . Hence, A is not M1S. $\square$

Example 86 (MXS $\not\Rightarrow$ PROPX for $n = 2$, [CGR⁺22]). *Let $t \in \{-1, 1\}$. Consider a fair division instance with 2 equally-entitled agents having identical additive valuations over 7 items: the first 2 items of value $4t$ and the last 5 items of value t . Then the allocation $A = (\{1, 3\}, \{2, 4, 5, 6, 7\})$ is not PROPX or EFX, but it is MXS because the agents have $([7] \setminus [2], [2])$ and $([2], [7] \setminus [2])$ as their MXS-certificates for A .*

Lemma 87 (M1S $\not\Rightarrow$ PROP1). *Consider a fair division instance with 2 equally-entitled agents having an identical additive valuation function v over 9 items. Let $t \in \{-1, 1\}$ and $v(9) = 4t$ and $v(j) = t$ for $j \in [8]$. Then allocation $A := (\{9\}, [8])$ is M1S but not PROP1.*

Proof. $v([9])/2 = 6t$. Let $B := ([4], [9] \setminus [4])$. For $t = 1$, B is agent 1's M1S-certificate for A , but A is not PROP1-fair to agent 1. For $t = -1$, B is agent 2's M1S-certificate for A , but A is not PROP1-fair to agent 2. $\square$

E.4 Three Equally-Entitled Agents

Example 88 (GAPS $\not\Rightarrow$ PROPX). *Consider a fair division instance with 3 equally-entitled agents having identical additive or unit-demand valuations. There are 2 goods of values 50 and 10. In every allocation, some agent doesn't get any good, and that agent is not PROPX-satisfied. The allocation where the first agent gets the good of value 50 and the second agent gets the good of value 10 is a groupwise APS allocation (set the price of the goods to 1.1 and 0.9).*

Lemma 89 (APS $>$ MMS). *Let $t \in \{-1, 1\}$. Consider a fair division instance with 3 equally-entitled agents having identical additive valuations over 15 items. The items' values are $65t, 31t, 31t, 31t, 23t, 23t, 23t, 17t, 11t, 7t, 7t, 7t, 5t, 5t, 5t$. Then the AnyPrice share is at least $97t$, the proportional share is $97t$, and the maximin share is less than $97t$.*

Proof. For $t = 1$, this follows from Lemma C.1 of [BEF23]. For $t = -1$, a similar argument tells us that the AnyPrice share is at least -97 . If the maximin share is at least -97 , then there must exist a partition P of the chores where each bundle has disutility 97. But then P would prove that the maximin share in the corresponding goods instance is at least 97, which is a contradiction. Hence, for $t = -1$, the maximin share is less than -97 . $\square$

Example 90 (GMMS $\not\Rightarrow$ APS, [BEF23]). *For the fair division instance in Lemma 89, the leximin allocation is GMMS (since on restricting to any subset of agents, the resulting allocation is still leximin, and is therefore MMS). However, no APS allocation exists, because $APS > MMS$, and the minimum value across all bundles is at most the MMS.*

Example 91 (PMMS $\not\Rightarrow$ MMS, Example 4.4 of [CKM⁺19]). *Let $t \in \{-1, 1\}$. Consider an instance with 3 equally-entitled agents having an identical valuation function v over 7 items where $v(1) = 6t$, $v(2) = 4t$, $v(3) = v(4) = 3t$, $v(5) = v(6) = 2t$, $v(7) = t$. Each agent's maximin share is $7t$ ($(\{1, 7\}, \{2, 3\}, \{4, 5, 6\})$ is a maximin partition). Allocation $(\{1\}, \{3, 4, 5\}, \{2, 6, 7\})$ is PMMS but not MMS.*

Example 92 (APS $\not\Rightarrow$ PROPm). *Consider a fair division instance with 3 equally-entitled agents having an identical additive valuation function v over 6 goods: $v(1) = 60$, $v(2) = 30$, and $v(3) = v(4) = v(5) = v(6) = 10$. The allocation $A := (\{2\}, \{3, 4, 5\}, \{1, 6\})$ is APS+MMS, since the MMS is 30, and the APS is at most 30 because of the price vector $(4, 3, 1, 1, 1, 1)$. However, A is not PROPm-fair to agent 1, because the proportional share is $130/3 > 40$.*

Example 93 (APS $\not\Rightarrow$ PROP1). *Consider an instance with 3 equally-entitled agents having an identical additive valuation function v over 6 chores: the first chore has disutility 18 (large chore) and the remaining chores have disutility 3 each (small chores). Then $X := ([6] \setminus \{1\}, \{1\}, \emptyset)$ is MMS+APS, since the MMS is -18 , and the APS is at most -18 due to the price vector $(1, 0, 0, 0, 0, 0)$. X is not PROP1-fair to agent 1, since the proportional share is -11 , and agent 1's disutility in X after removing any chore is 12. X is not EEF1-fair to agent 1 because even after redistributing chores among the remaining agents, someone will always have no chores.*

Lemma 94 (PROPm doesn't exist for mixed manna). *Consider an instance $([3], [6], (v_i)_{i=1}^3, -)$ where agents have identical additive valuations, and items have values $(-3, -3, -3, -3, -3, 3\varepsilon)$, where $0 < \varepsilon < 1/2$. Then there exists an EFX+GMMS+GAPS allocation but no PROPm allocation.*

Proof. The proportional share is $v([m])/3 = -5 + \varepsilon$. Without loss of generality, assume agent 1 receives the most number of chores, and agent 3 receives the least number of chores. Then agent 1 has at least 2 chores, and agent 3 has at most 1.

Case 1: agent 1 receives at least 3 chores.

Then even after removing one of her chores, and even if she receives the good, her value for her bundle is at most $-6 + 3\varepsilon < -5 + \varepsilon$. Hence, she is not PROPm-satisfied.

Case 2: agent 1 receives 2 chores.

Then agent 2 also receives 2 chores, and agent 3 receives 1 chore. Without loss of generality, assume agent 1 does not receive the good. Then this allocation is EFX and groupwise MMS. On pricing each chore at -3 and pricing the good at 3 , we get that the allocation is groupwise APS. However, if agent 1 adds the good to her bundle, her value becomes $-6 + 3\varepsilon < -5 + \varepsilon$, so she is not PROPm-satisfied. $\square$

Example 95 (PROPm $\not\Rightarrow$ PROPAvg). *Consider an instance with three equally-entitled agents having an identical additive valuation function v over three goods. Let $v(1) = v(2) = 60$ and*

$v(3) = 30$. Then allocation $A := (\{3\}, \{1, 2\}, \emptyset)$ is *PROP*_m, but agent 3 is not *PROPAvg*-satisfied by A .

E.5 Unequal Entitlements

Example 96 (*PROP*₁ $\not\Rightarrow$ *M1S*). Consider an instance with 2 agents having identical additive valuations. Let $t \in \{-1, 1\}$. Let there be 2 items, each of value t . Let the entitlement vector be $(2/3, 1/3)$. The allocation where the first agent gets both items is *PROP*₁ but not *M1S*.

Lemma 97 (*GAPS* $\Rightarrow$ *PROP*_x for $n = 2$). Let $t \in \{-1, 1\}$. Consider a fair division instance $([2], [12], (v, v), (0.4, 0.6))$, where $v(1) = v(2) = 10t$ and $v(j) = t$ for all $j \in [12] \setminus [2]$. Then allocation $A = (\{1\}, [12] \setminus \{1\})$ is *APS* but not *PROP*_x.

Proof. Agents 1 and 2 have proportional shares $12t$ and $18t$, respectively.

For $t = 1$, agent 1's *APS* is at most 10 (price items 1 and 2 at 0.45 each and the remaining items at 0.01 each), and agent 2's *APS* is at most 18 (price each item at its value). Hence, A is *APS* but not *PROP*_x-fair to agent 1.

For $t = -1$, agent 2's *APS* is at most -20 (price items 1 and 2 at -0.45 each and the remaining items at -0.01 each), and agent 1's *APS* is at most -12 (price each item at its value). Hence, A is *APS* but not *PROP*_x-fair to agent 2. $\square$

Example 98 (*EF*₁ $\not\Rightarrow$ *EFX* for binary mixed manna). Let $([2], [5], (v, v), (4/5, 1/5))$ be a fair division instance where $v(g) = 1$ for all $g \in [4]$ and $v(5) = -1$. Let $A = ([5] \setminus \{1\}, \{1\})$. Then A is *EF*₁ but not *EFX*.

Example 99 (*WMMS* and *M1S* are incompatible for chores). Let $([2], [2], (v, v), (0.9, 0.1))$ be a fair division instance, where $v(1) = v(2) = -1$. Then for all $i \in [2]$, we have

$$\frac{-\text{WMMS}_i}{w_i} = \min_X \max \left(\frac{10|X_1|}{9}, 10|X_2| \right),$$

so $-\text{WMMS}_1 = 2$ and $-\text{WMMS}_2 = 2/9$. Hence, allocation X is *WMMS* iff $|X_2| = 0$. On the other hand, X is *M1S* iff $|X_1| = |X_2| = 1$ iff X is *EFX*.

Lemma 100 (*PROP*₁+*M1S* is infeasible for goods). Consider a fair division instance $\mathcal{I} := ([3], [7], (v_i)_{i=1}^3, w)$, where the entitlement vector is $w := (7/12, 5/24, 5/24)$, the agents have identical additive valuations, and each good has value 1. Then

1. $\text{APS}_1 = 4$ and $\text{APS}_2 = \text{APS}_3 = 1$.
2. X is *APS* $\iff X$ is groupwise-*APS* (*GAPS*) $\iff X$ is *PROP*₁.
3. $\text{WMMS}_1 = 3$ and $\text{WMMS}_2 = \text{WMMS}_3 = 15/14$.
4. X is *WMMS* $\iff X$ is groupwise-*WMMS* (*GWMMMS*) $\iff X$ is *EFX* $\iff X$ is *M1S*.

Therefore,

1. *M1S*+*PROP*₁ is infeasible for this instance.
2. *GWMMMS*+*EFX* doesn't imply *PROP*₁.
3. *GAPS* doesn't imply *M1S*.

Proof. By Lemma 56, $\text{APS}_1 = \lfloor \frac{7 \times 7}{12} \rfloor = 4$ and $\text{APS}_2 = \text{APS}_3 = \lfloor \frac{5 \times 7}{24} \rfloor = 1$. By Lemmas 55 and 56, an allocation is *APS* iff it is *PROP*₁.

Any *GAPS* allocation is also *APS* by definition. We will now show that any *APS* allocation is also *GAPS*. Formally, let A be an *APS* allocation for $\mathcal{I}$. The cardinality vector of A , i.e., $c := (|A_1|, |A_2|, |A_3|)$, can have three possible values: $(5, 1, 1)$, $(4, 2, 1)$, $(4, 1, 2)$. For every possible value of c and $S \subseteq [3]$, we show that $(\widehat{\mathcal{I}}, \widehat{A}) := \text{restrict}(\mathcal{I}, A, S)$ is *APS* (c.f. Definition 11).

1. $c = (5, 1, 1)$ and $S = \{1, 2\}$: $\widehat{\mathcal{I}}$ has 6 goods and entitlement vector $(14/19, 5/19)$. $\text{APS}_1 = \lfloor \frac{14 \times 6}{19} \rfloor = 4$ and $\text{APS}_2 = \lfloor \frac{5 \times 6}{19} \rfloor = 1$. Hence, $\widehat{A}$ is APS for $\widehat{\mathcal{I}}$.
2. $c = (5, 1, 1)$, $S = \{1, 3\}$: Same as the $S = \{1, 2\}$ case.
3. $c = (5, 1, 1)$ and $S = \{2, 3\}$: $\widehat{\mathcal{I}}$ has 2 goods and entitlement vector $(1/2, 1/2)$. $\text{APS}_2 = \text{APS}_3 = 1$, so $\widehat{A}$ is APS for $\widehat{\mathcal{I}}$.
4. $c = (4, 2, 1)$ and $S = \{1, 2\}$: $\widehat{\mathcal{I}}$ has 6 goods and entitlement vector $(14/19, 5/19)$. $\text{APS}_1 = \lfloor \frac{14 \times 6}{19} \rfloor = 4$ and $\text{APS}_2 = \lfloor \frac{5 \times 6}{19} \rfloor = 1$. Hence, $\widehat{A}$ is APS for $\widehat{\mathcal{I}}$.
5. $c = (4, 2, 1)$ and $S = \{1, 3\}$: $\widehat{\mathcal{I}}$ has 5 goods and entitlement vector $(14/19, 5/19)$. $\text{APS}_1 = \lfloor \frac{14 \times 5}{19} \rfloor = 3$ and $\text{APS}_2 = \lfloor \frac{5 \times 5}{19} \rfloor = 1$. Hence, $\widehat{A}$ is APS for $\widehat{\mathcal{I}}$.
6. $c = (4, 2, 1)$ and $S = \{2, 3\}$: $\widehat{\mathcal{I}}$ has 3 goods and entitlement vector $(1/2, 1/2)$. $\text{APS}_2 = \text{APS}_3 = \lfloor \frac{1 \times 3}{2} \rfloor = 1$. Hence, $\widehat{A}$ is APS for $\widehat{\mathcal{I}}$.
7. $c = (4, 1, 2)$: Similar to the $c = (4, 2, 1)$ case.

Hence, any APS allocation for $\mathcal{I}$ is also GAPS.

For any allocation X , define

$$f(X) := \min_{j=1}^3 \frac{|X_j|}{w_j}.$$

Then $\text{WMMS}_i = w_i \max_X f(X)$ for all $i \in [3]$. If $|X_1| \leq 2$, then $f(X) \leq |X_1|/w_1 \leq 24/7 = 3 + 3/7$. If $|X_2| \leq 1$ or $|X_3| \leq 1$, $f(X) \leq 24/5 = 4 + 4/5$. Otherwise, $|X_1| = 3$ and $|X_2| = |X_3| = 2$, so $f(X) = \min(3 \times 12/7, 2 \times 24/5) = 36/7 = 5 + 1/6$. Hence, $\max_X f(X) = 36/7$, so $\text{WMMS}_1 = 3$ and $\text{WMMS}_2 = \text{WMMS}_3 = 15/14$. So, an allocation is WMMS iff it has cardinality vector $(3, 2, 2)$.

Any GWMMMS allocation is also WMMS by definition. We now prove the converse. For every possible value of $S \subseteq [3]$, we show that $(\widehat{\mathcal{I}}, \widehat{A}) := \text{restrict}(\mathcal{I}, A, S)$ is WMMS (c.f. Definition 11).

1. $S = \{1, 2\}$: $\widehat{\mathcal{I}}$ has 5 goods and entitlement vector $(14/19, 5/19)$. If $|X_1| \leq 2$, then $f(X) \leq |X_1|/w_1 \leq 38/14 = 2 + 10/14$. If $|X_2| \leq 1$, then $f(X) \leq |X_2|/w_2 \leq 19/5 = 3 + 10/14$. Otherwise, $|X_1| = 3$ and $|X_2| = 2$, so $f(X) = \min(3/w_1, 2/w_2) = \min(57/14, 38/5) = 57/14 = 4 + 1/14$. Hence, $\text{WMMS}_1 = 3$ and $\text{WMMS}_2 = \text{WMMS}_3 = 57/14 \times 5/19 = 1 + 19/266$. Hence, $\widehat{A}$ is WMMS for $\widehat{\mathcal{I}}$.
2. $S = \{1, 3\}$: Similar to the $S = \{1, 2\}$ case.
3. $S = \{2, 3\}$: $\widehat{\mathcal{I}}$ has 4 goods and entitlement vector $(1/2, 1/2)$. Then $\text{WMMS}_1 = \text{WMMS}_2 = 2$. Hence, $\widehat{A}$ is WMMS for $\widehat{\mathcal{I}}$.

Hence, any WMMS allocation for $\mathcal{I}$ is also GWMMMS.

Any GWMMMS allocation is EFX by Lemma 40, and any EFX allocation is M1S by Lemma 21. We will now show that any M1S allocation is WMMS.

Let X be an M1S allocation. Let A be agent 1's M1S certificate. If $|A_1| \leq 2$, then $|A_j| \geq 3$ for some $j \in \{2, 3\}$. Since agent 1 has higher entitlement, she would EF1-envy agent j , which is a contradiction. Hence, $|X_1| \geq |A_1| \geq 3$.

Let B be agent 2's M1S certificate. Suppose $|B_2| \leq 1$. Since 2 doesn't EF1-envy 3, we get $|B_3| \leq 2$. Since 2 doesn't EF1-envy 1, we get

$$\frac{|B_1| - 1}{w_1} \leq \frac{|B_2|}{w_2} \iff |B_1| \leq 1 + \frac{w_1}{w_2} = 3 + \frac{4}{5}.$$

Hence, $|B_1| + |B_2| + |B_3| \leq 3 + 1 + 2 = 6$, which is a contradiction. Hence, $|X_2| \geq |B_2| \geq 2$. Similarly, we can prove that $|X_3| \geq 2$.

Hence, $|X_i| \geq \text{WMMS}_i$ for all i , so X is WMMS. This proves that any M1S allocation is WMMS. $\square$

[CISZ21] also prove that EF1+PROP1 allocations may not exist for unequal entitlements. We use a different counterexample in Lemma 100, which allows us to also prove other non-implications.

Lemma 101. *Consider a fair division instance $\mathcal{I} := ([3], [7], (v_i)_{i=1}^3, w)$, where the entitlement vector is $w := (9/16, 7/32, 7/32)$, the agents have identical additive valuations, and each chore has disutility 1. Let X be an allocation. Then*

1. $\text{APS}_1 = -4$ and $\text{APS}_2 = \text{APS}_3 = -2$.
2. X is APS $\iff X$ is PROP1.
3. $\text{WMMS}_1 = -5$ and $\text{WMMS}_2 = \text{WMMS}_3 = -35/18$.
4. X is WMMS $\iff X$ is groupwise-WMMS $\iff (|X_1| = 5 \text{ and } |X_2| = |X_3| = 1)$.
5. X is EFX $\iff X$ is M1S $\iff (|X_1| = 3 \text{ and } |X_2| = |X_3| = 2)$.

Therefore,

1. PROP1+WMMS is infeasible for this instance.
2. M1S+WMMS is infeasible for this instance.
3. GWMMMS doesn't imply PROP1 or M1S.

Proof. Note that $\lfloor -x \rfloor = -\lceil x \rceil$ for all $x \in \mathbb{R}$.

By Lemma 56, $-\text{APS}_1 = \lceil \frac{9 \times 7}{16} \rceil = 4$, and $-\text{APS}_2 = -\text{APS}_3 = \lceil \frac{7 \times 7}{32} \rceil = 2$. By Lemmas 55 and 56, an allocation is APS iff it is PROP1.

For any allocation X , define

$$f(X) := \max_{j=1}^3 \frac{|X_j|}{w_j}.$$

Then $-\text{WMMS}_i = w_i \min_X f(X)$ for all $i \in [3]$. If $|X_1| \geq 6$, then $f(X) \geq 6/w_1 = 10 + 2/3$. If $|X_2| \geq 2$ or $|X_3| \geq 2$, then $f(X) \geq 2/w_2 = 9 + 1/7$. Otherwise, $|X_1| = 5$ and $|X_2| = |X_3| = 1$, so $f(X) = \max(5 \times 16/9, 1 \times 32/7) = 80/9 = 9 - 1/9$. Hence, $\min_X f(X) = 80/9$, so $-\text{WMMS}_1 = 5$ and $-\text{WMMS}_2 = -\text{WMMS}_3 = 35/18 < 2$. Hence, an allocation X is WMMS iff $|X_1| = 5$ and $|X_2| = |X_3| = 1$.

Any GWMMMS allocation is also WMMS by definition. We now prove the converse. Let A be a WMMS allocation. For every possible value of $S \subseteq [3]$, we show that $(\widehat{\mathcal{I}}, \widehat{A}) := \text{restrict}(\mathcal{I}, A, S)$ is WMMS (c.f. Definition 11).

1. $S = \{1, 2\}$: $\widehat{\mathcal{I}}$ has 6 chores and entitlement vector $\widehat{w} := (18/25, 7/25)$. Let X be an allocation for $\widehat{\mathcal{I}}$. Let $\widehat{f}(X) := \max_{j=1}^2 |X_j|/\widehat{w}_j$. If $|X_1| \geq 6$, then $\widehat{f}(X) \geq 6/\widehat{w}_1 = 25/3 = 8 + 1/3$. If $|X_2| \geq 2$, then $\widehat{f}(X) \geq 2/\widehat{w}_2 = 50/7 = 7 + 1/7$. Otherwise, $|X_1| = 5$ and $|X_2| = 1$, so $\widehat{f}(X) = \max(5/\widehat{w}_1, 1/\widehat{w}_2) = 125/18 = 7 - 1/18$. Hence, $-\text{WMMS}_1 = 5$ and $-\text{WMMS}_2 = 35/18 < 2$. Hence, $\widehat{A}$ is WMMS for $\widehat{\mathcal{I}}$.
2. $S = \{1, 3\}$: Similar to the $S = \{1, 2\}$ case.
3. $S = \{2, 3\}$: $\widehat{\mathcal{I}}$ has 2 chores and entitlement vector $(1/2, 1/2)$. Then $-\text{WMMS}_1 = -\text{WMMS}_2 = 1$. Hence, $\widehat{A}$ is WMMS for $\widehat{\mathcal{I}}$.

Hence, A is GWMMMS.

Suppose X is M1S. Let A be agent 1's M1S certificate. Then $|X_1| \leq |A_1|$ and A is EF1-fair to agent 1. Suppose $|A_1| \geq 4$. Then $|A_2| \leq 1$ or $|A_3| \leq 1$. Without loss of generality, let $|A_2| \leq 1$. Then

$$\frac{|A_1| - 1}{w_1} \geq \frac{3}{w_1} = 5 + \frac{1}{3} > 5 - \frac{3}{7} = \frac{1}{w_2} \geq \frac{|A_2|}{w_2}.$$

Hence, agent 1 EF1-envies agent 2, which is a contradiction. Hence, $|X_1| \leq |A_1| \leq 3$.

Let B be agent 3's M1S certificate. Then $|X_3| \leq |B_3|$ and B is EF1-fair to agent 3. Suppose $|B_3| \geq 3$. If $|B_2| \leq 1$, then agent 3 EF1-envies agent 2, so $|B_2| \geq 2$. Hence, $|B_1| \leq 2$. Then

$$\frac{|B_3| - 1}{w_3} \geq \frac{2}{w_3} = 9 + \frac{1}{7} > 4 - \frac{4}{9} = \frac{2}{w_1} \geq \frac{|B_1|}{w_1}.$$

Hence, agent 3 EF1-envies agent 2, which is a contradiction. Hence, $|X_3| \leq |B_3| \leq 2$. We can similarly prove that $|X_2| \leq 2$. Hence, if X is M1S, then $|X_1| = 3$ and $|X_2| = |X_3| = 2$. Moreover,

$$\begin{aligned} \frac{|X_1| - 1}{w_1} &= 4 - \frac{4}{9} < 9 + \frac{1}{7} = \frac{|X_2|}{w_2} = \frac{|X_3|}{w_3}, \\ \frac{|X_3| - 1}{w_3} &= \frac{|X_2| - 1}{w_2} = 5 - \frac{3}{7} < 5 + \frac{1}{3} = \frac{|X_1|}{w_1}. \end{aligned}$$

Hence, X is EFX. Also, for additive valuations, EFX implies M1S by Lemma 21. Hence, X is EFX iff X is M1S. $\square$

Lemma 102 (PROP $\not\Rightarrow$ M1S for chores). *Let $0 \leq \varepsilon \leq 2/7$. Let $([3], [7], (v_i)_{i=1}^3, w)$ be a fair division instance, where $w = (4/7, 3/14, 3/14)$ and v is given by the following table:*

c	1	2	3	4	5	6	7
$-v_1(c)$	1	1	1	1	1	1	1
$-v_2(c) = -v_3(c)$	1	1	1	1	ε	ε	ε

Let A be an allocation where $A_1 = [4]$. Then A is PROP but not M1S-fair to agent 1.

Proof. $-v_2(A_2) \leq 3\varepsilon \leq 4w_2$, so A is PROP-fair to agents 2 and 3. $-v_1(A_1) = 4 = 7w_1$, so A is PROP-fair to agent 1.

Suppose A is M1S-fair to agent 1. Let X be agent 1's M1S certificate for A . Then $|X_1| \geq 4$, so $|X_2| \leq 1$ or $|X_3| \leq 1$. Hence,

$$\frac{|X_1| - 1}{w_1} \geq \frac{3}{w_1} = 5 + \frac{1}{4} > 5 - \frac{1}{3} = \frac{1}{w_2} \geq \min \left(\frac{|X_2|}{w_2}, \frac{|X_3|}{w_3} \right).$$

Hence, X is not EF1-fair to agent 1, which is a contradiction. Hence, A is not M1S-fair to agent 1. $\square$

Example 103 (PROP $\not\Rightarrow$ MEFS for binary valuations). *Let $([3], [6], (v_i)_{i=1}^3, w)$ be a fair division instance, where $w = (1/2, 1/4, 1/4)$ and v is given by the following table:*

g	1	2	3	4	5	6
$v_1(g)$	1	1	1	1	1	1
$v_2(g)$	0	0	0	1	1	0
$v_3(g)$	0	0	0	0	0	1

Let $A := (\{1, 2, 3\}, \{4, 5\}, \{6\})$. Then A is PROP. A is not MEFS for agent 1. This is because in any allocation B such that $v_1(B_1) \leq v_1(A_1) = 3$, some agent $j \in \{2, 3\}$ must receive at least 2 goods, and then agent 1 would envy j in B .

F Non-Implications (Non-Additive Valuations)

F.1 Supermodular Valuations

Lemma 104 (EF+GAPS $\not\Rightarrow$ PROP1). *Let $0 \leq \varepsilon < 1/27$ and $k \in \mathbb{N}$. For any $x \geq 0$, let $f(x) := x\varepsilon + \max(0, x - k)$. Consider an instance with n equally-entitled agents having an identical valuation function v over $4n$ goods, where $v(X) = f(|X|)$ for all $X \subseteq [4n]$. Then v is supermodular.*

Let A be the allocation where each agent gets 4 goods. Then A is EF+GAPS+GMMS. For $n = 2$ and $k = 5$, no PROP1 or PROPm allocation exists. If $n \geq 3$ and $k = 8$, then A is PPROP but no PROP1 or PROPx allocation exists. Additionally if $\varepsilon = 0$, then A is PROPavg.

Proof. For any $X \subseteq [4n]$ and $g \in [4n] \setminus X$, we have $v(g \mid X) = \varepsilon$ if $|X| < k$ and $v(g \mid X) = 1 + \varepsilon$ otherwise. Since $v(g \mid X)$ is non-decreasing in $|X|$, v is supermodular.

Since all agents have the same bundle, A is EF and GMMS. Set the price of each good to 1 to get that A is GAPS.

For $n = 2$ and $k = 5$, the proportional share is $4 - k/n + 4\varepsilon$. For $k = 5$, the proportional share is at least $3/2$, but $f(5) = 5\varepsilon$ and $f(6) = 1 + 6\varepsilon < 3/2$. In any allocation, some agent will have at most 4 goods, and that agent would not be PROP1 or PROPm satisfied.

If $n \geq 3$ and $k = 8$, the proportional share is at least $4/3$. Since $f(8) = 8\varepsilon$ and $f(4) = 4\varepsilon$, A is PPROP. In any allocation, some agent gets at most 4 goods, and $f(5) = 5\varepsilon < 4/3$, so that agent is not PROP1-satisfied, and $f(9) = 1 + 9\varepsilon < 4/3$, so that agent is not PROPx-satisfied. Additionally, if $\varepsilon = 0$, for any two agents i and j , we have $v(A_j \mid A_i) = f(8) - f(4) = 0$. Hence, A is PROPavg. $\square$

F.2 Unit-Demand Valuations

It is known that unit-demand functions are submodular and cancelable. We first show how to perturb a unit-demand function such that submodularity and cancelability remain preserved and all marginals become positive.

Lemma 105. *Let $u : 2^{[m]} \rightarrow \mathbb{R}_{\geq 0}$ be a function and $\varepsilon \in \mathbb{R}_{\geq 0}$. For any $X \subseteq [m]$, define $\hat{u}(X) := u(X) + |X|\varepsilon$. If u is submodular, then $\hat{u}$ is also submodular. If $m \leq 3$ and u is cancelable, then $\hat{u}$ is also cancelable.*

Proof. If u is submodular, then for any $S, T \subseteq [m]$, we get $\hat{u}(S) + \hat{u}(T) - \hat{u}(S \cup T) - \hat{u}(S \cap T) = u(S) + u(T) - u(S \cup T) - u(S \cap T) \geq 0$. Hence, $\hat{u}$ is also submodular.

Let u be cancelable. For every $T \subseteq [m]$ and $S_1, S_2 \subseteq [m] \setminus T$, if $\hat{u}(S_1 \cup T) > \hat{u}(S_2 \cup T)$, we need to show that $\hat{u}(S_1) > \hat{u}(S_2)$ to prove that $\hat{u}$ is cancelable. This is trivial when $T = \emptyset$ or $\varepsilon = 0$. If $|S_1| = |S_2|$, then $u(S_1 \cup T) > u(S_2 \cup T)$. Since u is cancelable, we get $u(S_1) > u(S_2)$, which implies $\hat{u}(S_1) > \hat{u}(S_2)$. Let $\varepsilon > 0$. If $S_1 = \emptyset$, we get $\hat{u}(S_1 \cup T) > \hat{u}(S_2 \cup T) \implies u(S_2 \mid T) < -|S_2|\varepsilon$, which is a contradiction since u is non-negative. Hence, $S_1 \neq \emptyset$. If $S_2 = \emptyset$, we get $\hat{u}(S_1) \geq |S_1|\varepsilon > 0 = \hat{u}(S_2)$. When $m \leq 3$, we have considered all cases. Hence, if $m \leq 3$ and u is cancelable, then $\hat{u}$ is also cancelable. $\square$

Lemma 106 (PROP $\not\Rightarrow$ M1S). *Consider a unit-demand function v over 3 goods such that $v(1) = v(2) = 4$ and $v(3) = 3$. Let $0 \leq \varepsilon < 1/6$ and $\hat{v}(X) := v(X) + 2|X|\varepsilon$ for all $X \subseteq [3]$. Consider a fair division instance with 3 goods and 2 equally-entitled agents having valuation function $\hat{v}$. Then the allocation $A = (\{1, 2\}, \{3\})$ is PROP but not M1S.*

Proof. A is PROP since $\hat{v}(A_1) = 4 + 4\varepsilon$, $\hat{v}(A_2) = 3 + 2\varepsilon$, and the PROP share is $2 + 3\varepsilon$. Suppose A is M1S and agent 2's M1S certificate for A is B . $B_2 \neq \emptyset$, since B is EF1-fair to agent 2. Moreover,

$\widehat{v}(B_2) \leq \widehat{v}(A_2) = 3 + 2\varepsilon$, so $B_2 = \{3\}$. Hence, $B_1 = \{1, 2\}$. However, $\min_{g \in B_1} \widehat{v}(B_1 \setminus \{g\}) = 4 + 4\varepsilon > 3 + 2\varepsilon = \widehat{v}(B_2)$. Hence, agent 2 is not EF1-satisfied by B , which contradicts the fact that B is agent 2's M1S certificate for A . Hence, A is not M1S. $\square$

Example 107 ($\text{PROP1} \not\Rightarrow \text{PROPM}$). Consider a unit-demand function v over 2 goods such that $v(1) = 30$ and $v(2) = 10$. Let $0 \leq \varepsilon < 1$ and $\widehat{v}(X) := v(X) + |X|\varepsilon$ for all $X \subseteq [2]$. Consider a fair division instance with 2 goods and 2 equally-entitled agents having valuation function $\widehat{v}$. Then the allocation $A := (\{1, 2\}, \emptyset)$ is PROP1 but not PROPM.

Example 108 ($\text{PROP} \not\Rightarrow \text{PPROP}$). Consider a unit-demand function v over 3 goods such that $v(1) = 30$ and $v(2) = v(3) = 11$. Let $0 \leq \varepsilon < 1$ and $\widehat{v}(X) := v(X) + |X|\varepsilon$ for all $X \subseteq [3]$. Consider a fair division instance with 3 goods and 3 equally-entitled agents having valuation function $\widehat{v}$. Then the allocation $A := (\{1\}, \{2\}, \{3\})$ is PROP but no allocation is PPROP.

Example 109 ($\text{MMS} \not\Rightarrow \text{EF1}, \text{PROPM}$). Consider a unit-demand function v over 5 goods such that $v(1) = 200$ and $v(2) = 30$ and $v(3) = v(4) = v(5) = 10$. Let $0 \leq \varepsilon < 1$ and $\widehat{v}(X) := v(X) + |X|\varepsilon$ for all $X \subseteq [5]$. Consider a fair division instance with 5 goods and 4 equally-entitled agents having valuation function $\widehat{v}$. Then the allocation $A := (\{1, 2\}, \{3\}, \{4\}, \{5\})$ is MMS but not EF1 and not PROPM. (The maximin share is $10 + \varepsilon$. The proportional share is $50 + (5/4)\varepsilon$.)

Example 110 ($\text{PROPM} \not\Rightarrow \text{PROPavg}$). Consider a unit-demand function v over 3 goods such that $v(1) = 75$, $v(2) = 30$, and $v(3) = 10$. Let $0 \leq \varepsilon < 1$ and $\widehat{v}(X) := v(X) + |X|\varepsilon$ for all $X \subseteq [3]$. Consider a fair division instance with 3 goods and 3 equally-entitled agents having valuation function $\widehat{v}$. Then the allocation $A := (\{1, 3\}, \{2\}, \emptyset)$ is PROPM but not PROPavg. (The proportional share is $25 + \varepsilon$.)

F.3 Submodular Valuations

Definition 21 (PMRF). Let $P = (P_1, \dots, P_k)$ be a partition of a finite set M . Define $\text{PMRF}(P)$ as the function $f : 2^M \rightarrow [0, 1]$ where $f(X) := \sum_{i=1}^k \mathbf{1}(X \cap P_i \neq \emptyset)$. (Here $\mathbf{1}(C)$ is 1 if condition C is true and 0 otherwise.) For any $\varepsilon \in \mathbb{R}_{\geq 0}$, define $\text{PMRF}_{+\varepsilon}(P)$ as the function f where $f(X) = \text{PMRF}(P)(X) + |X|\varepsilon$.

Lemma 111. For any partition P over a finite set M , $f := \text{PMRF}_{+\varepsilon}(P)$ is submodular, and $f(g \mid X) \in \{\varepsilon, 1 + \varepsilon\}$ for any $X \subseteq M$ and $g \in M \setminus X$.

Proof sketch. $\text{PMRF}(P)$ is the rank function of a partition matroid, and hence, is submodular. The submodularity of f follows from Lemma 105. $\square$

Example 112 ($\text{EF} \not\Rightarrow \text{MMS}$). Let $P := (\{r_1, r_2\}, \{g_1, g_2\})$, $\varepsilon \in \mathbb{R}_{\geq 0}$, and $v := \text{PMRF}_{+\varepsilon}(P)$. Consider a fair division instance with 2 equally-entitled agents having identical valuation function v . Then the maximin share is at least $2 + 2\varepsilon$, as exemplified by the partition $(\{r_1, g_1\}, \{r_2, g_2\})$. The allocation P is EF but not MMS, since each agent's value for her own bundle is $1 + 2\varepsilon$.

Example 113 ($\text{EF1} \not\Rightarrow \text{MXS}$). Let $P := (\{r_1, r_2\}, \{g_1, g_2\}, \{b\})$ and $v := \text{PMRF}(P)$. Consider a fair division instance with 2 equally-entitled agents having identical valuation function v . Then the allocation $A := (\{r_1, r_2\}, \{g_1, g_2, b\})$ is EF1 but agent 1 is not MXS-satisfied by A .

Example 114 ($\text{MEFS} \not\Rightarrow \text{EF1}$). Let $P := (\{a\}, \{b\}, \{c\}, \{d\}, \{e_1, e_2\}, \{f_1, f_2\}, \{g_1, g_2\})$. Let $v := \text{PMRF}_{+\varepsilon}(P)$ for $0 \leq \varepsilon \leq 1/2$. Consider a fair division instance with 2 equally-entitled agents having identical valuation function v . Let $A := (\{a, e_1, f_1, g_1\}, \{b, c, d, e_2, f_2, g_2\})$. Since $v(A_2) = 6 + 6\varepsilon$ and

$v(A_1) = 4 + 4\varepsilon$, agent 2 is envy-free in A but agent 1 is not EF1-satisfied by A . However, agent 1 is MEFS-satisfied by A , and her MEFS-certificate for A is $B := (\{a, b, c, d\}, \{e_1, e_2, f_1, f_2, g_1, g_2\})$, since $v(B_1) = v(A_1) = 4 + 4\varepsilon$ and $v(B_2) = 3 + 6\varepsilon$.

Example 115. Let $M := \{a_j\}_{j=1}^6 \cup \{b_j\}_{j=1}^6 \cup \{c_j\}_{j=1}^4$. Let P be a partition of M where $X \in P$ iff $X = \{a_j, b_j\}$ for some $j \in [6]$ or $X = \{c_j\}$ for some $j \in [4]$. Let $v := \text{PMRF}_{+\varepsilon}(P)$ for $\varepsilon \in [0, 1/6]$. Consider an instance with 3 equally-entitled agents having identical valuation function v .

Let $A := (\{a_j\}_{j=1}^6, \{b_j\}_{j=1}^6, \{c_j\}_{j=1}^4)$. Then $v(A_1) = v(A_2) = 6(1 + \varepsilon)$ and $v(A_3) = 4(1 + \varepsilon)$. Then A is not EF1 and not PPROP. But A is EEF since agents 1 and 2 are envy-free, and agent 3's EEF-certificate is B where $B_1 := \{a_1, a_2, a_3, b_1, b_2, b_3\}$, $B_2 := \{a_4, a_5, a_6, b_4, b_5, b_6\}$, and $B_3 = A_3$. Then $v(B_3) = 4(1 + \varepsilon)$ and $v(B_1) = v(B_2) = 3 + 6\varepsilon < 4$.

Example 116 (PROP $\not\Rightarrow$ M1S). Let $v : 2^{[10]} \rightarrow \mathbb{R}_{\geq 0}$ be a function defined as $v(X) := \min(6, |X|)$. Then v is submodular (since v is the rank function of a uniform matroid) and v has binary marginals (i.e., $v(g \mid X) \in \{0, 1\}$ for all $X \subseteq [10]$ and $g \in [m] \setminus X$). Let $0 \leq \varepsilon < 1$ and $\hat{v}(X) := v(X) + |X|\varepsilon$.

Consider a fair division instance with 2 equally-entitled agents having identical valuation function $\hat{v}$. Then $A := ([4], [10] \setminus [4])$ is PROP but agent 1 is not M1S-satisfied.

Lemma 117 (MMS $\not\Rightarrow$ APS for $n = 2$, Remark 2 of [BEF23]). Let $C := \{\{1, 2, 3\}, \{1, 5, 6\}, \{2, 4, 6\}, \{3, 4, 5\}\}$. For $\varepsilon \in [0, 1/3]$, define $v : 2^{[6]} \rightarrow \mathbb{R}_{\geq 0}$ as

$$v(X) := |X|\varepsilon + \begin{cases} 2|X| & \text{if } |X| \leq 2 \\ 5 & \text{if } |X| = 3 \text{ and } X \notin C \\ 6 & \text{if } X \in C \text{ or } |X| \geq 4 \end{cases}$$

Then v is submodular. Consider an instance with 2 equally-entitled agents having identical valuation function v . The maximin share is $5 + 3\varepsilon$, but the APS is $6 + 3\varepsilon$. Moreover, there exists an MMS allocation but no allocation is APS.

Proof. For any $X \subseteq [6]$, and $g \in [6] \setminus X$, we have

1. $|X| \leq 1 \implies v(g \mid X) = 2 + \varepsilon$.
2. $|X| = 2 \implies v(g \mid X) \in \{1 + \varepsilon, 2 + \varepsilon\}$.
3. $|X| = 3 \implies v(g \mid X) \in \{\varepsilon, 1 + \varepsilon\}$.
4. $|X| = 4 \implies v(g \mid X) = \varepsilon$.

Hence, $v(g \mid X)$ decreases with $|X|$, so v is submodular.

The MMS is less than 6 because for every set in C , its complement is not in C . The APS is $6 + 3\varepsilon$ by the dual definition of APS (Definition 18, set $x_S = 1/4$ for $S \in C$).

For identical valuations, an MMS allocation always exists. For identical valuations, in any allocation, some agent gets a bundle of value at most the MMS. Since the APS is strictly greater than the MMS, that agent is not APS-satisfied. Hence, no allocation is APS. $\square$

F.4 Subadditive Valuations

Lemma 118. For any $k \in \mathbb{N}$, define the set functions $f_1, f_2 : 2^{[m]} \rightarrow \mathbb{R}$ as $f_1(S) = \lfloor |S|/k \rfloor$ and $f_2(S) = \lceil |S|/k \rceil$, respectively. Then f_1 is superadditive, f_2 is subadditive, and for any $S \subseteq [m]$, $g \in [m] \setminus S$, and $i \in [2]$, we have $f_i(g \mid S) \in \{0, 1\}$.

Example 119. Let $([2], [9], (v_i)_{i=1}^2, -)$ be an instance where $v_i(S) := \lceil |S|/4 \rceil$ for all $i \in [2]$ and $S \subseteq [9]$. Then v_i is subadditive for $i \in [2]$ by Lemma 118. Let A be an allocation where agent 1 has 3 goods and agent 2 has 6 goods. Then A is GAPS+EFX+PROPx but not EF1 or PROP1.

Each agent's APS is ≤ 1 , because if we price each good at $1/9$, then a bundle is affordable iff its value is at most 1. Each agent's proportional share is $3/2$. Note that A is not EFX₀ (c.f. Definition 13); the difference between EFX and EFX₀ is crucial here.

Example 120. Let $([3], [7], (v_i)_{i=1}^3, -)$ be a fair division instance where $v(S) := \lceil |S|/2 \rceil$ for all $S \subseteq [7]$. Then v is subadditive by Lemma 118. Let $A := (\{1\}, \{2, 3, 4\}, \{5, 6, 7\})$. Then A is EF1 but is not PROP1 to agent 1.

Lemma 121 (optimal bin packing is subadditive). Let $(s_j)_{j=1}^m$ be a sequence where $0 \leq s_j \leq 1$ for all $j \in [m]$ (s_j is called the size of item j). For any $X \subseteq [m]$, define $\text{optBP}_s(X)$ as the smallest integer k such that a partition $(P_1, \dots, P_k)$ of X exists such that $\sum_{j \in P_i} s_j \leq 1$ for all $i \in [k]$. Then optBP_s is subadditive and has binary marginals, i.e., $\text{optBP}_s(X \cup \{g\}) - \text{optBP}_s(X) \in \{0, 1\}$.

Proof. Pick any two disjoint sets $X, Y \subseteq [m]$. Let $k_X := \text{optBP}_s(X)$ and $k_Y := \text{optBP}_s(Y)$. Let $(P_1, \dots, P_{k_X})$ and $(Q_1, \dots, Q_{k_Y})$ be the corresponding partitions. Then $(P_1, \dots, P_{k_X}, Q_1, \dots, Q_{k_Y})$ is a feasible partition of $X \cup Y$, so $\text{optBP}_s(X \cup Y) \leq k_X + k_Y$. $\square$

Example 122. Let $s = (1/5, 1/5, 3/5, 3/5, 3/5, 3/5)$ and $\varepsilon \in [0, 1/3]$. Consider an instance with 3 equally-entitled agents having an identical valuation function v over 6 goods. $v(X) := \text{optBP}_s(X) + |X|\varepsilon$ (optBP_s is defined in Lemma 121). Then by Lemma 121, v is subadditive, and for every $g \in [6]$, we have $v(g \mid \cdot) \in \{\varepsilon, 1 + \varepsilon\}$.

Then allocation $A = (\{1, 2\}, \{3, 4\}, \{5, 6\})$ is EFX, pairwise PROP, and groupwise APS (set the price of the first two goods to $1/2$ and the last four goods to 1). However, agent 1 is not PROP1-satisfied in A , since $v([6]) = 4 + 6\varepsilon$, and $v(A_1 \cup \{g\}) = 1 + 3\varepsilon$ for all $g \in [6] \setminus A_1$.

Lemma 123 ($\text{PAPS} + \text{PPROP} \not\Rightarrow \text{PROPM}$ or MIS for binary subadd). Let A_1, A_2 , and A_3 be pairwise-disjoint sets where A_1 contains 7 goods of size 0.6, A_2 contains 21 goods of size 0.4, and A_3 contains 21 goods of size 0.4. Let $M := A_1 \cup A_2 \cup A_3$. Let $v(X) := \text{optBP}(X)$ for all $X \subseteq M$. Consider a fair division instance with 3 equally-entitled agents having the same valuation function v over M . Then allocation $A := (A_1, A_2, A_3)$ is EFX+PPROP+PAPS, but A is not PROPM-fair or MIS-fair to agent 1.

Proof. First, we show that $v(S) = \max(|S \cap A_1|, \lceil |S|/2 \rceil)$ for any $S \subseteq M$. To see this, suppose S contains a items of size 0.6 and b items of size 0.4. If $b \leq a$, we require a to pack the items and $\lceil (a+b)/2 \rceil \leq a$. If $b \geq a$, then we require $\lceil (a+b)/2 \rceil$ bins to pack the items and $a \leq (a+b)/2$. As a corollary, we get that for all $S \subseteq M \setminus A_1$, we have $v(S \mid A_1) = \max(0, \lceil (|S| - |A_1|)/2 \rceil)$.

$v(A_2) = v(A_3) = 11$ and $v(A_1) = 7$, so agents 2 and 3 are envy-free. For any $S \subseteq A_2$, $v(S \mid A_1) > 0 \iff |S| \geq 8$, and $|S| \geq 8 \implies |A_2 \setminus S| \leq 13 \implies v(A_2 \setminus S) \leq 7 = v(A_1)$. Hence, agent 1 doesn't EFX-envy agent 2. Similarly, she doesn't EFX-envy agent 3. Hence, A is EFX.

$v(A_2 \cup A_3) = 21$, and $v(A_1 \cup A_2) = v(A_1 \cup A_3) = 14$. Hence, A is pairwise PROP. We will now show that A is pairwise APS. Set the price of each item to 1. It's easy to see that the allocation restricted to agents 2 and 3 is APS. Now suppose we restrict the allocation to agents 1 and 2. Then $S \subseteq A_1 \cup A_2$ is affordable iff $|S| \leq |A_1 \cup A_2|/2 = 14$. For $|S| = 14$, we get $v(S) = 7$, so the APS is at most 7. Hence, A is pairwise APS satisfied.

$v(M)/3 = 25/3 = 8 + 1/3$. For $S \subseteq A_2$ such that $|S| = 8$, we get that $v(S \mid A_1) = 1$, but $v(A_1 \cup S) = 8 < v(M)/3$. Hence, A is not PROPM-fair to agent 1. Also, $v(A_1 \cup \{g\}) = v(A_1) = 7$ for all $g \in M \setminus A_1$, so A is also not PROP1-fair to agent 1.

Suppose A is M1S-fair to agent 1. Let B be her M1S-certificate for A . Then $v(B_1) \leq v(A_1) = 7$ and B is EF1-fair to agent 1. $v(B_1) \leq 7$ implies $|B_1| \leq 14$. Since $|M| = 49$, we get that $\max(|B_2|, |B_3|) \geq \lceil |M - B_1|/2 \rceil \geq 18$. Hence, some agent $i \in \{2, 3\}$ has a bundle of value at least 9. If one good is removed from their bundle, its value can drop by at most 8, so B is not EF1-fair to agent 1, which is a contradiction. Hence, A is not M1S-fair to agent 1. $\square$

Lemma 124 (GMMS $\not\Rightarrow$ APS). *Let there be 15 items of sizes $65/96, 31/96, 31/96, 31/96, 23/96, 23/96, 23/96, 17/96, 11/96, 7/96, 7/96, 7/96, 5/96, 5/96, 5/96$. Consider a fair division instance with 3 equally-entitled agents having the same valuation function $v := \text{optBP}_s$ over the 15 items (c.f. Lemma 121). Then v is subadditive and has binary marginals. Also, the AnyPrice share is at least 2, but the maximin share is 1. Hence, no APS allocation exists, but the leximin allocation is GMMS.*

Proof. The total size of the items is $3 \times 97/96$. By Lemma C.1 of [BEF23], no 3-partition of the items exists such that the total size of each partition is at least $97/96$. Hence, in every 3-partition M , some bundle has total size at most 1, so that bundle has value at most 1. So, the maximin share is 1.

By Lemma C.1 of [BEF23], there exist 6 sets $(S_j)_{j=1}^6$ of items such that each set has total size $97/96$ (and hence value 2) and each item appears in exactly two sets. Hence, the APS is at least 2 (c.f. Definition 18). $\square$

G Feasibility of Fairness Notions

We list results regarding the feasibility and infeasibility of fairness notions in Tables 6 and 7, respectively.

Example 125 (PROP_x is infeasible). *An instance with 3 equally-entitled agents having identical additive valuations over 2 goods with values 10 and 1, respectively.*

Lemma 126 (WMXS is infeasible for goods, Theorem 8 of [SHY24]). *Let $0 < \varepsilon \leq 1/4$, and $\phi := (\sqrt{5} + 1)/2$. No MXS allocation exists for the fair division instance $([2], [4], (v_i)_{i=1}^2, w)$, where $w_1 = 1/(\sqrt{\phi} + 1)$, $w_2 = \sqrt{\phi}/(\sqrt{\phi} + 1)$, and agents have additive valuations given by the following table:*

g	1	2	3	4
$v_1(g)$	ε	1	ϕ	ϕ
$v_2(g)$	ε	ε	1	1

Proof sketch. Note that $1 + \varepsilon < \sqrt{\phi} = w_2/w_1$. The only bundles agent 2 is EFX-satisfied with are $\{1, 2, 3\}$, $\{1, 2, 4\}$, $\{3, 4\}$, and their supersets. Hence, $\text{MXS}_2 = 1 + 2\varepsilon$. The only bundles agent 1 is EFX-satisfied with are $\{1, 3\}$, $\{1, 4\}$, $\{2, 3\}$, $\{2, 4\}$, $\{3, 4\}$, and their supersets. Hence, $\text{MXS}_1 = \phi + \varepsilon$. One can check that no allocation is MXS. $\square$

Lemma 127 (WMXS is infeasible for chores, Theorem 17 of [SHY24]). *Let $0 < \varepsilon \leq 1/4$, and $\phi := (\sqrt{5} + 1)/2$. No MXS allocation exists for the fair division instance $([2], [4], (v_i)_{i=1}^2, w)$, where $w_1 = \sqrt{\phi}/(\sqrt{\phi} + 1)$, $w_2 = 1/(\sqrt{\phi} + 1)$, and agents have additive disutilities given by the following table:*

c	1	2	3	4
$-v_1(c)$	ε	1	ϕ	ϕ
$-v_2(c)$	ε	ε	1	1

Table 6: Feasibility of fairness notions

	notion	valuation	marginals	identical	n	entitlements	
1	EF1	–	dbl-mono*	–	–	equal	Theorem 4 of [BSV21]
2	EF1	additive	goods	–	–	–	Theorem 3.3 of [CISZ21]
3	EF1	additive	chores	–	–	–	Theorem 19 of [SHY24]
4	EF1+PO	additive	goods	–	–	–	[CKM ⁺ 19]
5	EF1+PO	additive	chores	–	–	–	[Mah26]
6	EF1+PO	additive	–	–	$n = 2$	–	Theorem 14 of [GS24]
7	MMS	additive	–	–	$n = 2$	equal	Cut-and-choose
8	WMMS	–	–	yes	–	–	Trivial
9	APS	additive	–	–	$n = 2$	–	Proposition 6 of [BEF23]
10	PROPx	additive	chores	–	–	–	Theorem 4.1 of [LLW22] [†]
11	PROPm	additive	goods	–	–	equal	[BGG21b]
12	PROPavg	additive	goods	–	–	equal	[KM25] [‡]
13	PROP1	additive	–	–	–	–	[AMS20]
14	EFX	additive	$\geq 0, \leq 0$	yes	–	–	Theorem 5 of [SHY24]
15	EEFX	cancelable	$\geq 0, \leq 0$	–	–	equal	[CGR ⁺ 22]
16	EEFX	–	$\geq 0, \leq 0$	–	–	equal	[AR25]
17	EFX	additive	bival goods	–	–	equal	Theorem 4.1 of [ABFR ⁺ 21]
18	MMS	additive	bival goods	–	–	equal	[Fei22]
19	MMS	additive	bival chores	–	–	equal	[Fei22]
20	MMS	submodular	$\{0, 1\}$	–	–	equal	Theorem 3.4 of [BV21]
21	EFX+PO	submodular	$\{0, 1\}$	–	–	equal	Theorem 1 of [BEF21]
22	MMS	submodular	$\{0, -1\}$	–	–	equal	Theorem 9 of [BNV23]
23	GMMS	–	–	yes	–	equal	Leximin is GMMS

* A function $v : 2^M \rightarrow \mathbb{R}$ is *doubly monotone* if there is a partition (G, C) of M such that $v(g \mid \cdot) \geq 0 \ \forall g \in G$ and $v(c \mid \cdot) \leq 0 \ \forall c \in C$.

† Algorithm 2 of [LLW22] must be slightly modified to meet our slightly stricter definition of PROPx. Change line 5 from ‘if $|v_i(X_i)| > w_i$ ’ to ‘if $|v_i(X_i)| \geq w_i$ ’.

‡ Algorithm 4.1 of [KM25] must be slightly modified to meet our slightly stricter definition of PROPavg. Change the definition of the PROPavg-graph to use a strict inequality instead.

Proof sketch. Note that $1 + \varepsilon < \sqrt{\phi} = w_1/w_2$. The only bundles agent 2 is EFX-satisfied with are $\{1, 2\}$, $\{3\}$, $\{4\}$, and their subsets. Hence, $\text{MXS}_2 = -1$. The only bundles agent 1 is EFX-satisfied with are $\{2, 3\}$, $\{2, 4\}$, $\{1, 3\}$, $\{1, 4\}$, $\{1, 2\}$, and their subsets. Hence, $\text{MXS}_1 = -\phi^2$. One can check that no allocation is MXS. $\square$

Example 128 (MMS is infeasible). Let $0 \leq a < b$. Let $([2], [4], (v_i)_{i=1}^2, -)$ be a fair division instance where $v_1(S) = |S|a + (b - a)(\mathbf{1}(S \supseteq \{1, 2\}) + \mathbf{1}(S \supseteq \{3, 4\}))$ and $v_2(S) = |S|a + (b - a)(\mathbf{1}(S \supseteq \{1, 3\}) + \mathbf{1}(S \supseteq \{2, 4\}))$. Then v_1 and v_2 are supermodular, each agent’s MMS is $a + b$, and no MMS allocation exists. In fact, the best multiplicative approximation to the MMS one can achieve is $2a/(a + b)$.

Example 129 (MMS is infeasible). Let $0 \leq a < b$. Let $([2], [4], (v_i)_{i=1}^2, -)$ be a fair division instance where $-v_1(S) = |S|b - (b - a)(\mathbf{1}(S \supseteq \{1, 2\}) + \mathbf{1}(S \supseteq \{3, 4\}))$ and $-v_2(S) = |S|b - (b - a)(\mathbf{1}(S \supseteq \{1, 3\}) + \mathbf{1}(S \supseteq \{2, 4\}))$. Then v_1 and v_2 are supermodular, each agent has MMS $-(a + b)$, and no

Table 7: Infeasibility of fairness notions

	notion	valuation	marginals	identical	n	entitlements	
1	PROP	$m = 1$	$1, -1$	yes	any	equal	Example 74
2	APS	submod	> 0	yes	$n = 2$	equal	Lemma 117
3	APS	additive	$> 0, < 0$	yes	$n = 3$	equal	Lemma C.1 of [BEF23]
4	APS	subadd	$\{0, 1\}$	yes	$n = 3$	equal	Lemma 124
5	MMS	additive	$> 0, < 0$	no	$n = 3$	equal	[FST21]
6	MMS	XOS	$\{0, 1\}$	no	$n = 2$	equal	Theorem 3.5 of [BV21]
7	PROPx	additive	> 0 bival	yes	$n = 3$	equal	Example 125
8	PROPM	additive	mixed bival	yes	$n = 3$	equal	Lemma 94
9	MXS	additive	> 0	no	$n = 2$	unequal	Lemma 126
10	MXS	additive	< 0	no	$n = 2$	unequal	Lemma 127
11	PROP1+M1S	additive	1	yes	$n = 3$	unequal	Lemma 100
12	PROP1	supermod	≥ 0 bival	yes	$n = 2$	equal	Lemma 104
13	PROPM	supermod	≥ 0 bival	yes	$n = 2$	equal	Lemma 104
14	MMS	supermod	≥ 0 bival	no	$n = 2$	equal	Lemma 128
15	MMS	supermod	≤ 0 bival	no	$n = 2$	equal	Lemma 129
16	MMS	submod	> 0	no	$n = 2$	equal	Lemma 130
17	PMMS	–	≥ 0	no	$n = 3$	equal	Theorem 1 of [BMP26]

MMS allocation exists. In fact, the best multiplicative approximation to the MMS one can achieve is $2b/(a+b)$.

Lemma 130 (MMS is infeasible, Theorem 4.1 of [GHS⁺18]). *Let $([2], [4], (v_i)_{i=1}^2, -)$ be a fair division instance, where v_1 and v_2 are as follows ($\varepsilon \in \mathbb{R}_{\geq 0}$):*

S	$v_1(S) - S \varepsilon$	$v_2(S) - S \varepsilon$
$ S \leq 1$	$ S $	$ S $
$\{1, 2\}, \{3, 4\}$	2	$3/2$
$\{2, 3\}, \{1, 4\}$	$3/2$	2
$\{1, 3\}, \{2, 4\}$	$3/2$	$3/2$
$ S \geq 3$	2	2

Then v_1 and v_2 are submodular, each agent's MMS is $2 + 2\varepsilon$, and no MMS allocation exists.

Proof. For any $i \in [2]$, $S \subseteq [4]$, and $g \in [4] \setminus S$, we have

1. $S = \emptyset \implies v_i(g \mid S) = 1 + \varepsilon$.
2. $|S| = 1 \implies v_i(g \mid S) \in \{1/2 + \varepsilon, 1 + \varepsilon\}$.
3. $|S| = 2 \implies v_i(g \mid S) \in \{\varepsilon, 1/2 + \varepsilon\}$.
4. $|S| = 3 \implies v_i(g \mid S) = \varepsilon$.

Hence, $v_i(g \mid S)$ decreases with $|S|$, so v_i is submodular. It is easy to verify that each agent's MMS is $2 + 2\varepsilon$, and that no allocation is MMS. $\square$

H Implicit Representation of Set Families

Section 5 introduced the concept of *conditional predicate implications*, and presented an algorithm for inferring additional implications and counterexamples. In fair division, we defined Ω to be the

set of all pairs $(\mathcal{I}, A)$, where $\mathcal{I}$ is a fair division instance and A is an allocation for $\mathcal{I}$. We defined $\mathcal{F}$ to be a set family over Ω , and in fair division, each set in $\mathcal{F}$ represents a setting. But Ω and the sets in $\mathcal{F}$ are uncountable, so how do we compute with them? This is the question we answer formally in this section.

H.1 Mappings from Partial Orders

Definition 22. A set family $\mathcal{F} \subseteq 2^\Omega$ is represented by a partial order $(P, \preceq)$ if there exists an order-preserving surjective mapping $f : P \rightarrow \mathcal{F}$, i.e., for all $S \in \mathcal{F}$, there exists $x \in P$ such that $f(x) = S$, and for all $x, y \in P$, we have $x \preceq y \implies f(x) \subseteq f(y)$.

Note that the converse is not required to be true, i.e., $f(x) \subseteq f(y)$ need not imply $x \preceq y$. Hence, if P is an antichain, then P trivially represents $\mathcal{F}$. However, such a representation is useless. The more a representation captures the subset relations in $\mathcal{F}$, the better that representation is.

Example 131. Let E and O be the sets of even and odd integers, respectively, i.e., $E := \{2i : i \in \mathbb{Z}\}$ and $O := \{2i + 1 : i \in \mathbb{Z}\}$. Then the set family $\mathcal{F} := \{E, O, \mathbb{Z}\}$ can be represented by the partial order $(\{e, o, a\}, \{e \preceq e, a \preceq a, o \preceq o, e \preceq a, o \preceq a\})$ over three elements, where the corresponding mapping f is given by $f(e) = E$, $f(o) = O$, and $f(a) = \mathbb{Z}$.

Hence, for the conditional predicate implication problem, if we can represent a set family $\mathcal{F} \subseteq 2^\Omega$ by a finite partial order $(P, \preceq)$, then we can indirectly specify the sets that implications and counterexamples are conditioned on by elements in P . In fact, for computation, we do not even need to know the set $\mathcal{F}$ and the mapping f ; we can just work with elements in P instead. In the algorithm for inferring additional implications and counterexamples, we perform several checks of the form $S \subseteq T$, where $S, T \in \mathcal{F}$. We replace them with checks of the form $x \preceq y$, where $f(x) = S$ and $f(y) = T$.

H.2 Partial Order for Fair Division Settings

We represent a fair division setting as a 5-tuple, as specified in Section 2.2. We now explain how to define a partial order on these 5-tuples, and how to map each 5-tuple to a subset of Ω .

Definition 23 (Product order). Let $((P_i, \preceq_i))_{i=1}^k$ be a sequence of partial orders. Their product is another partial order $(P, \preceq)$, where $P := \prod_{i=1}^k P_i := \{(p_i)_{i=1}^k : p_j \in P_j \forall j \in [k]\}$ and $(p_1, \dots, p_k) \preceq (q_1, \dots, q_k)$ iff $p_i \preceq_i q_i$ for all $i \in [k]$.

Example 132. The product of $(\mathbb{N}, \leq)$ with itself is $(\mathbb{N}^2, \preceq)$, where $(a_1, a_2) \preceq (b_1, b_2)$ iff $a_1 \leq b_1$ and $a_2 \leq b_2$.

Recall the 5 features of fair division from Section 2.2: whether entitlements are equal, whether there are only two agents, whether agents have identical valuations, valuation function type, and marginal values. We define a partial order for each of these 5 features. The first three features are represented by the *boolean* partial order: $(\{\text{true}, \text{unknown}\}, \{\text{true} \preceq \text{unknown}, \text{true} \preceq \text{true}, \text{unknown} \preceq \text{unknown}\})$. The partial orders for the last two features are represented as DAGs in Figs. 9a and 9b, respectively, in Appendix A. (Formally, for a DAG $G = (V, E)$, the corresponding partial order is $(V, \preceq)$, where $u \preceq v$ iff there is a path from u to v in G .) Let $(P, \preceq)$ be the product of these 5 partial orders. For a fair division setting $s \in P$, let $f(s) := \{(\mathcal{I}, A) : \mathcal{I} \text{ is an instance consistent with } s, A \text{ is an allocation for } \mathcal{I}\}$, and $\mathcal{F} := \{f(s) : s \in P\}$. It is easy to check that f is order-preserving and surjective. This completes our description of how to map fair division settings to subsets of Ω .

Note that f is not injective. The settings $s_1 := (\text{unknown}, \text{unknown}, \text{true}, \text{additive}, \{1\})$ and $s_2 := (\text{unknown}, \text{unknown}, \text{unknown}, \text{general}, \{1\})$ map to the same set in $\mathcal{F}$, because if each item's marginal value is 1, then valuations are identical and additive. Querying the inference engine with s_2 may fail to infer implications that rely on additivity or identical valuations. Note that $s_1 \preceq s_2$. Among equivalent settings, querying the inference engine with a minimal setting gives the most informative results, provided that counterexamples fed to the engine are also conditioned on minimal settings.